

The Physics of Physiology: Part 1

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Physics of Physiology - Physics of Physiology lectures - Google Drive



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Lecture 1: *Physical units for physiology & introduction to multiscale modeling*

Lecture 1: *Physical units for physiology*

Life depends on the **flow of energy and increase in entropy** associated with mechanical, electromagnetic and chemical processes. By the end of this lecture you should understand why we choose the **six fundamental units** needed for all the biophysical mechanisms that underpin physiology. We also show how processes at the macroscale of physiology are linked with atomic level physics and that the small number of physical constants needed for physiology (e.g. Faraday's constant and the gas constant) use Avogadro's number to bridge the scale from atoms to cells and tissues.

The overall approach to multiscale modeling of human physiology is also described briefly here, in order to motivate the modeling techniques presented in upcoming lectures.

7 units:

- **Joule (J)**
 - **Second (s)**
 - **Meter (m)**
- } space-time + energy
- **Coulomb (C)** - count electrons
 - **Candela (Cd)*** - count photons
 - **Mole (mol)** - count atoms
 - **Entropy (e)** - count probable states

*A photon, with energy $h\nu$, is a quantum of an electromagnetic field. The Candela is only needed when a model requires the number of discrete photons, rather than just their summed energy. Mammalian eyes can detect 1 photon ([ref](#)).

Physiological processes

1. **Mechanics (J,s,m,e):** (i) Solids; (ii) Fluids
2. **Heat transfer (J,s,e)**
3. **Electro-physiology (J,s,C)**
4. **Electro-magnetic (J,s,m,C)**
5. **Membrane transporters (J,s,mol,C)**
6. **Metabolic pathways (J,s,mol)**

Physical constants

$$c \approx 3.10^8 \text{ m.s}^{-1} (=299\,792\,458 \text{ m.s}^{-1})$$

$$N_A = 6.022 \times 10^{23} \text{ particles.mol}^{-1} + k_B = 1.38 \times 10^{-23} \text{ J.K}^{-1} \Rightarrow R = k_B \cdot N_A = 8.314 \text{ J.mol}^{-1}.\text{K}^{-1} \Rightarrow RT \cong 2.5 \text{ kJ.mol}^{-1} \text{ at } 300\text{K}$$

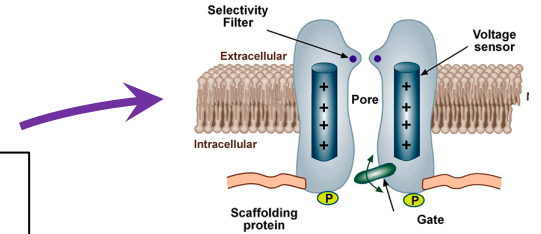
$$+ q_e = 1.602 \times 10^{-19} \text{ C} \Rightarrow F = q_e \cdot N_A = 0.965 \times 10^5 \text{ C.mol}^{-1}$$

For n moles of an ideal gas, RT (J.mol^{-1}) $\times n$ (mol) gives nRT (J), which is expressed by the ideal gas law as $PV=nRT$ (J) – see later.

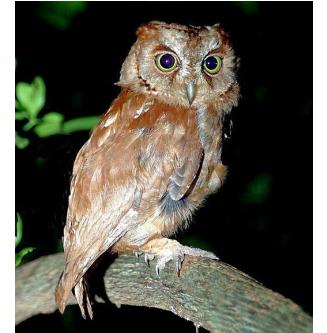
$$\Delta u^q = RT \ln \frac{q_o}{q_i} + z_q F \Delta u^{membrane}$$

$\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
 J.mol^{-1} J.mol^{-1} C.mol^{-1} J.C^{-1}

Equation 5-6 in Boron & Boulpaep textbook (3rd Edⁿ).



Detect 1 electron

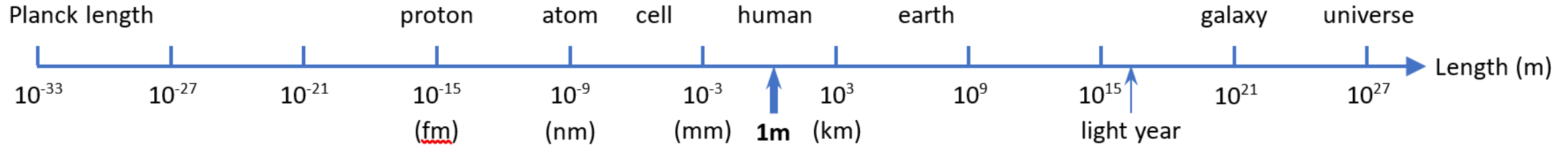


Detect 1 photon

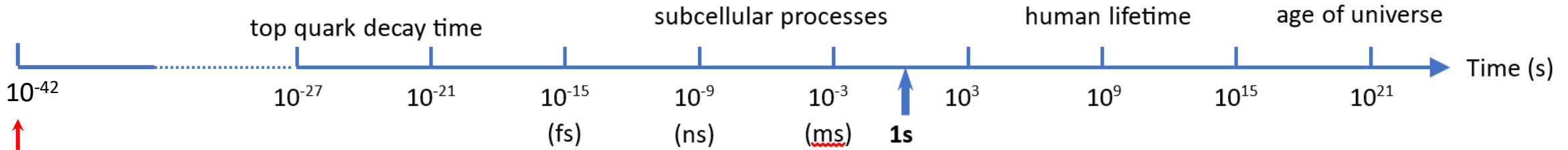


300m olfactory sensors
1 part in 10¹²

Length in meters



Time in seconds



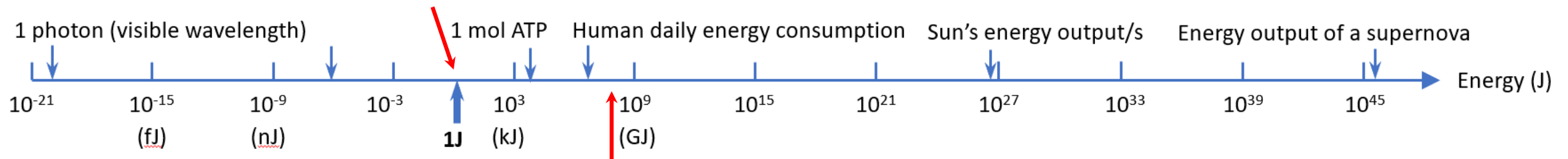
‘Planck time’ (time for photon to traverse a Planck length) = $10^{-33}/3 \times 10^8 \text{ m.s}^{-1}$

1J is energy of a 0.1 kg apple dropping 1m to give Newton a bump on the head.

1J is also the energy output by the heart in one beat.

4J per gram per degree Kelvin is the heat capacity of water (= 4 kJ.kg⁻¹.K⁻¹).

Energy in Joules



300 MJ is the energy contained in a fully charged Tesla car battery
.. and is the energy stored in 10 kg of body fat (30 MJ/kg)

1 kJ.s⁻¹ (1 kW) is power output of an electric kettle (1L or 1kg).
Heating 1 kg of water from 0-100 degK takes 400s (→ 400kJ).

Respiration: ΔG for ATP $\approx 30 \text{ kJ.mol}^{-1}$. 1 mol glucose (180g) generates 36 mol ATP i.e. $\Delta G \approx 1 \text{ MJ.mol}^{-1}$, so 180g bar of chocolate is 1MJ.

Note on unit prefixes and frequently used constants

Prefix

milli-	10^{-3}
micro-	10^{-6}
nano-	10^{-9}
pico-	10^{-12}
femto-	10^{-15}

- The volume of a cell is about 1 pL ($=10^{-12}$ L or 10^{-15} m³) which is $(10^{-5}$ m)³ or $(10\text{ }\mu\text{m})^3$.

A concentration of 1 mM (= millimoles per Liter) is 10^{-3} mol.L⁻¹ or 1 mol.m⁻³.

1 mM x 1 pL is 1 mol.m⁻³ x 10^{-15} m³ or 10^{-15} mol or 1 fmol (femto is 10^{-15}).

So if we assume a volume of **1 pL**, then **1 mM** is equivalent to a molar quantity of **1 fmol**.

- $R = k_B \cdot N_A = 8.314 \text{ J.mol}^{-1}.\text{K}^{-1}$

Therefore at 300K, **$RT \approx 2.5 \text{ kJ.mol}^{-1}$** x 10^{12} nJ.kJ⁻¹ x 10^{-15} mol.fmol⁻¹ gives **$RT \approx 2.5 \times 10^{-3} \text{ nJ.fmol}^{-1}$**

- Faraday's constant is approx $F = 0.965 \times 10^5 \text{ C.mol}^{-1}$ or $0.965 \times 10^{-10} \text{ C.fmol}^{-1}$ or 0.0965 nC.fmol⁻¹

1 V is 1 J.C⁻¹ or 1 nJ.nC⁻¹ and 1 nA is 1 nC.s⁻¹.

$$\frac{RT}{F} \approx 2.5 \text{ kJ.mol}^{-1} / 10^5 \text{ C.mol}^{-1} = 0.025 \text{ J.C}^{-1} \quad \text{or} \quad \frac{RT}{F} \approx \mathbf{25\text{mV}}$$

Thermodynamics[†]:

Work and heat are two different faces of a single concept - *energy*.

.. the English physicist James Joule figured this out and hence the unit of energy is the Joule (J).

Energy is the *capacity to do work*.

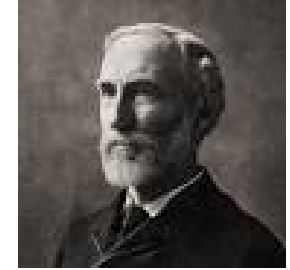
Work is *motion against an opposing force*.

Heat (thermal vibration of atoms) is intimately linked with entropy (see next slide).

An **open system** can exchange both energy and matter with its surroundings.

A **closed system** can exchange energy but not matter with its surroundings

An **isolated system** is cannot exchange either energy or matter with its surroundings



Josiah Willard Gibbs

The conservation laws of physics:

Conservation of mass, **Conservation of charge** and **Conservation of energy**

First law of thermodynamics: *The internal energy of an isolated system is constant* (Conservation of energy)

Second law of thermodynamics: *The entropy of an isolated system tends to increase* (Constraint on energy flow)

[†] Peter Atkins, Physical Chemistry: Thermodynamics, Structure, and Change. 11th Edn 2018

Energy and entropy (measures distribution of energy)



$$\text{Energy} = h\nu$$

0.5 μm
Highly directed
(low entropy)

1.3 $\text{kJ.s}^{-1}.\text{m}^{-2}$
(3.5 $10^{14} \text{ kJ.s}^{-1}$)



10 μm
Undirected
(high entropy)

For every 1 photon at 0.5 μm reaching the earth, 20 photons at 10 μm radiate away. Energy is conserved, but entropy increases.

Energy flow

q_i - quantity (qty) in units of m or m³, C, mol, or entropy

$v_i = \frac{dq_i}{dt}$ - rate of change ('flow' or 'flux'). Quantity per second (qty.s⁻¹).

u_i - potential (drives the flow v_i), and is measured in J.(qty)⁻¹

mechanical potential (J.m ⁻¹ , J.m ⁻² , J.m ⁻³)	drives movement (m.s ⁻¹ , m ² .s ⁻¹ , m ³ .s ⁻¹)
electrical potential (J.C ⁻¹)	drives electrical current (C.s ⁻¹)
chemical potential (J.mol ⁻¹)	drives molar flux (mol.s ⁻¹)
thermal potential (J.entropy ⁻¹ = K)	drives heat flow (entropy.s ⁻¹)

The product: potential (J.qty⁻¹) x flow (qty.s⁻¹) = power (J.s⁻¹)

We will study some protein families

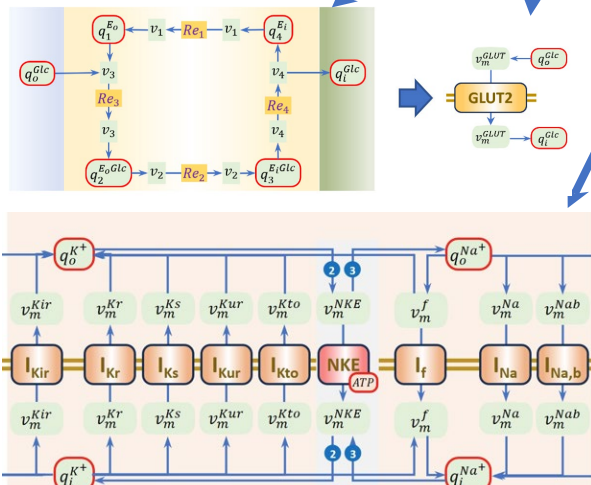
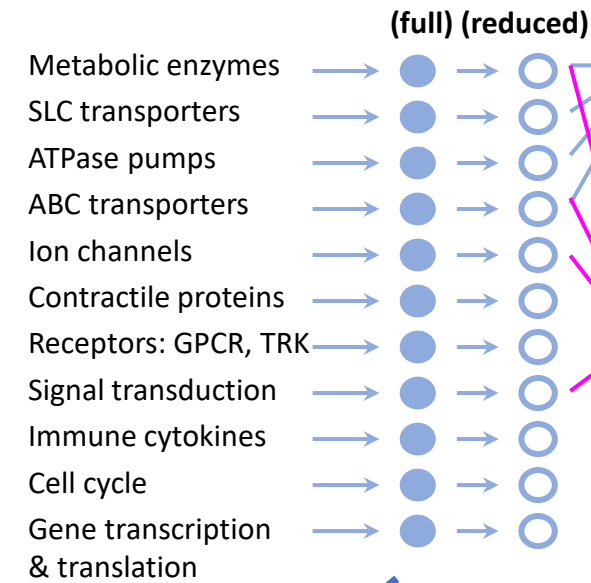
- **SLC transporters**
 - transport of small molecules across membranes via ‘facilitated’ diffusion or using an ion gradient (usually sodium)
- **ATPase pumps**
 - energy-dependent transport
- **ABC transporters**
 - (ATP-Binding Cassette) energy-dependent transport
- **Ion channels**
 - high volume driven by concentration gradient but usually transient with voltage-dependent gating
- **Metabolic enzymes**
 - tissue-specific control of metabolism
- **Contractile proteins**
 - actin & myosin interact to generate mechanical force
- **Aquaporins**
 - water transport through membrane channels
- **Receptors: GPCR, TRK, ...**
 - convey external signals to intracellular signalling pathways

And functional cell units (FCUs)

- **Glucose transport & Na/K regulation**
- **Intracellular pH regulation**
- **Metabolic pathways**
- **Action potentials**
- **Myofilament mechanics & calcium handling**

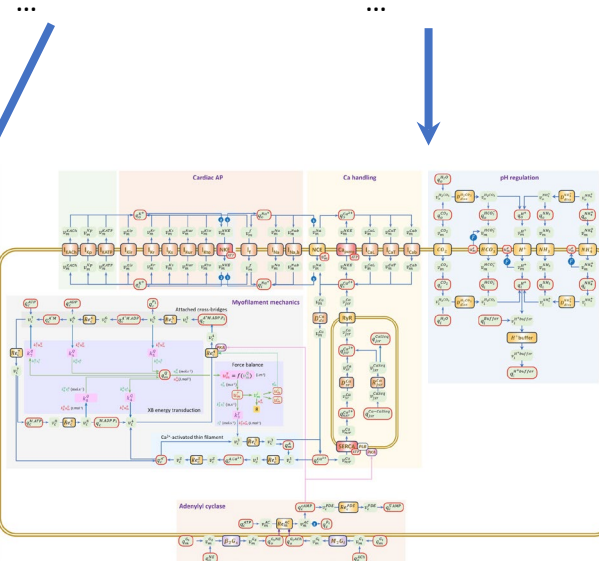
Overview of physics-based multiscale modelling

1. BG protein templates



2. FCUs

Metabolism
Action potentials
Calcium handling
pH regulation
Myofilament mechs
Receptors & signaling
Glucose transport
Na/K regulation
Cell cycle
Cell adhesion
Gene regulation



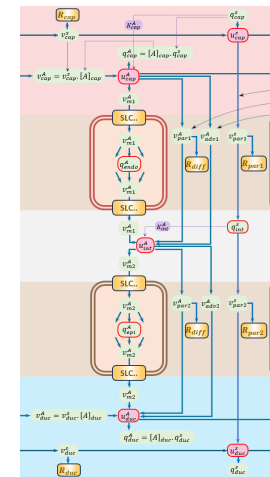
3. Cells

Endothelial cell
Epithelial cell
Cardiomyocyte
Muscle fiber
Fibroblast
Smooth muscle cell
Osteocyte
Enterochromaffin cell
Hepatocyte
Goblet cell
Red blood cell

4. FTUs

(including 3D structure)

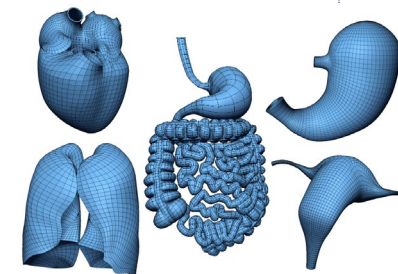
Glomerulus, ...
Lung alveoli
Myocardial sheet
Liver lobule
Bone osteon
Fibre, etc
...



5. Organs

● =surrogate

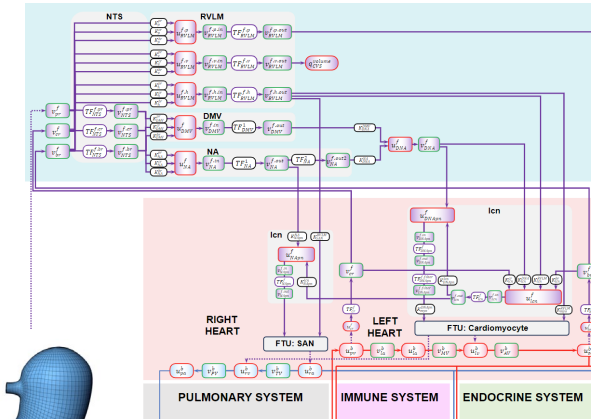
nephron → Kidney
acinus → Lung
block → Heart
lobules → Liver
osteon → Bone
fascicle → Muscle
... → ...



6. Systems

(3D anatomical model of the body)

Regulation of fluid volume
Regulation of arterial blood pressure
Regulation of pH
Regulation of electrolyte balance
Regulation of glucose
Regulation of O₂ and CO₂
Regulation of calcium
Regulation of phosphate
Regulation of body temperature
...



Finite element models that are subsequently reduced (with AI methods) to surrogate models.

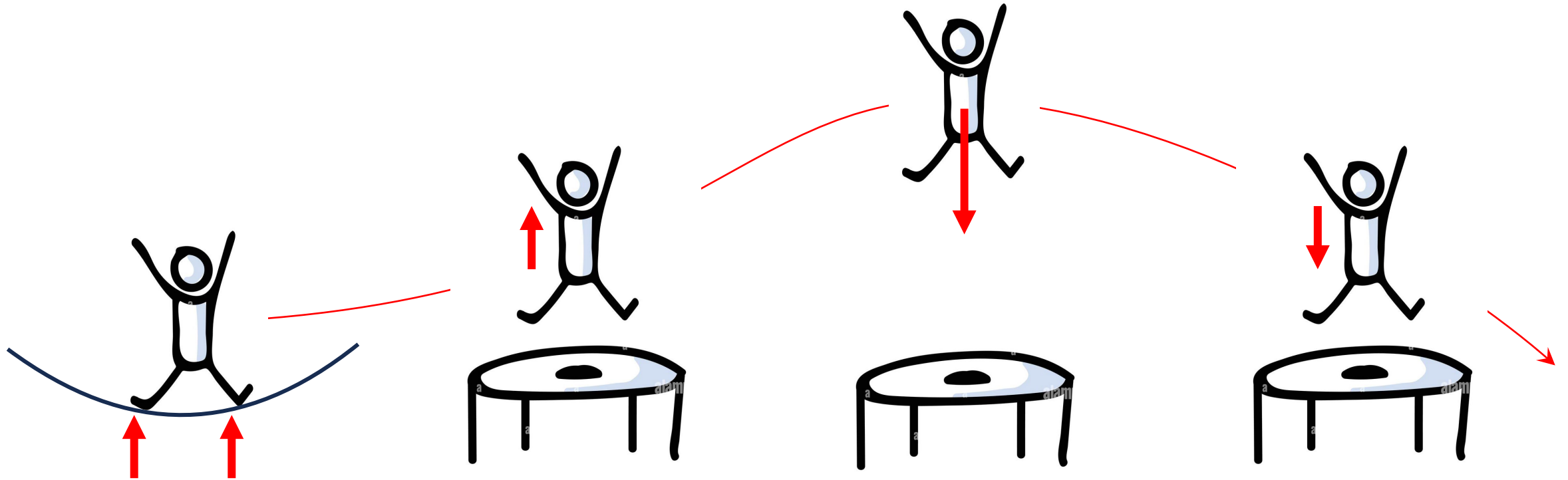
Lecture 2: *Physical laws and bond graphs*

Lecture 2: *Physical laws and bond graphs*

Energy can be **transmitted**, **stored**, and **converted** between **mechanical energy**, **electromagnetic energy** and **chemical energy** (including heat) ... nothing else! Almost every physiology process uses ALL of these 3 energetic mechanisms ... think of a muscle cell: chemical processes generate an electrical current to activate the cell, the mechanical energy for contraction (and the generated heat from all three helps to maintain body temperature). By the end of this lecture you should understand the physical laws (**conservation of mass**, **conservation of charge**, and **conservation of energy** – all coming from one underlying principle .. **conservation of power**) that govern the flow and storage of energy and its conversion between the 3 physical processes.

You should also understand how all physiological processes can be described by one simple but very powerful way of modelling them, called 'bond graphs'. You will learn that energy can be stored mechanically with a spring, electrically with a capacitor, or chemically as a solute in a solution – but that these are just variations on the common process of *energy storage*. The wide variety of ad-hoc descriptions of physical processes often presented in physiology text books, such as Fick's law of diffusion, Ohms' law & Fourier's law of heat conduction, are just variations on the underlying principle of *a potential gradient driving a flow of a quantity (that has the units described in Lecture 1)*, through a resistive element that is either a mechanical damper, an electrical resistance, or a chemical reaction. But they unhelpfully confuse laws of physics with empirical material properties.

Energy is either potential (due to position) or kinetic (due to motion)



Potential energy
(elastic trampoline)

Kinetic energy
 $\left(\frac{1}{2}mv^2\right)$

Potential energy
(gravity .. mgh)

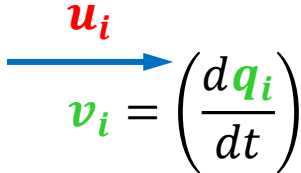
Kinetic energy
 $\left(\frac{1}{2}mv^2\right)$

Energy can be **transmitted**, **stored** and **converted** between **mechanical energy**, **electromagnetic energy** & **chemical energy** (including **heat**) ... nothing else! (at the physiology scale).

Energy transmission

mechanical potential (J.m ⁻¹ , J.m ⁻² , J.m ⁻³)	drives movement (m.s ⁻¹ , m ² .s ⁻¹ , m ³ .s ⁻¹)
electrical potential (J.C ⁻¹)	drives electrical current (C.s ⁻¹)
chemical potential (J.mol ⁻¹)	drives molar flux (mol.s ⁻¹)
thermal potential (J.entropy ⁻¹)	drives heat flow (entropy.s ⁻¹)

Let's represent that transmission by a line 

We need to also represent the flow v_i and the potential driving that flow u_i : 

where v_i is in **m.s⁻¹** **m³.s⁻¹** **C.s⁻¹** (Amp) **mol.s⁻¹** or **e.s⁻¹**

and u_i is in **J.m⁻¹** (Newton) **J.m⁻³** (Pascal) **J.C⁻¹** (Volt) **J.mol⁻¹** or **J.e⁻¹** (deg Kelvin)

$u_i \times v_i$ is **J.s⁻¹** **J.s⁻¹** **J.s⁻¹** **J.s⁻¹** **J.s⁻¹**

$u_i \times v_i$ is always power!


is the 'bond' in a 'bond graph'.

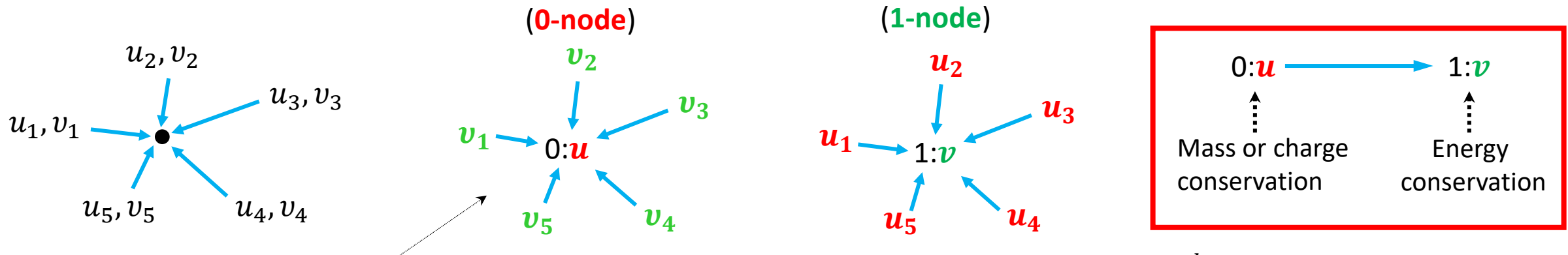
Bond graph 0:node and 1:node



Henry Paynter, MIT

Conservation of power: $u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4 + u_5 v_5 + \dots = 0$

$$\sum u_i v_i = 0 \left\{ \begin{array}{l} \text{If } u_1 = u_2 = u_3 = u_4 = u_5 = \mathbf{u} \text{ then } \sum \mathbf{v}_i = 0 \text{ .. This is conservation of mass or charge}^\# \\ \text{If } v_1 = v_2 = v_3 = v_4 = v_5 = \mathbf{v} \text{ then } \sum \mathbf{u}_i = 0 \text{ .. This is conservation of energy} \end{array} \right.$$



Note: To sum to zero, some of these flows must be negative! $^\# \sum v_i = \frac{d}{dt} (\sum q_i) = 0$, so $\sum q_i$ is constant.

Note: $\sum \mathbf{v}_i = 0$ only contains $\mathbf{v}_i$
And $\sum \mathbf{u}_i = 0$ only contains $\mathbf{u}_i$

We also need to find a relationship between $\mathbf{u}_i$ and $\mathbf{v}_i$ (or $\mathbf{q}_i$ or $\frac{d\mathbf{v}_i}{dt}$)

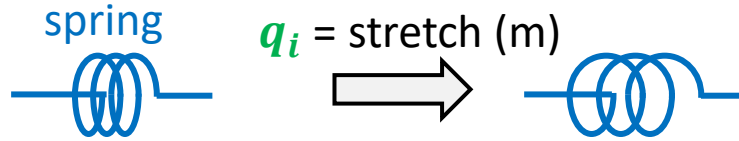
Material property or 'constitutive law' or 'phenomenological relation'

Energy storage

Potential energy (static)

Kinetic energy (dynamic)

Mechanical



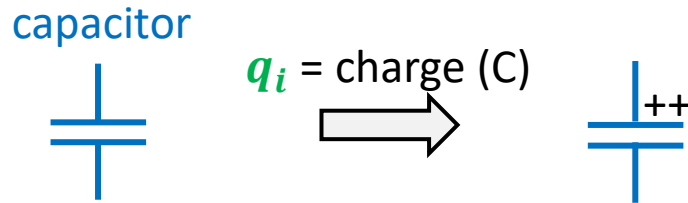
potential $u_i = f(q_i)$

$u_i = k \cdot q_i$

or $u_i = k \cdot \left(\frac{q_i - q_0}{q_0} \right) = k \bar{q}_i$

Hooke's law ' $f=kx$ '
where k is stiffness

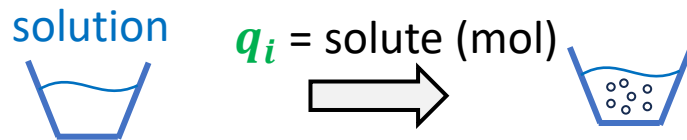
Electrical



$u_i = \frac{1}{C} \cdot q_i$

C is capacitance

Chemical



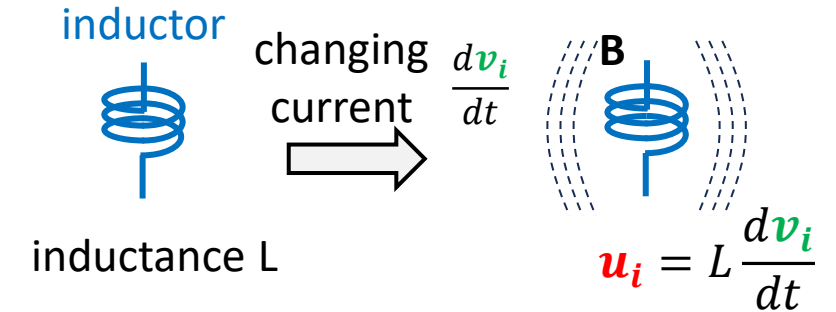
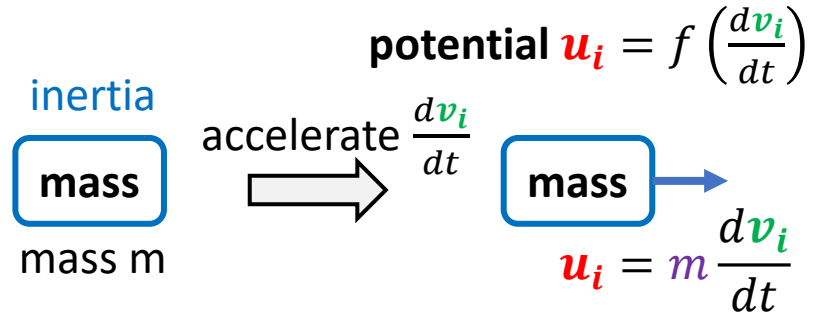
$u_i = RT \ln(K_i q_i)$

or $u_i = RT \ln\left(\frac{q_i}{q_0}\right)$

or $u_i = RT \ln(\bar{q}_i)$

Boltzmann eqn

Note that the static potential u_i is always defined relative to a reference.



heat

Heat is kinetic energy of particles
i.e. reflects average velocity of particles

Gas law: $PV=nRT$

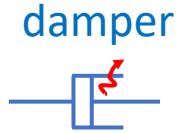
These are 'constitutive laws' and 'material constants' – found by experiment
(e.g. elasticity, electrical capacitance, thermodynamic constant)

Energy dissipation (conversion to heat)



Aharon Katchalsky

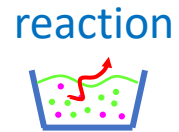
Mechanical



Electrical

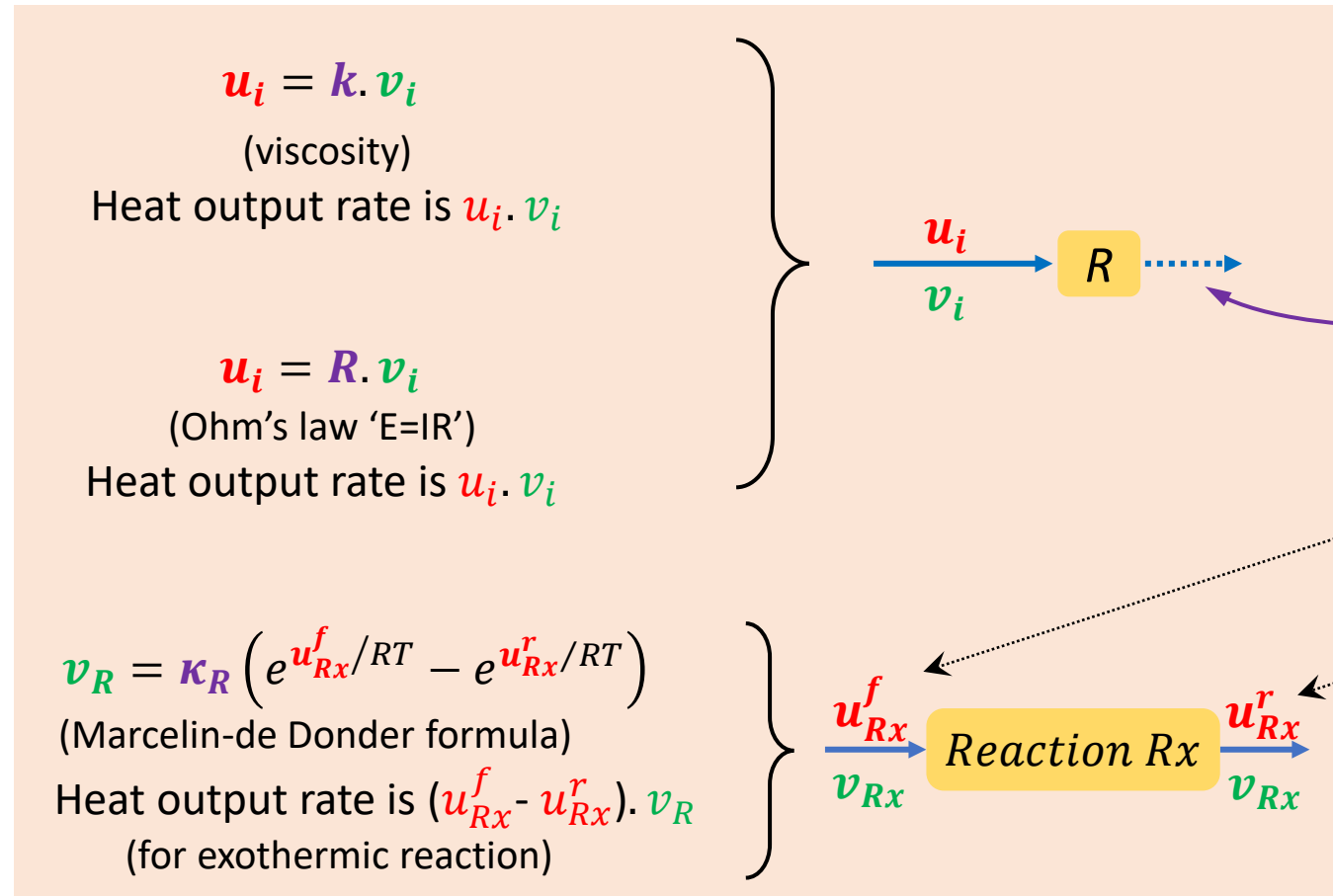


Chemical



Potential $u_i = f(v_i)$

Bond graph



(1-port node)

.. but only because we ignore the heat output!

forward potential
(sum of reactant potentials)

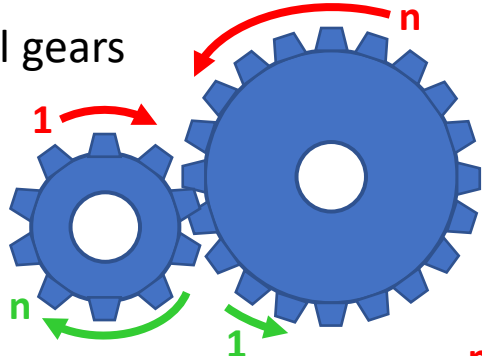
reverse potential
(sum of product potentials)

(2-port node)

These are 'constitutive laws' and 'material constants' – found by experiment
(e.g. viscosity, resistance, reaction rate)

Energy conversion: Bond graph 'transformers'

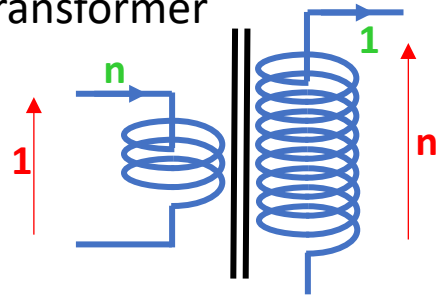
Mechanical gears
(ratio n)



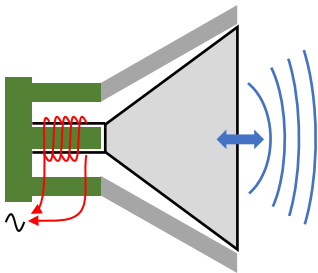
Mechanical lever
(ratio n)



Electrical transformer
(ratio n)

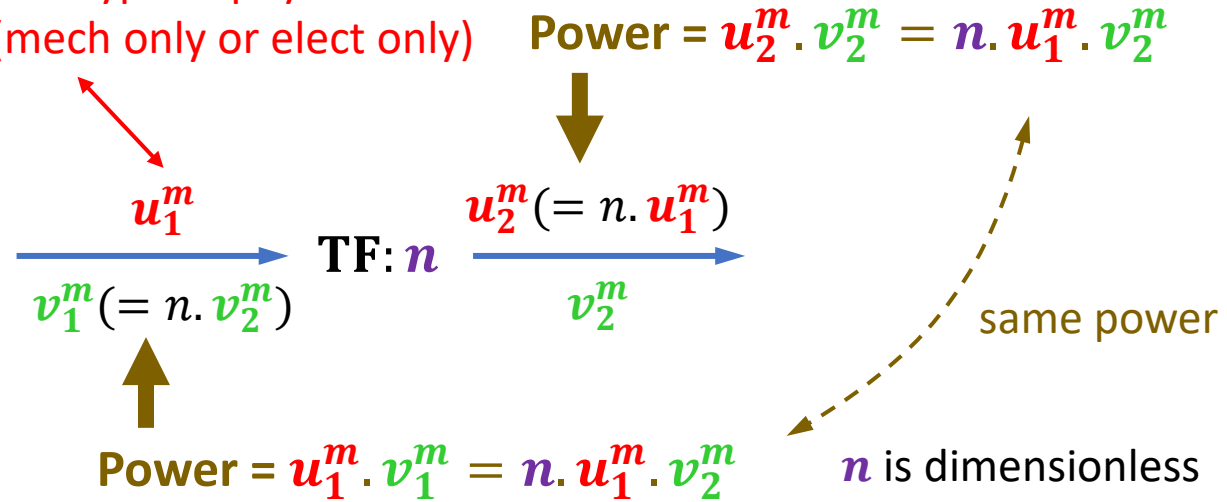


Mechano-electric coupling

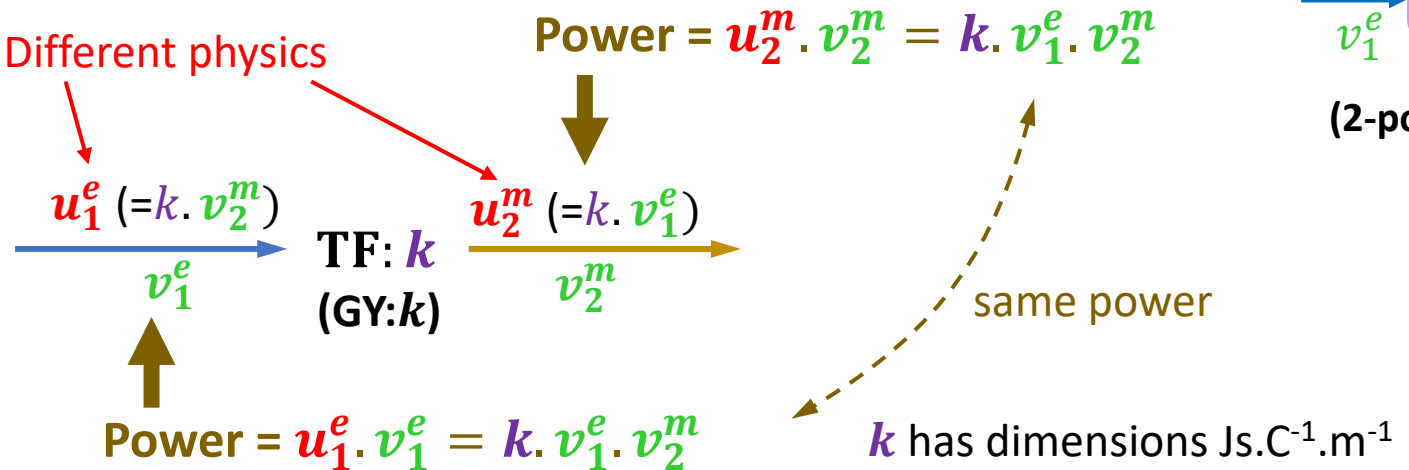


(voice coil, coupling via magnetic field)

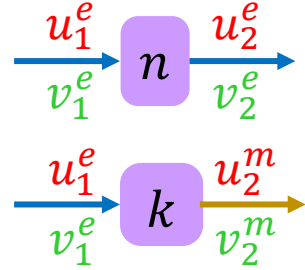
One type of physics
(mech only or elect only)



Different physics



Bond graph



(2-port node)

Summary

Energy is either *potential energy* (associated with position) or *kinetic energy* (associated with movement).

Energy *transmission* is indicated by a *bond* $\xrightarrow[v_i]{u_i}$ that carries the *flow* $v_i = \left(\frac{dq_i}{dt}\right)$, driven by *potential* u_i .

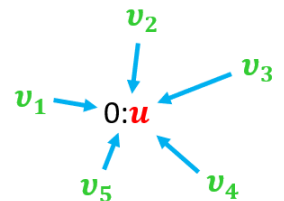
The quantity q_i has units m (or m³), C, mol, or e, encompassing *mechanics* (solid & fluid), *electromagnetics* and *chemistry*. Sometimes useful to express as a nondimensional quantity. E.g. strain $\bar{q}_i = \frac{q_i - q_0}{q_0}$ in mechanics, or $\bar{q}_i = \frac{q_i}{q_0}$ for biochemistry.

The potential (energy gradient) u_i has units J.m⁻¹ (N), J.m⁻³ (Pa), J.C⁻¹ (V), J.mol⁻¹, J.e⁻¹ (K).

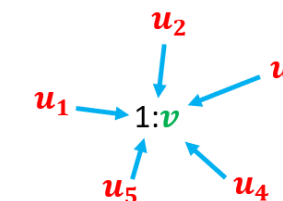
The product $u_i.v_i$ has units of power (J.s⁻¹). i.e. J.m⁻¹.m.s⁻¹, J.m⁻³.m³.s⁻¹, J.C⁻¹.C.s⁻¹, J.mol⁻¹.mol.s⁻¹, or J.e⁻¹.e.s⁻¹.

Power conservation is *mass or charge conservation* at a 0:node (0: u) and *energy conservation* at a 1:node (1: v).

These are both in multiport junctions:

 $\sum u_i v_i = u. \sum v_i = 0$

Mass/charge conservation

 $\sum u_i v_i = v. \sum u_i = 0$

Energy conservation

The potential u_i and flow v_i are connected by an empirical *material law* for energy storage, dissipation or conversion.

Energy *storage* is either *static* (stretched *spring*, charged *capacitor*, or solute in *solution*) or dynamic (*mass*, *inductance*)

Energy *dissipation* (conversion to heat) is via an electrical or mechanical *resistance* R (a 1-port device): $\longrightarrow R$

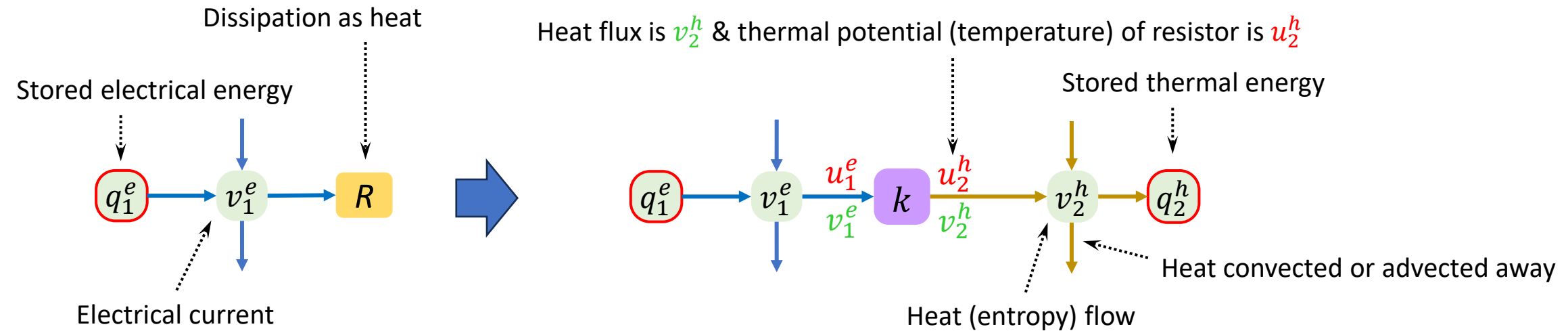
or via a chemical *reaction* R (a 2-port device): $\longrightarrow \text{Reaction Rx} \longrightarrow$

Energy *conversion* (mechanical-electrical-chemical) is handled via a 2-port *transforming factor* (TF): $\longrightarrow k \longrightarrow$

Accounting explicitly for heat flow

The heat output from a 1-port dissipation mechanism is carried away into the surroundings. Usually there is no need to further consider this heat (other than knowing its magnitude).

But if necessary, the 1-port electrical resistor or mechanical damper can be replaced by a 2-port TF node in order to explicitly model the heat flow, as shown below.



The conversion factor k depends both on the material through which the current is passing and how the heat is carried away via conduction, advection or radiation.

Heat from a biochemical reaction in a cell is *conducted* by diffusion through the tissue to the blood stream, where it is then *advected* away and either stored elsewhere in the body or conveyed to the skin where it is *radiated* away (often using the latent heat of evaporation).

For a black body radiator the radiated power is proportional to the 4th power of temperature. i.e. $u_2^h \cdot v_2^h = \sigma \cdot (u_2^h)^4$ (Stefan's law).
 The electrical resistance of most metals increases linearly with temperature: $R = R_0(1 + \alpha(T - T_0))$, where $T = u_2^h$ is temperature in Kelvin.

Lecture 3 *Simple bond graph models*

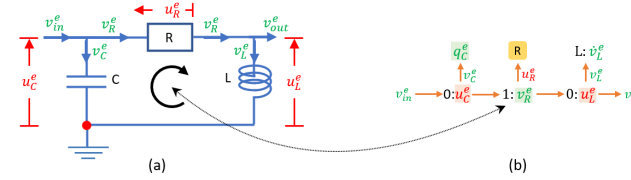
Lecture 3: *Simple bond graph models*

To show the power of **bond graphs** we use them to model a wide range of simple processes used in physiology, including reactions with multiple reactants and products, enzyme-catalysed reactions, and the electro-diffusion of a charged species (such as H^+ or HCO_3^-). For enzyme catalysed reactions we focus on the steady state approximations that are commonly used in physiology texts, and we discuss when those approximations are valid.

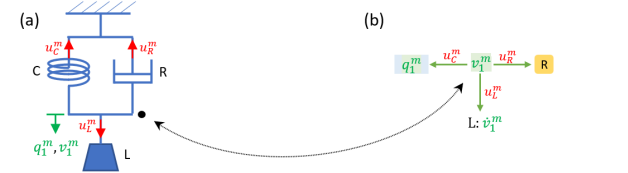
We show you how to solve the bond graph models using an open-source software package called **OpenCOR**. By the end of this lecture and the accompanying tutorials you should understand how to build simple models using bond graphs and how to solve them and interpret the results using OpenCOR.

Simple bond graph models

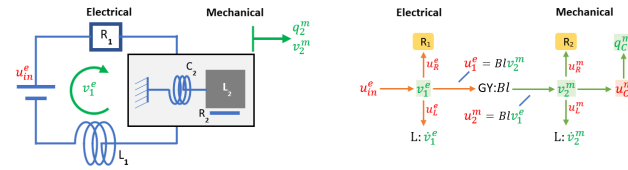
1. An electrical circuit



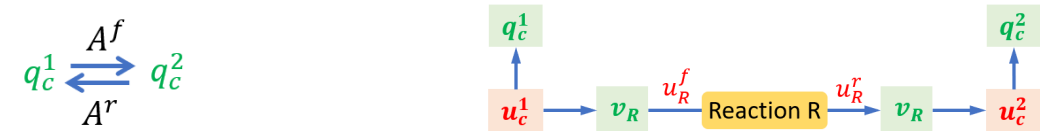
2. A mechanical system



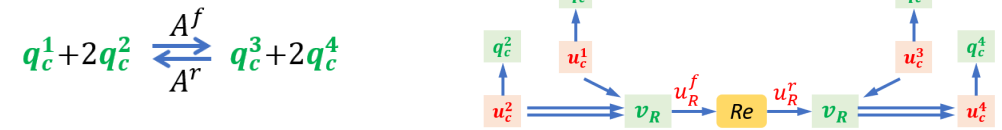
3. An electromechanical system



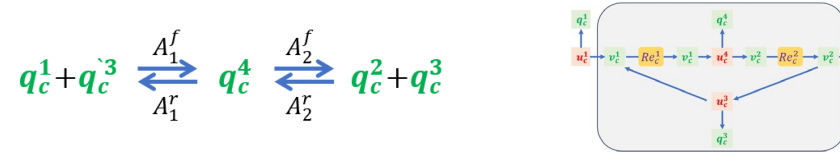
4. A simple biochemical reaction



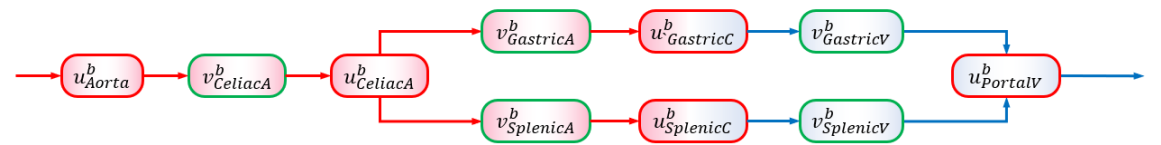
5. A reaction with mixed stoichiometry



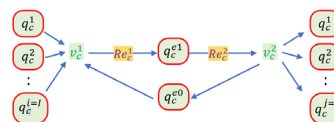
6. An enzyme-catalyzed reaction



7. Blood flow in the cardiovascular system



8. A reaction with multiple reactants & products



Defining units in CellML

$J.C^{-1}$ { `def unit J_per_C as`
 `unit joule;`
 `unit coulomb {expo: -1};`
 `enddef;`

$C.S^{-1}$ { `def unit C_per_s as`
 `unit coulomb;`
 `unit second {expo: -1};`
 `enddef;`

$C.S^{-2}$ { `def unit C_per_s2 as`
 `unit coulomb;`
 `unit second {expo: -2};`
 `enddef;`

$C^2.J^{-1}$ { `def unit C2_per_J as`
 `unit coulomb {expo: 2};`
 `unit joule {expo: -1};`
 `enddef;`

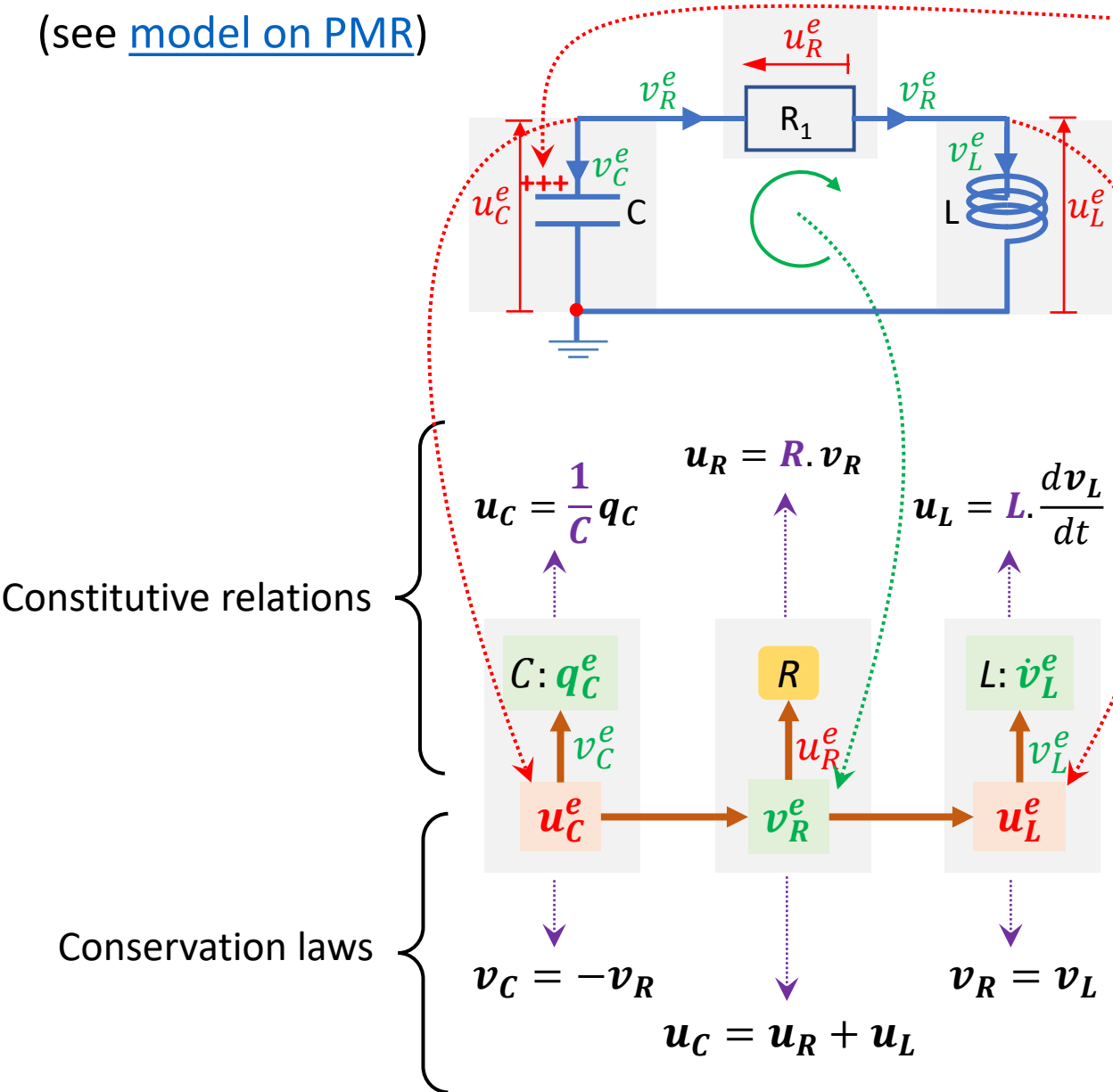
$J.S.C^{-2}$ { `def unit Js_per_C2 as`
 `unit joule;`
 `unit second;`
 `unit coulomb {expo: -2};`
 `enddef;`

$J.S^2.C^{-2}$ { `def unit Js2_per_C2 as`
 `unit joule;`
 `unit second {expo: 2};`
 `unit coulomb {expo: -2};`
 `enddef;`

`def unit dim as`
 `unit dimensionless;`
 `enddef;`

1. An electrical circuit

(see [model on PMR](#))



```
def comp main as
  var t: second {init: 0};

  // state variables
  var q_C: coulomb {init: 1};
  var v_C: C_per_s;
  var v_R: C_per_s;
  var v_L: C_per_s {init: 0};
  var a_L: C_per_s2;
  var u_C: J_per_C;

  var u_R: J_per_C;
  var u_L: J_per_C;

  // constitutive parameters
  var C: C2_per_J {init: 20};
  var R: Js_per_C2 {init: 2};
  var L: Js2_per_C2 {init: 10};

  // constitutive relations
  u_C = q_C/C;
  u_R = v_R*R;
  u_L = a_L*L;

  // conservation laws
  ode(q_C, t) = v_C;
  ode(v_L, t) = a_L;

  v_C = -v_R;
  v_R = v_L;
  u_C = u_R+u_L;

enddef;
```

Power & energy

Capacitor

$$Power = \frac{dE}{dt} = u \cdot v = \frac{1}{C} q \cdot \frac{dq}{dt} = \frac{1}{C} \cdot \frac{1}{2} \frac{dq^2}{dt} \quad \text{gives } E = \frac{1}{2} \cdot \frac{1}{C} \cdot q^2$$

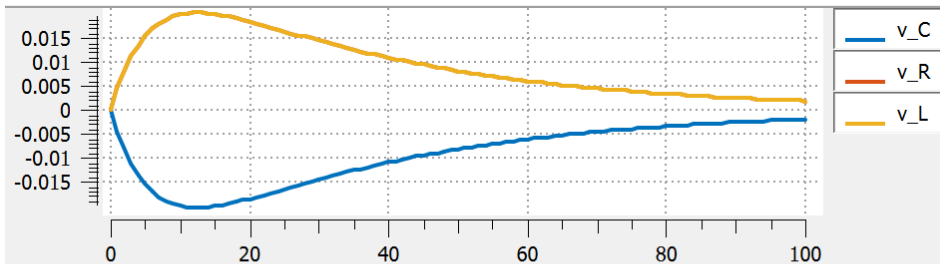
Inductor

$$Power = \frac{dE}{dt} = u \cdot v = L \frac{dv}{dt} \cdot v = L \cdot \frac{1}{2} \frac{dv^2}{dt} \quad \text{gives } E = \frac{1}{2} L \cdot v^2$$

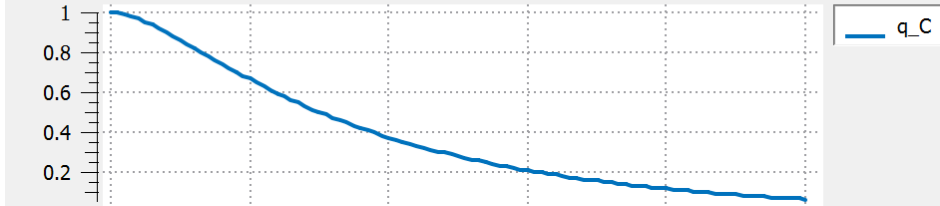
Resistor

$$Power = \frac{dE}{dt} = u \cdot v = R \cdot v \cdot v = R \cdot v^2 \quad \text{gives } E = \int R \cdot v^2 dt$$

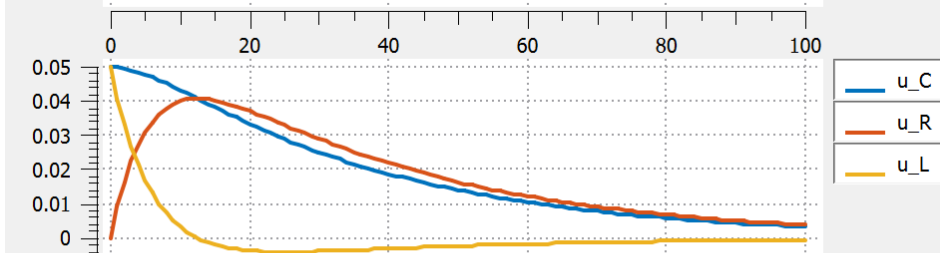
Currents



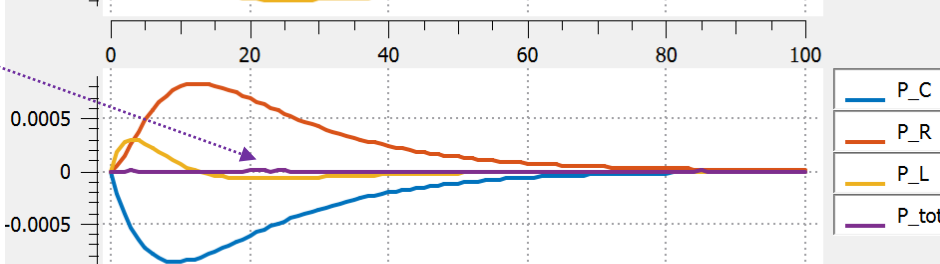
Charge



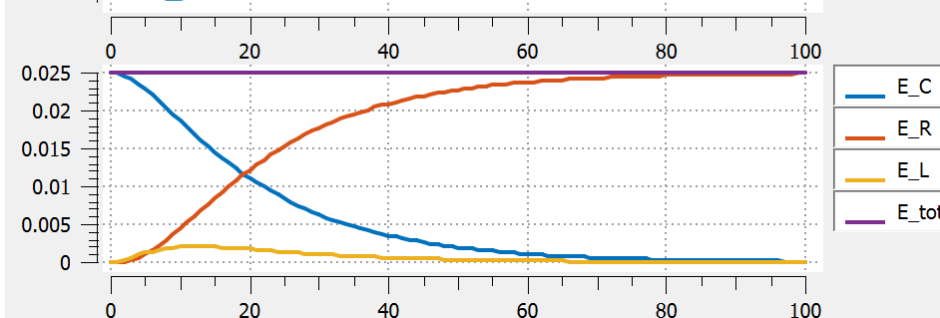
Potentials



Power



Energy



// conservation laws

```
ode(v_L, t) = a_L;  
v_C = -v_R;  
v_R = v_L;
```

```
ode(q_C, t) = v_C;
```

```
u_C = u_R+u_L;
```

// constitutive relations

```
u_C = q_C/C;  
u_R = v_R*R;  
u_L = a_L*L;
```

// Energy calculations

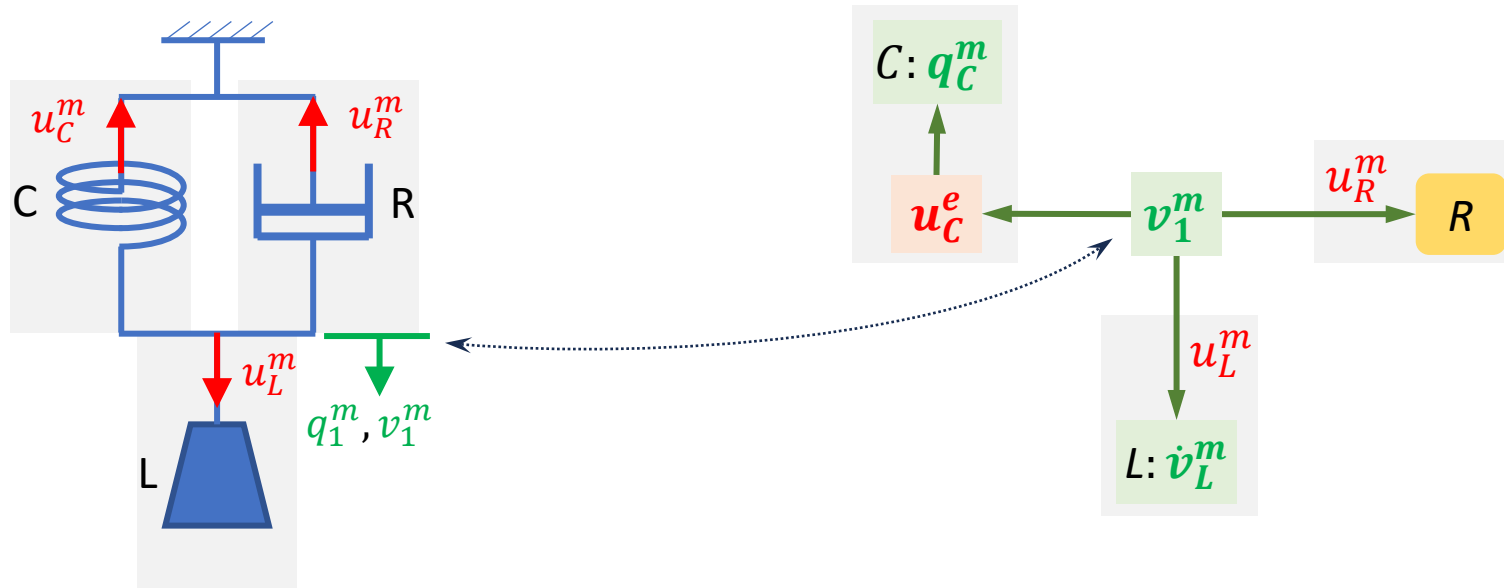
```
P_C = u_C*v_C;  
P_R = u_R*v_R;  
P_L = u_L*v_L;  
P_tot = P_C+P_R+P_L;
```

```
ode(E_R, t) = P_R;  
E_C = 0.5*1/C*sqr(q_C);  
E_L = 0.5*L*sqr(v_L);  
E_tot = E_C+E_L+E_R;
```

Model on PMR

2. A mechanical system

(see [model on PMR](#))



Conservation of energy at 1:node: $u_C^m + u_R^m + u_L^m = 0$

Constitutive relations $u_C^m = \frac{1}{C} q_1^m$; $u_R^m = R v_1^m$; $u_L^m = L \frac{dv_1^m}{dt}$.

Note that substituting the constitutive relations into the conservation eqn with $v_1^m = \frac{dq_1^m}{dt}$ gives the 2nd order eqn $L \frac{d^2 q_1^m}{dt^2} + R \frac{dq_1^m}{dt} + \frac{1}{C} q_1^m = 0$, but bond graphs always uses the first order eqns.

3. An electromechanical system

(see [model on PMR](#))

Electromechanical energy exchange

Faraday's law

$$u_1^e (=k \cdot v_2^m)$$

Lorentz force

$$u_2^m (=k \cdot v_1^e)$$

TF: k

(GY: k)

Units of k : $\text{J.C}^{-1}.\text{s.m}^{-1} = \text{J.m}^{-1}.\text{s.C}^{-1}$

Fixed potential (battery)
delivers current into system

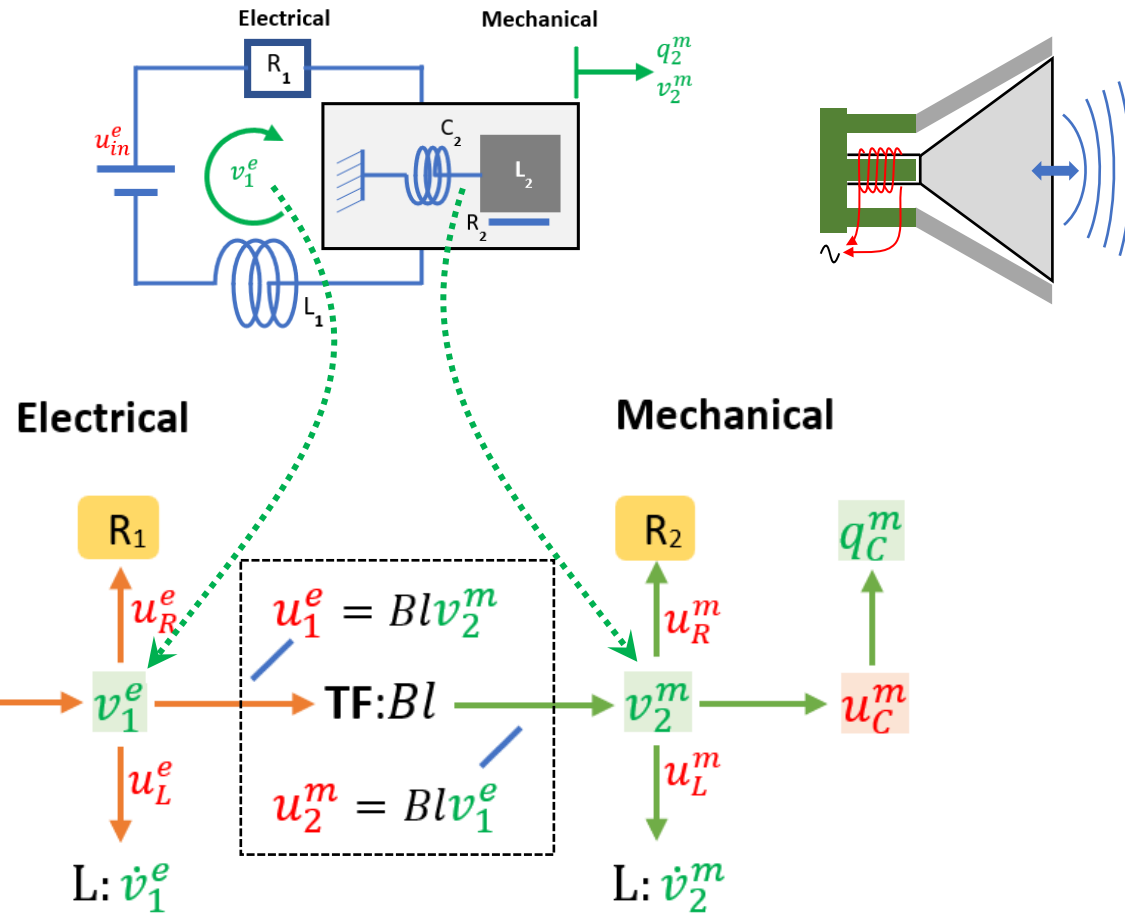
$$(u_C^m) \quad \frac{dq_C^m}{dt} = v_2^m$$

$$(v_1^e) \quad u_{in}^e = u_R^e + u_L^e + u_1^e$$

$$(v_2^e) \quad u_2^m = u_R^m + u_L^m + u_C^m$$

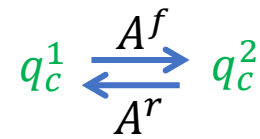
$$(TF) \quad u_1^e = Blv_2^m; \quad u_2^m = Blv_1^e$$

$$(CRs) \quad u_R^e = R_1 v_1^e; \quad u_L^e = L_1 \frac{dv_1^e}{dt}; \quad u_R^m = R_2 v_2^m; \quad u_L^m = L_2 \frac{dv_2^m}{dt}; \quad u_C^m = q_C^m / C_2;$$



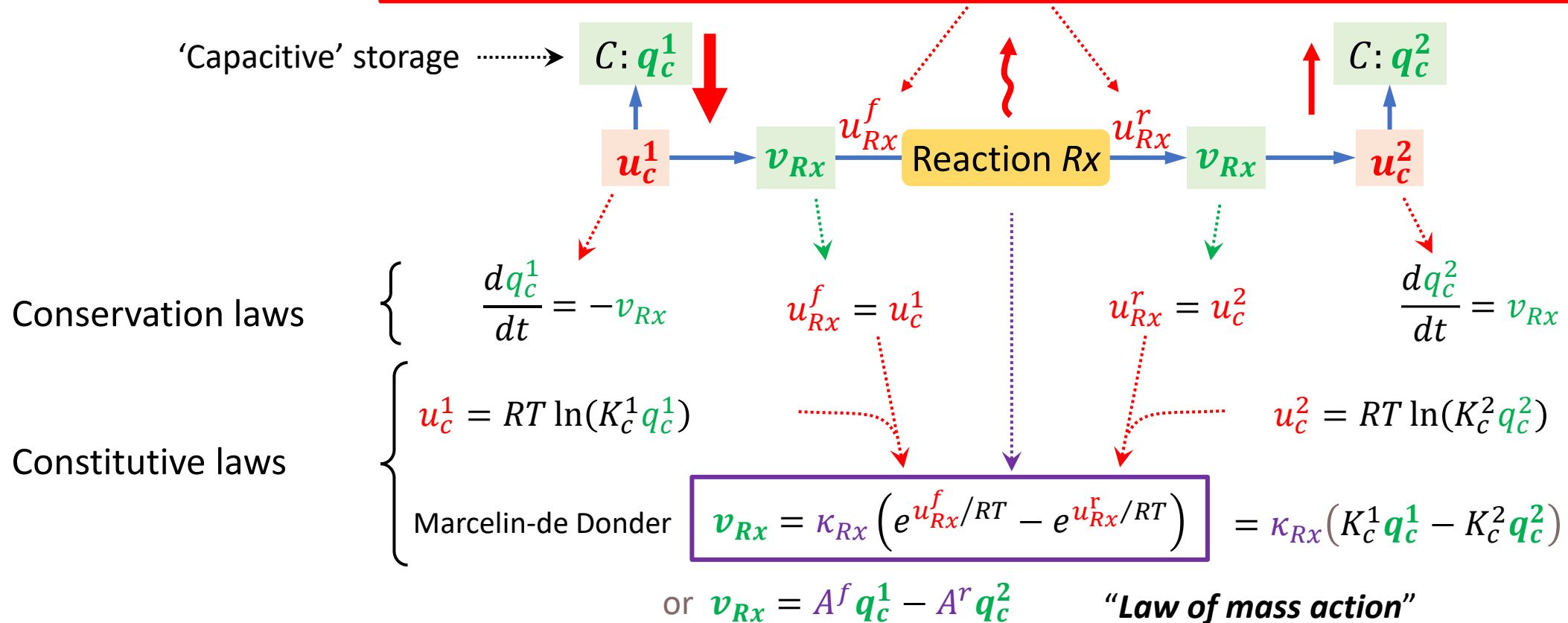
4. A simple biochemical reaction

(see [model on PMR](#))



The change in **Gibbs 'free energy'** (per mole) is the difference between these two chemical potentials and is the energy available to do work (i.e. convert the internal energy of the reactant stored as q_c^1 into the internal energy of the product stored as q_c^2). The heat production rate from the reaction is $(u_{Rx}^f - u_{Rx}^r) v_{Rx}$.

i.e. Reactant internal energy flux $u_{Rx}^f \cdot v_{Rx}$ = Product internal energy flux $u_{Rx}^r \cdot v_{Rx}$ + Heat flux $\Delta u \cdot v_{Rx}$.



where the parameters $A^f = \kappa_{Rx} K_c^1$ and $A^r = \kappa_{Rx} K_c^2$ combine thermodynamic and reaction kinetic parameters.

Note on thermodynamic parameters

For a dilute system the chemical potential is given by the Boltzmann thermodynamic relation

$$u_c^i = u_0^i + RT \ln \frac{q_c^i}{q_c^{tot}} \quad (\text{J.mol}^{-1})$$

where q_c^i is the number of moles of chemical species i and q_c^{tot} is the total number of moles of all substances in the mixture in compartment c (Sauro, 2012). u_0^i is the (reference) potential when $q_c^i = q_c^{tot}$.

More compactly, $u_c^i = RT \ln K_c^i q_c^i$ (J.mol⁻¹), where $K_c^i = \frac{1}{q_c^{tot}} e^{u_0^i/RT}$ (mol⁻¹),

is the constitutive law for biochemical energy storage, and K_c^i (mol⁻¹) is a thermodynamic parameter.

Note: $pH = -\log_{10}[H^+]$ where $[H^+]$ is expressed in mol.L⁻¹. i.e. pH of 7 is $[H^+] = 10^{-7}$ mol.L⁻¹

Note on concentrations

q_1^A represents the quantity in moles of solute A at location 1

$u_1^A = RT \ln(K_1^A q_1^A)$ is the corresponding chemical potential

K_1^A is the thermodynamic constant that provides a reference for u_1^A

i.e. when $q_1^A = \frac{1}{K_1^A}$, $u_1^A = RT \ln(1) = 0$

The concentration of solute A in a volume V_1 at location 1 is $c_1^A = \frac{q_1^A}{V_1}$.

K_1^A is proportional to $\frac{1}{V_1}$ since a larger volume V_1 requires a larger molar amount q_1^A to maintain the same u_1^A

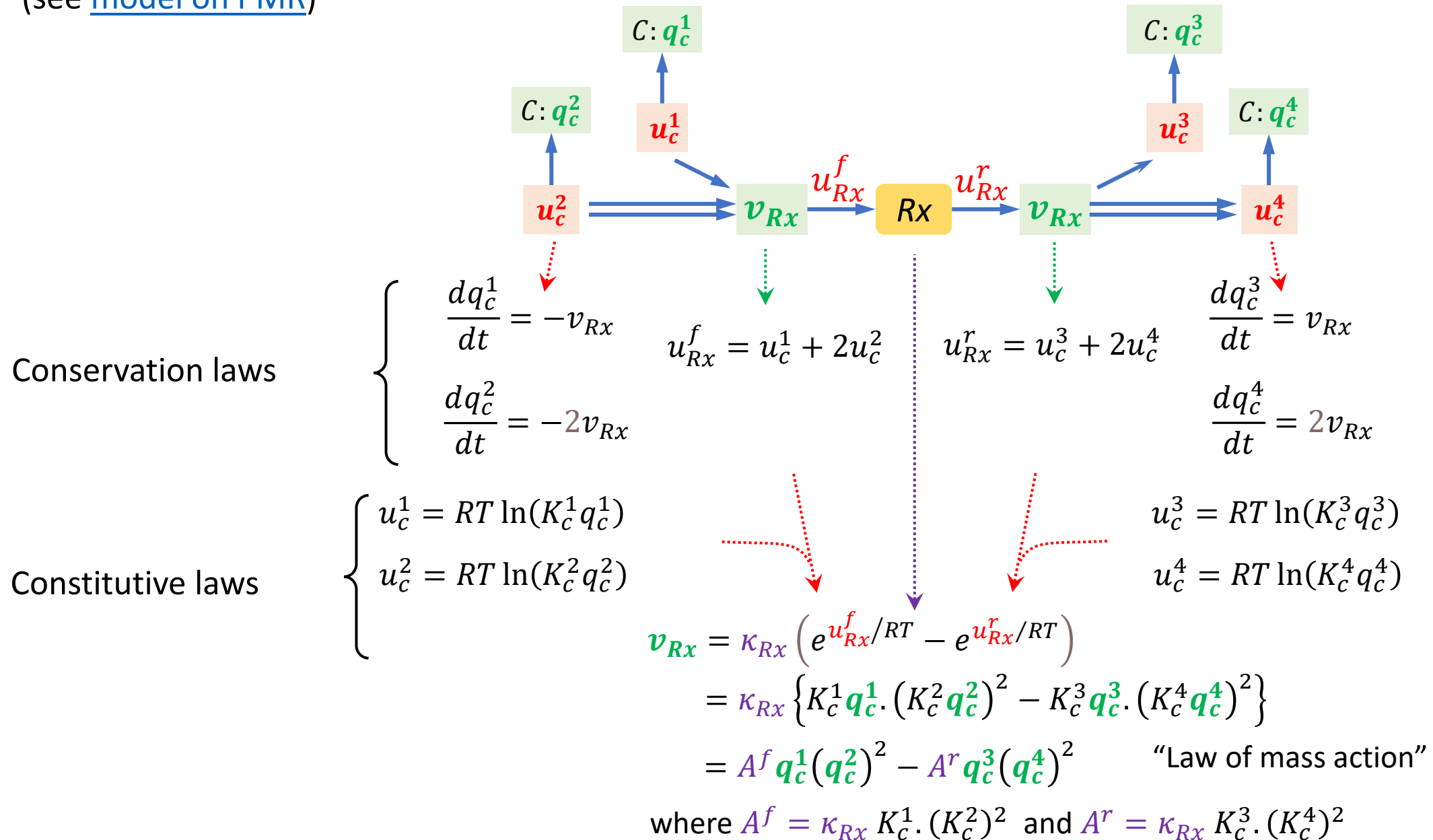
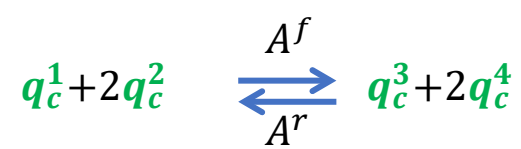
So $\frac{K_2^A}{K_1^A} = \frac{V_1}{V_2}$.

And hence $\frac{K_2^A q_2^A}{K_1^A q_1^A} = \frac{V_1 q_2^A}{V_2 q_1^A} = \frac{q_2^A/V_2}{q_1^A/V_1} = \frac{c_2^A}{c_1^A}$.

We later use this to express the Nernst potential in terms of concentrations: $u_m^e = \frac{RT}{z_q F} \ln \left(\frac{K_2^i q_2^i}{K_1^i q_1^i} \right) = \frac{RT}{z_q F} \ln \left(\frac{c_2^i}{c_1^i} \right)$.

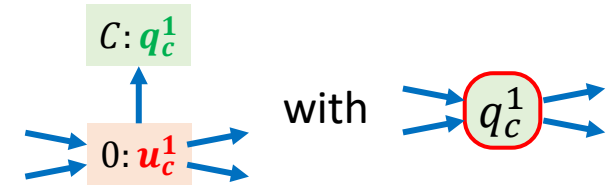
5. A biochemical reaction with mixed stoichiometry

(see [model on PMR](#))



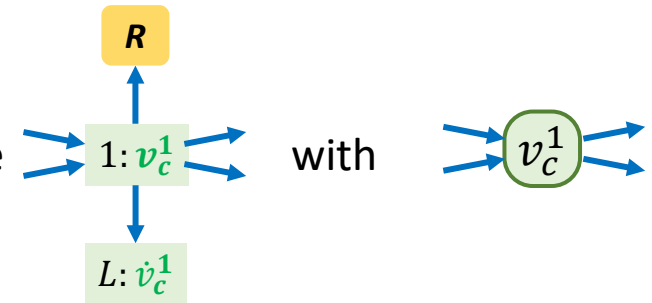
Simplification of bond graph diagrams

It is often convenient to combine (1-port) static storage with a 0:node and replace



The red border indicates the 0:node (mass or charge balance) and the q_c^1 term reflects both the static storage and that the potential u_c^1 is given by a storage relationship $u_c^1 = f(q_c^1)$.

Similarly, 1-port dissipation can sometimes be combined with a 1:node to replace



The green border indicates the 1:node (energy balance) together with the dissipation function $f(v_c^1)$ (e.g. $f(v_c^1) = Rv_c^1$) and the kinetic energy storage term $L\dot{v}_c^1$. All of these depend on v_c^1 .

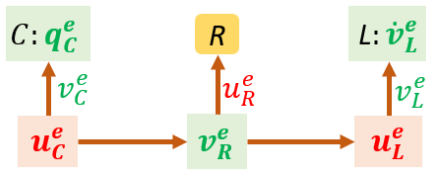
Note that the 1:node v_{Rx} is retained in its current form in chemical reactions where the stoichiometry needs to be shown.

We also replace $\Rightarrow v_{Rx}$ with $\textcircled{2} \rightarrow v_{Rx}$

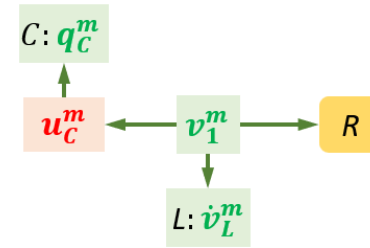
A further simplification is to put $\bar{q}_c^i = K_c^i q_c^i$ where $\bar{q}_c^i$ is now a dimensionless quantity and is the relative molar quantity that provides the energy to drive reactions (i.e. $u_c^i = RT \ln \bar{q}_c^i$).

Note that from now on we use the superscript i and subscript j on q_j^i , v_j^i & u_j^i to denote species i and location j , respectively.

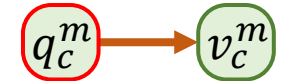
Examples ..



becomes

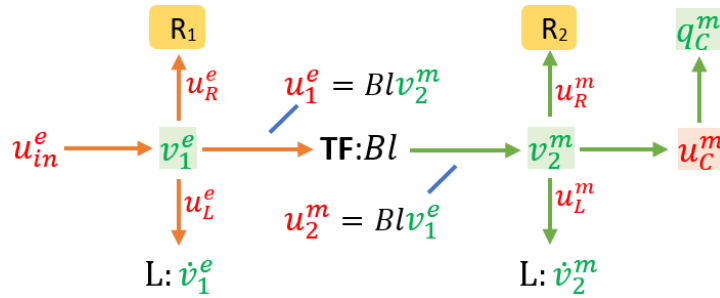


becomes

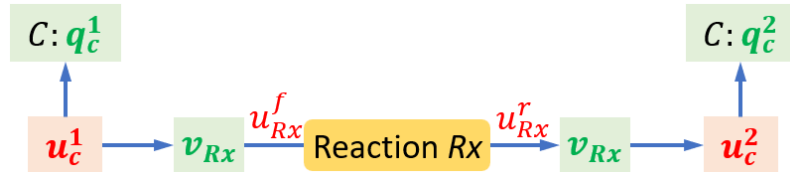


Electrical

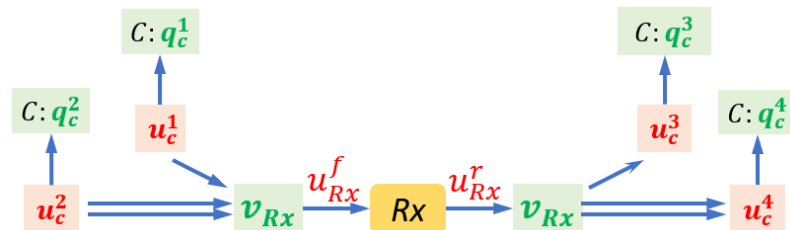
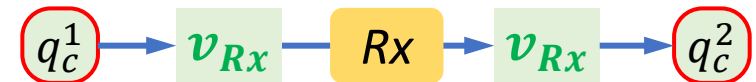
Mechanical



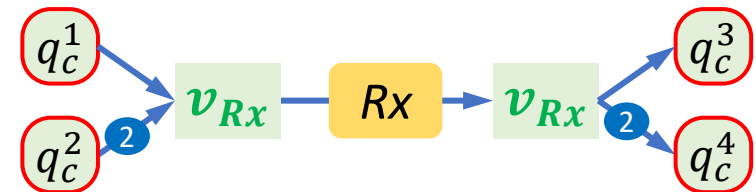
becomes



becomes

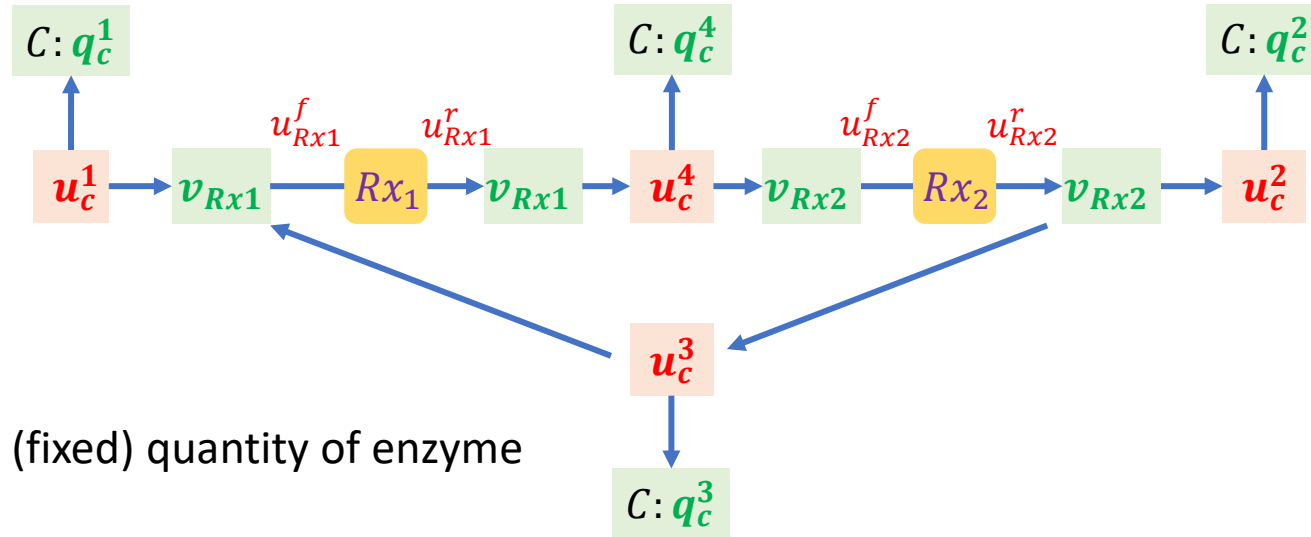
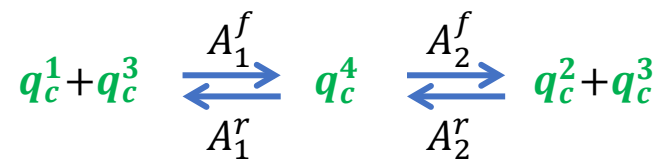


becomes



6. An enzyme-catalysed reaction

(see [model on PMR](#))



$q_c^3 + q_c^4 = E_0$, where E_0 is the initial (fixed) quantity of enzyme

Mass conservation at the 0:nodes:

$$(u_c^1) \quad \frac{d}{dt} q_c^1 = -v_{Rx1}; \quad (u_c^2) \quad \frac{d}{dt} q_c^2 = v_{Rx2}; \quad (u_c^3) \quad \frac{d}{dt} q_c^3 = v_{Rx2} - v_{Rx1}; \quad (u_c^4) \quad \frac{d}{dt} q_c^4 = v_{Rx1} - v_{Rx2};$$

$$(v_{Rx1}) \quad u_{Rx1}^f = u_c^1 + u_c^3; \quad u_{Rx1}^r = u_c^4; \quad (v_{Rx2}) \quad u_{Rx2}^f = u_c^4; \quad u_{Rx2}^r = u_c^2 + u_c^3.$$

Substitute for u_{Rx}^f & u_{Rx}^r and use Boltzmann equations

$$(Rx_1) \quad v_{Rx1} = \kappa_{Rx1} \left(e^{u_{Rx1}^f/RT} - e^{u_{Rx1}^r/RT} \right) = \kappa_{Rx1} (K_c^1 q_c^1 K_c^3 q_c^3 - K_c^4 q_c^4) = A_1^f q_c^1 q_c^3 - A_1^r q_c^4,$$

$$\text{where } A_1^f = \kappa_{Rx1} \cdot K_c^1 K_c^3 \text{ (mol}^{-1} \cdot \text{s}^{-1}\text{); } A_1^r = \kappa_{Rx1} \cdot K_c^4 \text{ (s}^{-1}\text{)}$$

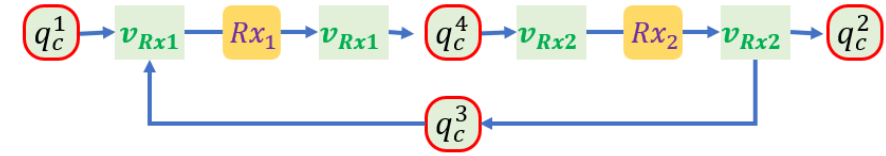
$$(Rx_2) \quad v_{Rx2} = \kappa_{Rx2} \left(e^{u_{Rx2}^f/RT} - e^{u_{Rx2}^r/RT} \right) = \kappa_{Rx2} (K_c^4 q_c^4 - K_c^2 q_c^2 K_c^3 q_c^3) = A_2^f q_c^4 - A_2^r q_c^2 q_c^3$$

$$\text{where } A_2^f = \kappa_{Rx2} \cdot K_c^4 \text{ (s}^{-1}\text{) and } A_2^r = \kappa_{Rx2} \cdot K_c^2 K_c^3 \text{ (mol}^{-1} \cdot \text{s}^{-1}\text{)}$$

Rewriting using more condensed notation:

$$(Rx_1) \quad v_{Rx1} = \kappa_{Rx1} (K_c^1 q_c^1 K_c^3 q_c^3 - K_c^4 q_c^4) = \kappa_{Rx1} (\bar{q}_c^1 \bar{q}_c^3 - \bar{q}_c^4)$$

$$(Rx_2) \quad v_{Rx2} = \kappa_{Rx2} (K_c^4 q_c^4 - K_c^2 q_c^2 K_c^3 q_c^3) = \kappa_{Rx2} (\bar{q}_c^4 - \bar{q}_c^2 \bar{q}_c^3)$$



The **Briggs-Haldane assumption** is that the amount of enzyme ($q_c^3 + q_c^4$) is much less than the amounts of substrate (q_c^1 & q_c^2), so that the unbound enzyme q_c^3 and the complex q_c^4 therefore quickly reach a steady state ($\frac{d}{dt} q_c^3 = 0$ and $\frac{d}{dt} q_c^4 = 0$).

Hence $\frac{d}{dt} q_c^3 = v_{Rx2} - v_{Rx1} = 0$ gives $v_{Rx1} = v_{Rx2}$ and $\frac{d}{dt} q_c^4 = v_{Rx1} - v_{Rx2} = 0$ gives $v_{Rx1} = v_{Rx2}$

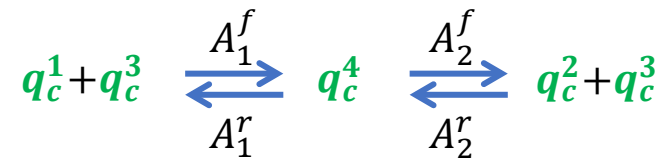
The other two mass-balance relations: $\frac{d}{dt} q_c^1 = -v_{Rx1} = \kappa_{Rx1} (\bar{q}_c^1 \bar{q}_c^3 - \bar{q}_c^4)$ and $\frac{d}{dt} q_c^2 = v_{Rx2} = \kappa_{Rx2} (\bar{q}_c^4 - \bar{q}_c^2 \bar{q}_c^3)$,

together with the constraint $q_c^3 + q_c^4 = E_0$, can be solved to eliminate q_c^3 & q_c^4 and give a steady-state flux:

$$v = v_{Rx1} = v_{Rx2} = E_0 \cdot \frac{\kappa_{Rx1} \kappa_{Rx2} (\bar{q}_c^1 - \bar{q}_c^2)}{\frac{\kappa_{Rx1} + \kappa_{Rx2}}{K_c^3} + \frac{\kappa_{Rx1}}{K_c^4} \bar{q}_c^1 + \frac{\kappa_{Rx2}}{K_c^4} \bar{q}_c^2}$$

or $v = v_m \frac{\bar{q}_c^1 - \bar{q}_c^2}{1 + \frac{\bar{q}_c^1}{k_m^1} + \frac{\bar{q}_c^2}{k_m^2}}$ where $v_m = E_0 \cdot \frac{K_c^3 \cdot \kappa_{Rx1} \kappa_{Rx2}}{\kappa_{Rx1} + \kappa_{Rx2}}$, $k_m^1 = \frac{K_c^4}{K_c^3} \cdot \frac{\kappa_{Rx1} + \kappa_{Rx2}}{\kappa_{Rx1}}$, $k_m^2 = \frac{K_c^4}{K_c^3} \cdot \frac{\kappa_{Rx1} + \kappa_{Rx2}}{\kappa_{Rx2}}$

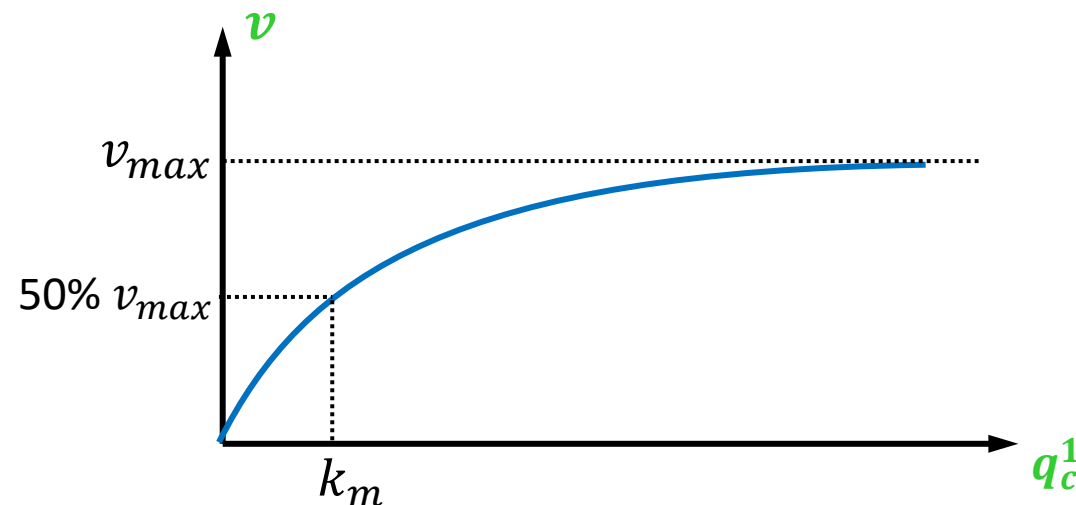
Michaelis-Menten (MM)



If $q_c^2 \ll q_c^1$, the MM assumption puts $q_c^2 = 0$ (which violates the laws of thermodynamics!).

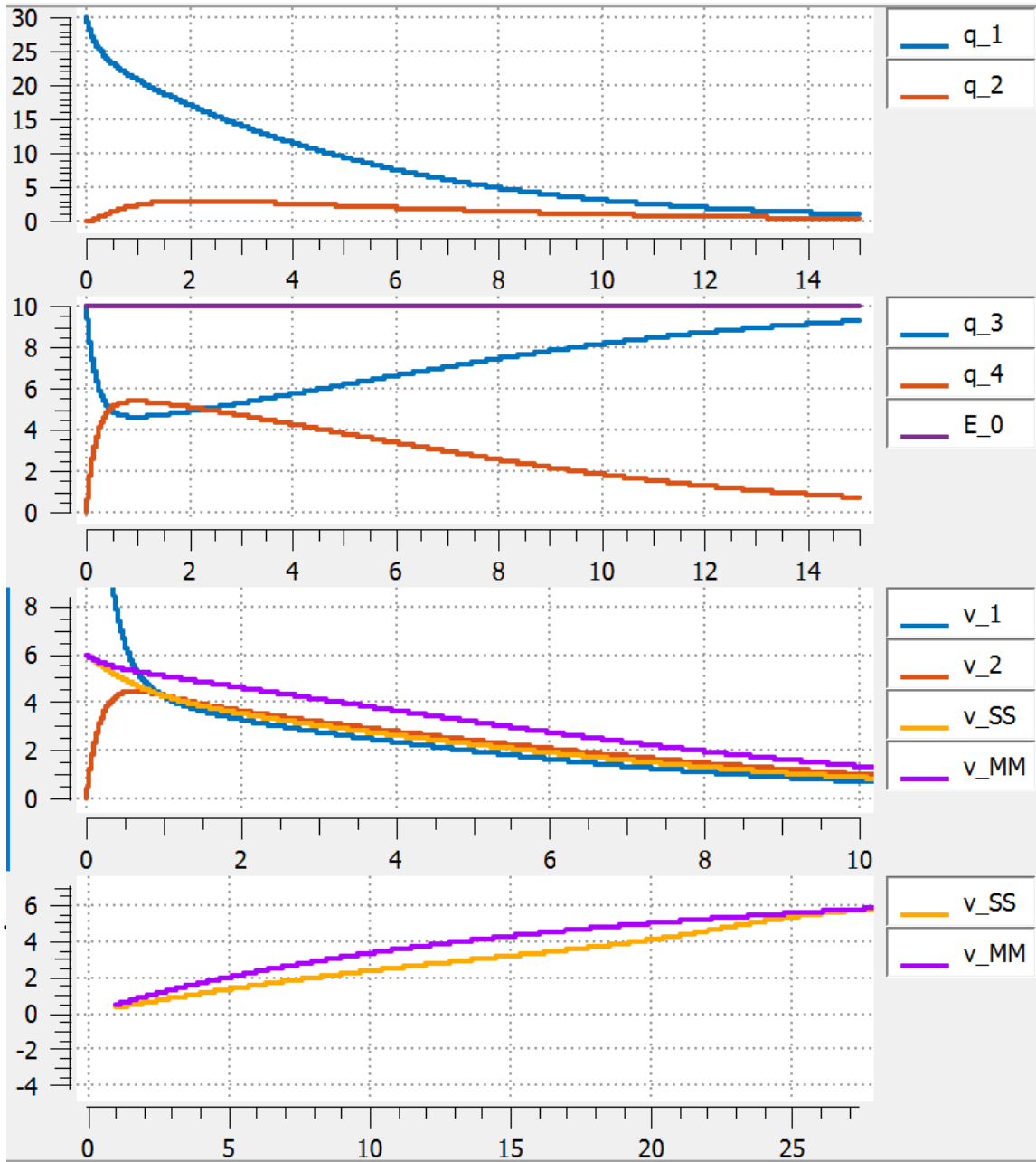
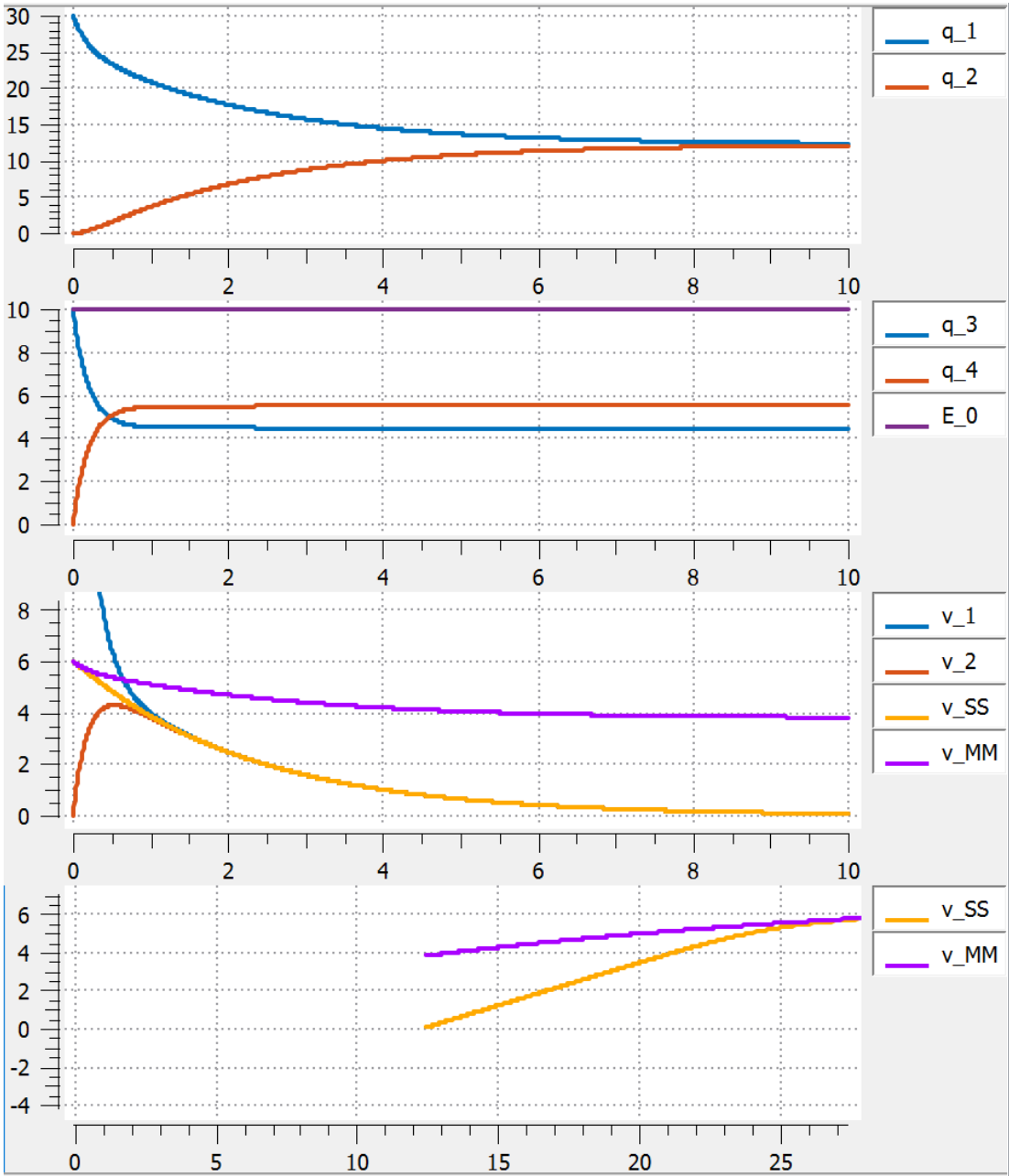
With that assumption, $v = v_m \frac{\bar{q}_c^1 - \bar{q}_c^2}{1 + \frac{\bar{q}_c^1}{k_m^1} + \frac{\bar{q}_c^2}{k_m^2}}$ reduces to $v = v_m \frac{\bar{q}_c^1}{1 + \frac{\bar{q}_c^1}{k_m^1}} = v_m k_m^1 \cdot \frac{q_c^1}{\frac{k_m^1}{K_c^1} + q_c^1}$,

which can be rewritten as $v = v_{max} \cdot \frac{q_c^1}{k_m + q_c^1}$ (where $v_{max} = v_m \cdot k_m^1$ and $k_m = \frac{k_m^1}{K_c^1}$)



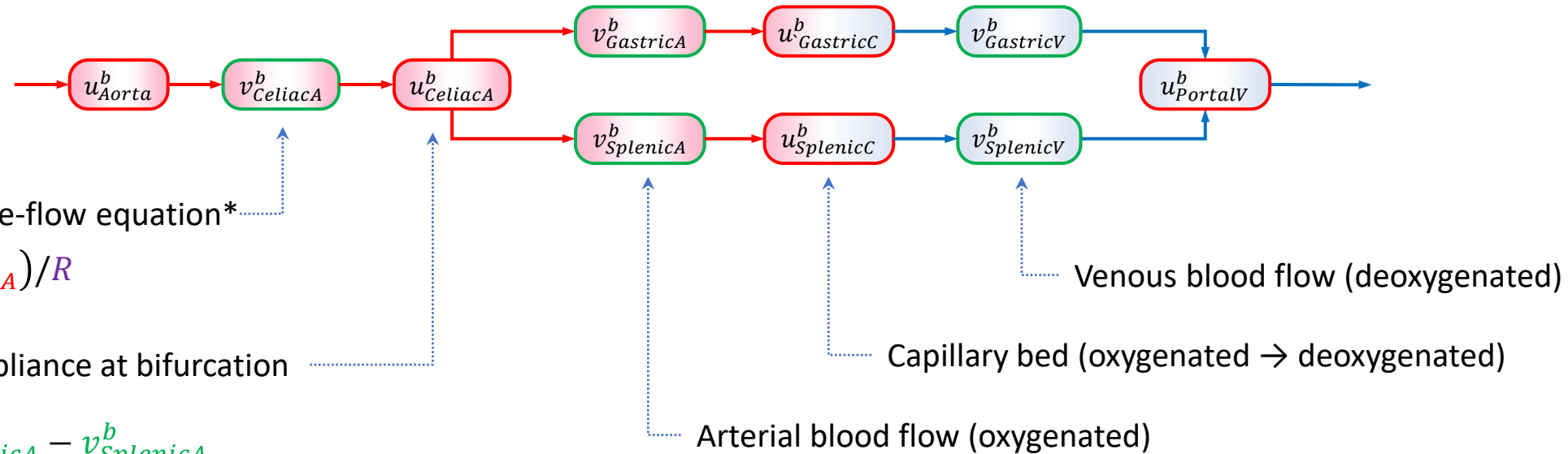
Comparing solutions under different conditions

Add a sink for q_c^2



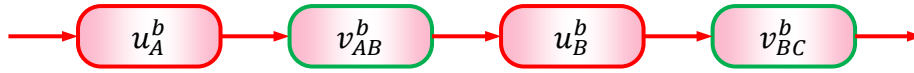
7. Blood flow in the cardiovascular system

The following example of blood flow through the stomach & spleen illustrates the use of both of these new symbols:



*We could also add inertia in here, in which case $u_{Aorta}^b = u_{CeliacA}^b + R \cdot v_{CeliacA}^b + L \frac{dq_{CeliacA}^b}{dt}$

Calculation of material constants for blood flow model



Mass conservation at the 0:node u_B^b is $\frac{dq_B^b}{dt} = v_{AB}^b - v_{BC}^b$

with the vascular compliance given by $u_B^b = q_B^b / C_B$

To calculate the compliance C_B we assume a thin vascular wall (thickness h).

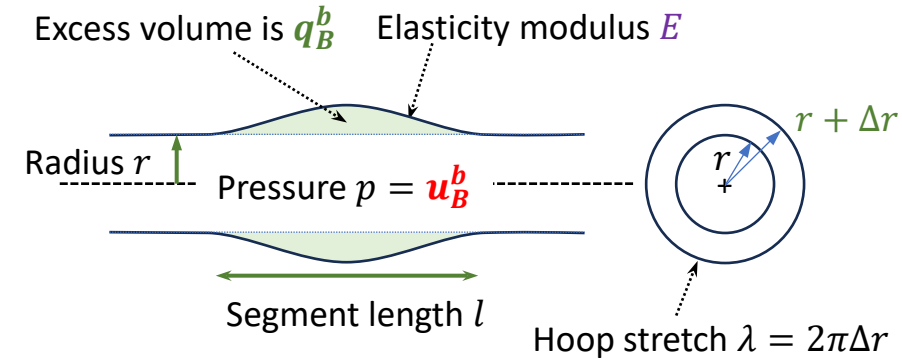
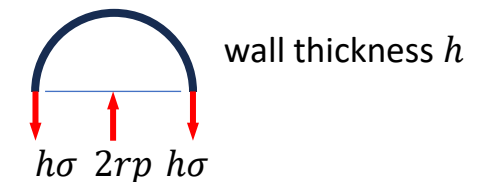
At pressure p the radius r increases to $r + \Delta r$, giving a hoop stretch $\lambda = 2\pi(r + \Delta r) - 2\pi r = 2\pi\Delta r$.

Assuming a linear elastic material (modulus E), this generates a stress $\sigma = E\lambda = E \cdot 2\pi\Delta r$, giving $\Delta r = \frac{\sigma}{2\pi E}$

and the increase in volume for the cylinder is $q_B^b = \Delta V = 2\pi(r + \Delta r)l - 2\pi rl = 2\pi\Delta r l = 2\pi \cdot \frac{\sigma}{2\pi E} l = \frac{\sigma l}{E}$.

The force balance for a thin-walled cylinder (Laplace) is: $2rp = 2h\sigma$ or $\sigma = \frac{r}{h}p = \frac{r}{h}$,

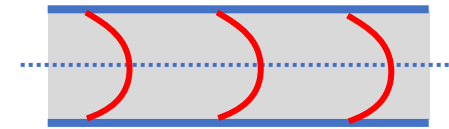
and therefore $q_B^b = \frac{rl}{Eh}p \cdot u_B^b$ or $u_B^b = q_B^b / C_B$, where the compliance $C_B = \frac{rl}{Eh}$.



Finally energy conservation at the 1:node v_{AB}^b is given by $v_{AB}^b = (u_A^b - u_B^b) / R_{AB}$

where, assuming steady fully developed Poiseuille flow, the viscous resistance is $R_{AB} = \frac{8\mu l}{\pi r^4}$.

Derivation of Poiseuille relation for steady flow in a tube



Steady fully-developed flow

The Navier-Stokes (NS) equations are for incompressible fluid with density ρ and dynamic viscosity μ :

$$\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} + \frac{1}{\rho} \nabla p = \frac{\mu}{\rho} \nabla^2 \mathbf{v} \quad \text{with} \quad \nabla \cdot \mathbf{v} = 0.$$

Assuming $v = v_x(r, x, t)$ in the x -direction only and putting $p = p(x, t)$, the NS eqn in cylindrical polar coordinates is

$$\frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + \frac{1}{\rho} \frac{\partial p}{\partial x} = \frac{\mu}{\rho} \left\{ \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_x}{\partial r} \right) + \frac{\partial^2 v_x}{\partial x^2} \right\} \quad \text{with incompressibility condition} \quad \frac{\partial v_x}{\partial x} = 0.$$

For steady flow in a straight tube there is no change in x or t and therefore $v(r)$ is determined by $\frac{1}{\rho} \frac{\partial p}{\partial x} = \frac{\mu}{\rho} \cdot \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v}{\partial r} \right)$,

where the constant pressure gradient balances the viscous forces.

Putting $\frac{1}{\mu} \frac{\partial p}{\partial x} = k$, this becomes $\frac{1}{r} \frac{d}{dr} \left(r \frac{dv}{dr} \right) = k$ or $r \frac{dv}{dr} = \frac{1}{2} k r^2 + a$ or $\frac{dv}{dr} = \frac{1}{2} k r + \frac{a}{r}$.

But, since $\left. \frac{dv}{dr} \right|_{r=0} = 0$, $a = 0$ and $\frac{dv}{dr} = \frac{1}{2} k r$ gives $v(r) = \frac{1}{4} k r^2 + b$

At the tube inner radial boundary, where $r = R$, $v(R) = \frac{1}{4} k R^2 + b = 0$. Therefore $b = -\frac{1}{4} k R^2$, and $v(r) = \frac{1}{4} k (r^2 - R^2)$

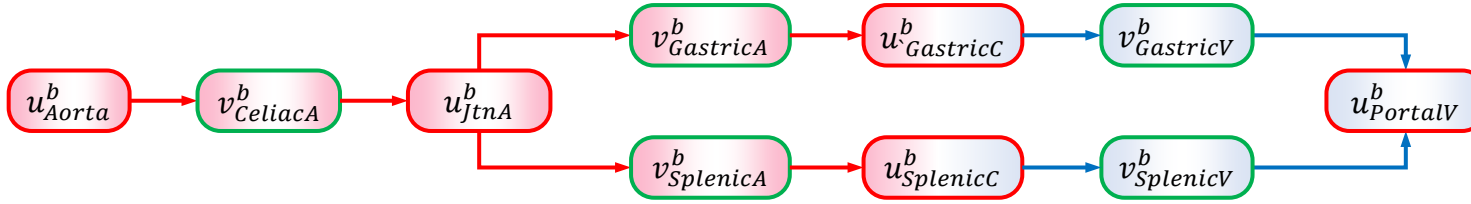
If we put $v(0) = V = -\frac{1}{4} k R^2$. With $k = \frac{1}{\mu} \frac{dp}{dx}$, $\frac{dp}{dx} = -\frac{4\mu V}{R^2}$ and the flow through the tube is

$$Q = \int_0^R v(r) \cdot 2\pi r \cdot dr = \frac{\pi}{2} k \int_0^R (r^3 - R^2 r) \cdot dr = \frac{\pi}{2} k \left(\frac{R^4}{4} - \frac{R^4}{2} \right) = -\frac{\pi R^4}{8\mu} \cdot \frac{dp}{dx}, \quad \text{or} \quad k = -\frac{8}{\pi R^4} Q \quad \text{and} \quad \frac{dp}{dx} = -\frac{8\mu}{\pi R^4} Q.$$

i.e. the pressure drop between the two ends of the constant radius tube of length l is $\Delta p = \frac{8l\mu}{\pi R^4} Q$

or $\mathbf{u}_A^b - \mathbf{u}_B^b = R_{AB} \cdot \mathbf{v}_{AB}^b$, where $R_{AB} = \frac{8l\mu}{\pi r^4}$ is the resistance for the vessel AB of length l and radius r with blood of viscosity μ .

Bond graph equations and solution for blood flow model



Assuming the pressures u_{Aorta}^b and $u_{PortalV}^b$ are specified as bdy conditions.

Energy balance at 1-nodes

$$v_{CeliacA}^b = (u_{Aorta}^b - u_{CeliacA}^b) / R_{CeliacA}^b$$

$$v_{GastricA}^b = (u_{CeliacA}^b - u_{GastricC}^b) / R_{GastricA}^b$$

$$v_{GastricV}^b = (u_{GastricC}^b - u_{PortalV}^b) / R_{GastricV}^b$$

$$v_{SplenicA}^b = (u_{CeliacA}^b - u_{SplenicC}^b) / R_{SplenicA}^b$$

$$v_{SplenicV}^b = (u_{SplenicC}^b - u_{PortalV}^b) / R_{SplenicV}^b$$

Mass balance & compliance relations at 1-nodes

$$\frac{dq_{CeliacA}^b}{dt} = v_{CeliacA}^b - v_{GastricA}^b - v_{SplenicA}^b$$

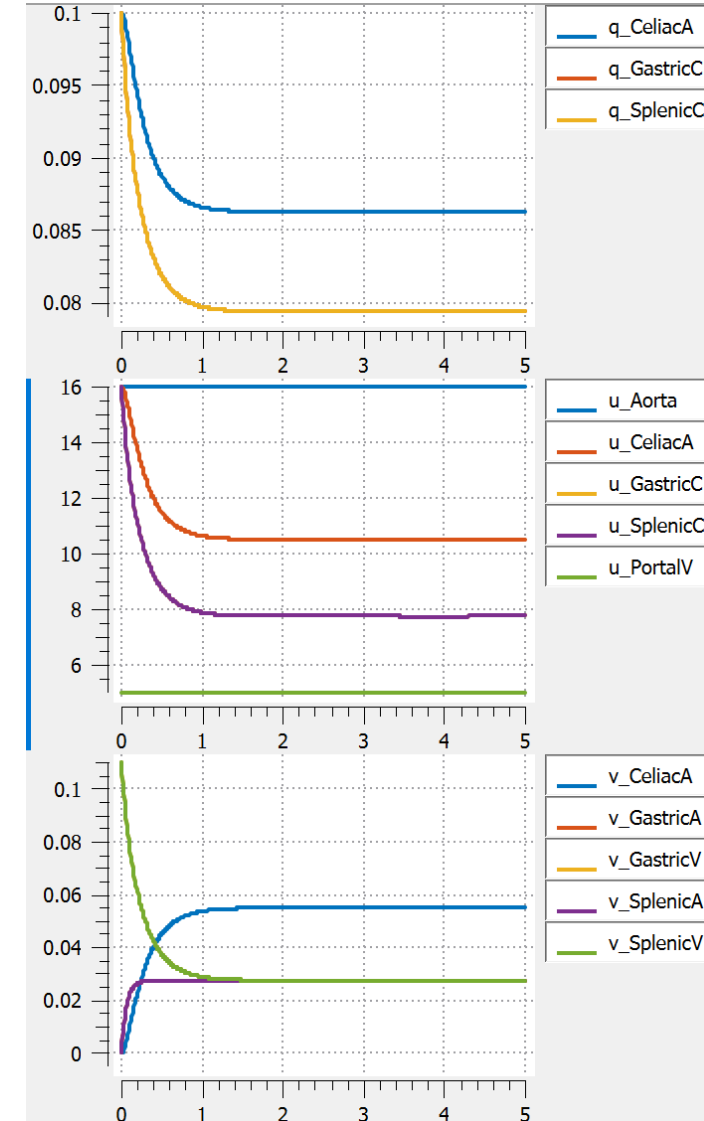
$$u_{CeliacA}^b = q_{CeliacA}^b / C_{CeliacA}^b$$

$$\frac{dq_{CeliacA}^b}{dt} = v_{CeliacA}^b - v_{GastricA}^b - v_{SplenicA}^b$$

$$u_{CeliacA}^b = q_{CeliacA}^b / C_{CeliacA}^b$$

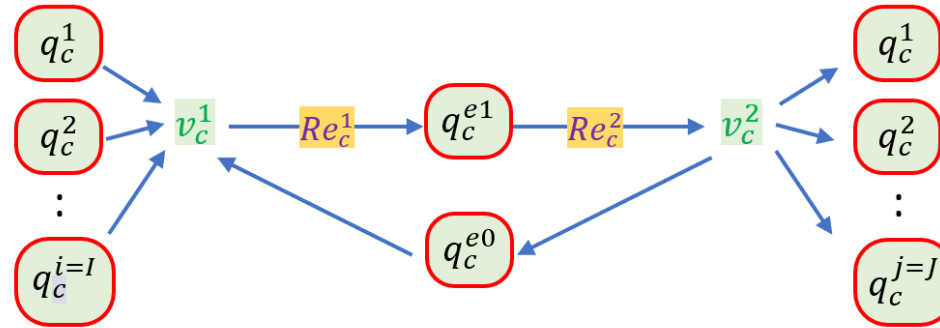
$$\frac{dq_{CeliacA}^b}{dt} = v_{CeliacA}^b - v_{GastricA}^b - v_{SplenicA}^b$$

$$u_{CeliacA}^b = q_{CeliacA}^b / C_{CeliacA}^b$$



8. An enzyme-catalysed reaction with multiple reactants and products

(see [model on PMR](#))



$$v = \frac{A^f \prod_i q_c^i - A^r \prod_j q_c^j}{1 + B^f \prod_i q_c^i + B^r \prod_j q_c^j} \quad \text{where } \prod_i q_c^i = q_c^1 \cdot q_c^2 \cdot q_c^3 \cdot \dots \cdot q_c^I$$

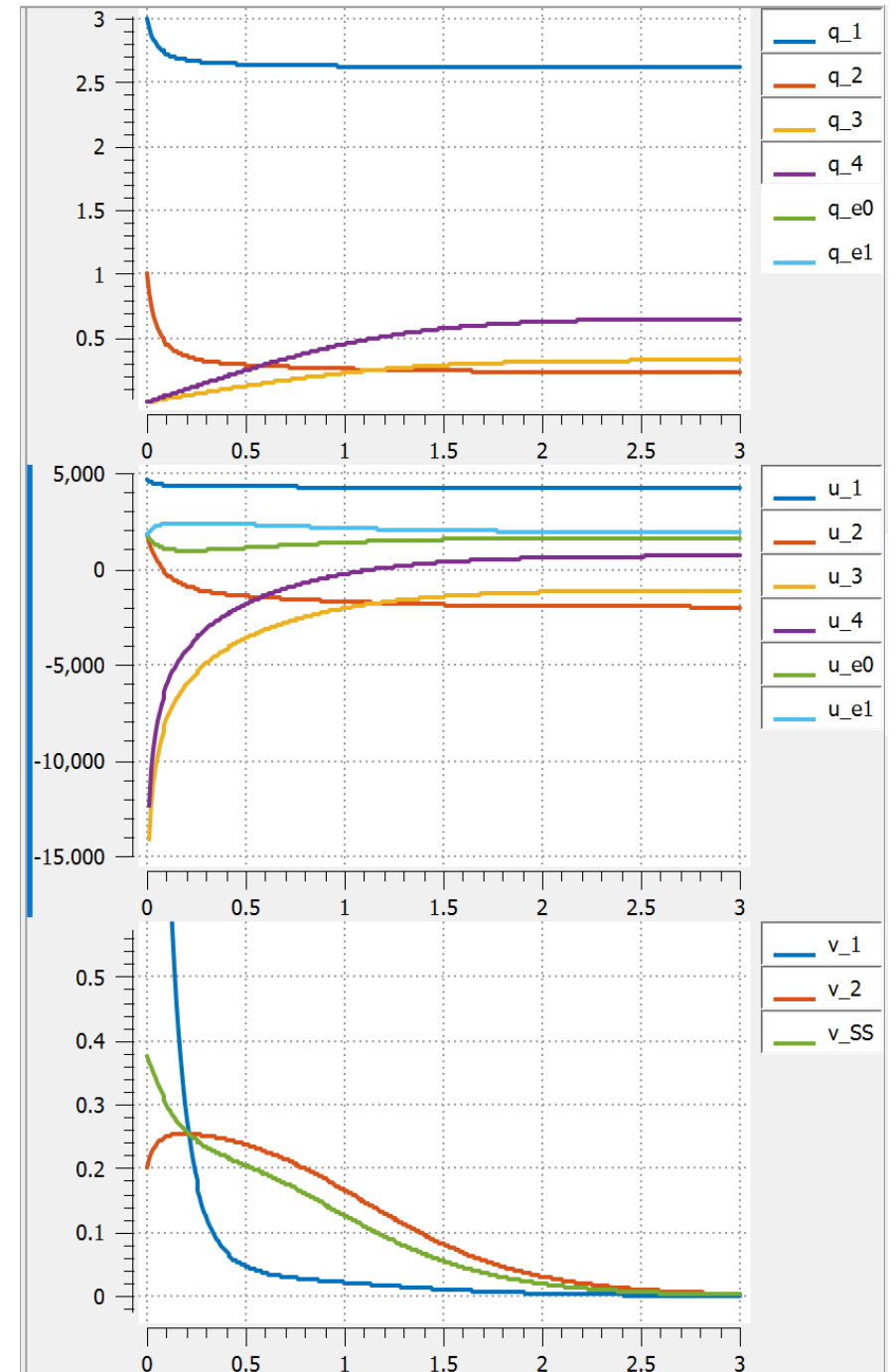
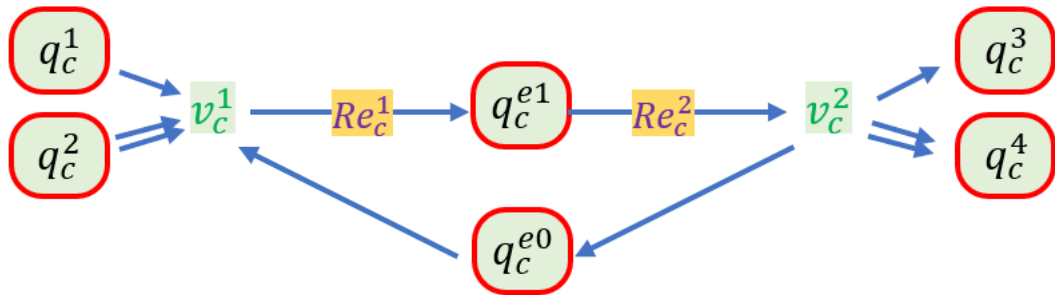
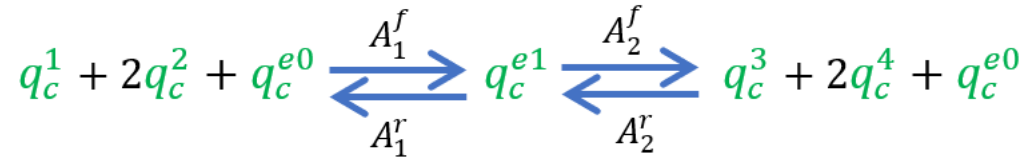
and $\prod_j q_c^j = q_c^1 \cdot q_c^2 \cdot q_c^3 \cdot \dots \cdot q_c^J$

$$\text{where } A^f = E_0 K_c^{e0} \cdot \frac{\kappa_1 \kappa_2}{\kappa_1 + \kappa_2} \cdot \prod_i K_c^i \quad (\text{mol}^{-I} \cdot \text{s}^{-1}); \quad A^r = E_0 K_c^{e0} \cdot \frac{\kappa_1 \kappa_2}{(\kappa_1 + \kappa_2)} \cdot \prod_j K_c^j \quad (\text{mol}^{-J} \cdot \text{s}^{-1});$$

$$B^f = \frac{K_c^{e0}}{K_c^{e1}} \cdot \frac{\kappa_1}{(\kappa_1 + \kappa_2)} \prod_i K_c^i \quad (\text{mol}^{-I}); \quad B^r = \frac{K_c^{e0}}{K_c^{e1}} \cdot \frac{\kappa_2}{(\kappa_1 + \kappa_2)} \prod_j K_c^j \quad (\text{mol}^{-J});$$

An example

(see [model on PMR](#))



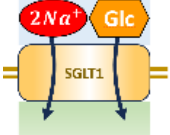
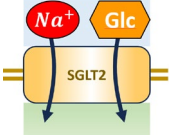
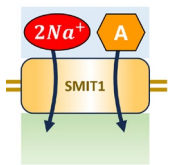
Lecture 4: The SLC family

(Transport of small molecules across cell membranes)

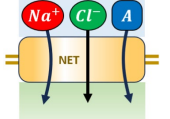
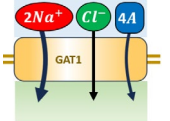
The primary goal of this lecture is to show you how to model **facilitated diffusion** – the process by which a small molecule passes into a cell – and how to handle **electrogenic transporters**. The notes describe many more **transporters** (**from the SLC family**), and once you understand the bond graph approach to modelling the reactions involved in these transporters, with enough practice you should be able to tackle any of these. The other key message in this lecture is how to **reduce the complexity of a model in a way that continues to satisfy the physical conservation laws**. The model for steady state flux of a small molecule through a cell membrane can usefully be reduced to depend on a small number of parameters, but unlike standard Hill-type models that use only 2 parameters (v_{max} and k_m), these can now be understood in terms of the underlying chemical process parameters. Most importantly, the model deals with situations where the Michaelis-Menten assumption that the concentration of a solute on one side of a reaction (whether in the cytosol or extracellular space) is much less than on the other side (and therefore can be ignored) is not appropriate – as it violates thermodynamic principles.

The SLC (solute carrier) group of proteins, encoded by 458 mammalian genes, mediate the passive transport of small molecules across cell membranes in nearly all types of tissues (i.e., does not include ATPase pumps). They are categorised under 66 gene families, labelled SLC1 to SLC66 ([SLCtables \(bioparadigms.org\)](http://bioparadigms.org)). Each family deals with a specific type of transported molecule.

SLC5 family: Na⁺/glucose cotransporters

SLC5A1	P13866	SGLT1	Na ⁺ -driven glucose transporter	
SLC5A2	P31639	SGLT2		
SLC5A3	P53794	SMIT1	Na ⁺ /myo-inositol cotransporter A = myo-inositol	

SLC6 family: Na⁺- and Cl⁻-dependent neurotransmitter symporters

SLC6A1	P30531	GAT1	A = Aminobutanoate	
SLC6A2	P23975	NET	A = Norepinephrine	

SLC transporters: 458 genes in 66 gene families

SLC1: High-affinity glutamate and neutral amino acid transporters
SLC2: Facilitative GLUT transporters
SLC3: Heavy subunits heterodimeric amino acid transporters
SLC4: Bicarbonate cotransporters
SLC5: Na⁺/glucose cotransporters
SLC6: Na⁺- and Cl⁻-dependent neurotransmitter symporters
SLC7: Cationic AA transporter/glycoprotein - assoc AA transporters
SLC8: Na⁺/Ca²⁺ exchangers
SLC9: Na⁺/H⁺ exchangers
SLC10: Na⁺/bile salt co-transporters
SLC11: Proton-coupled metal ion transporters
SLC12: Electroneutral cation/Cl⁻ co-transporters
SLC13: Na⁺/SO₄²⁻/carboxylate co-transporters
SLC14: Urea transporters
SLC15: Proton-driven oligopeptide cotransporters
SLC16: Monocarboxylate transporters
SLC17: Type I Na⁺/phosphate cotransporters
& vesicular glutamate transporters
SLC18: Vesicular amine transporters
SLC19: Folate/thiamine transporters
SLC20: Type-III Na⁺/phosphate co-transporters
SLC21: Organic anion and cation transporters
SLC22: Organic cation/anion/zwitterion transporters
SLC23: Na⁺-dependent ascorbic acid transporters
SLC24: Na⁺/(Ca²⁺-K⁺) exchangers
SLC25: Mitochondrial carriers
SLC26: Multifunctional anion exchangers
SLC27: Fatty acid transporters
SLC28: Na⁺-coupled nucleoside transporters
SLC29: Facilitative nucleoside transporters
SLC30: Zinc efflux transporters
SLC31: Copper transporters
SLC32: Vesicular inhibitory amino acid transporters
SLC33: Acetyl-CoA transporters

SLC34: Type II Na⁺-phosphate cotransporters
SLC35: Nucleoside-sugar transporters
SLC36: Proton-coupled amino acid transporters
SLC37: Sugar-phosphate/phosphate exchangers
SLC38: System A and System N Na⁺-coupled neutral AA transporters
SLC39: Metal ion transporters
SLC40: Basolateral Fe²⁺ transporters
SLC41: MgtE-like magnesium transporters
SLC42: NH₃/NH₄⁺, CO₂ channels
SLC43: Na⁺-independent, system-L-like amino acid transporters
SLC44: Choline-like transporters
SLC45: H⁺/sugar cotransporters
SLC46: Folate transporters
SLC47: Multidrug and toxin extrusion (MATE) transporters
SLC48: Heme transporters
SLC49: FLVCR-related transporters
SLC50: Sugar efflux transporters
SLC51: Steroid-derived molecules transporters
SLC52: Riboflavin transporters
SLC53: Phosphate carriers
SLC54: Mitochondrial pyruvate carriers
SLC55: Mitochondrial cation/proton exchangers
SLC56: Sideroflexins
SLC57: NiPA-like magnesium transporters
SLC58: MagT-like magnesium transporters
SLC59: Sodium-dependent lysophosphatidylcholine symporters
SLC60: Glucose transporters
SLC61: Molybdate transporters
SLC62: Pyrophosphate transporters
SLC63: Sphingosine-phosphate transporters
SLC64: Golgi Ca²⁺/H⁺ exchangers
SLC65: NPC-type cholesterol transporters
SLC66: PQ-loop amino acid transporters

458 genes



66 gene families



bond graph templates

Examples:

SLC1: (EAAT) Glutamate transport

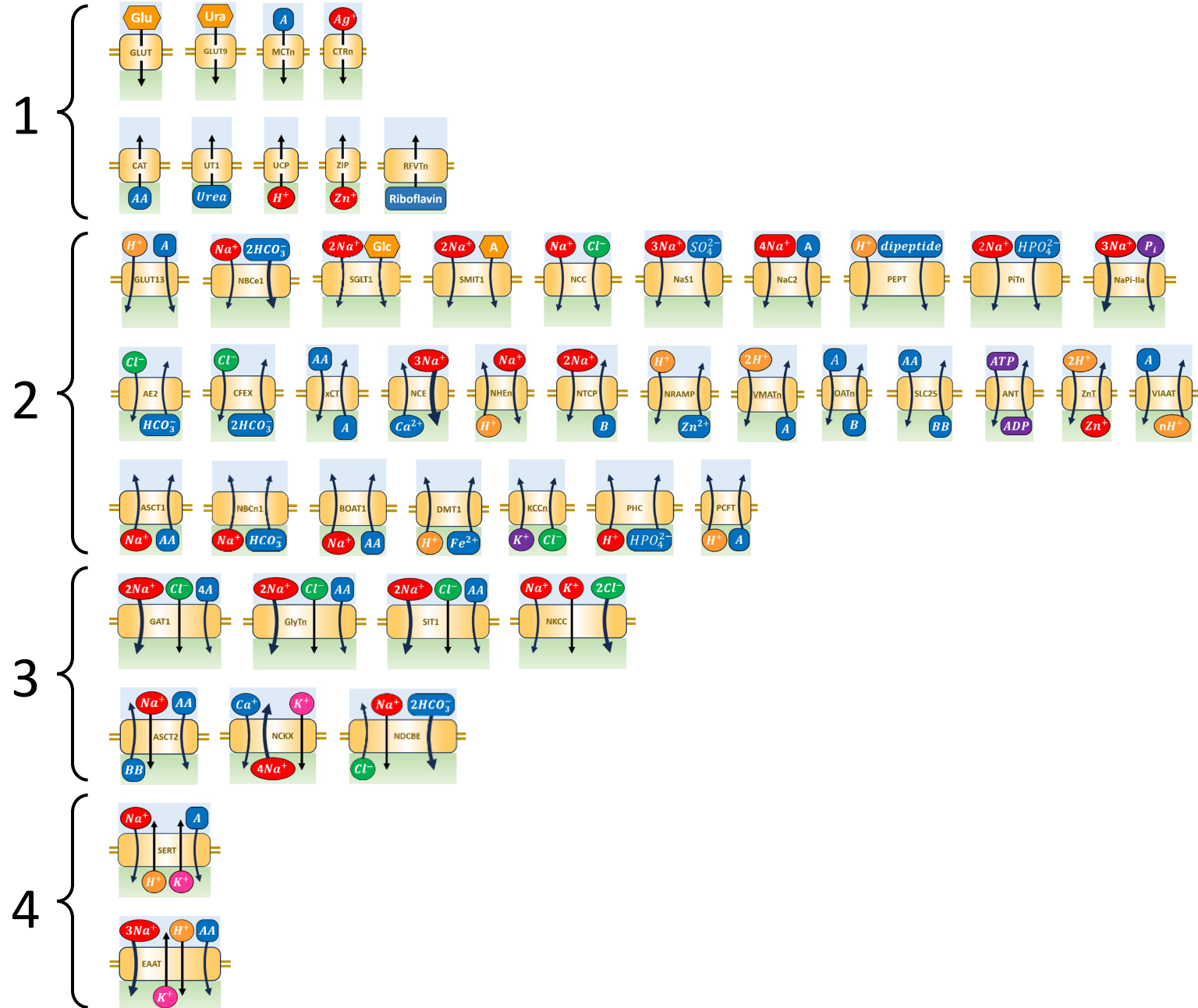
SLC2: (GLUT) Facilitated glucose transport

SLC4: (AE, NBC) Bicarbonate cotransporters

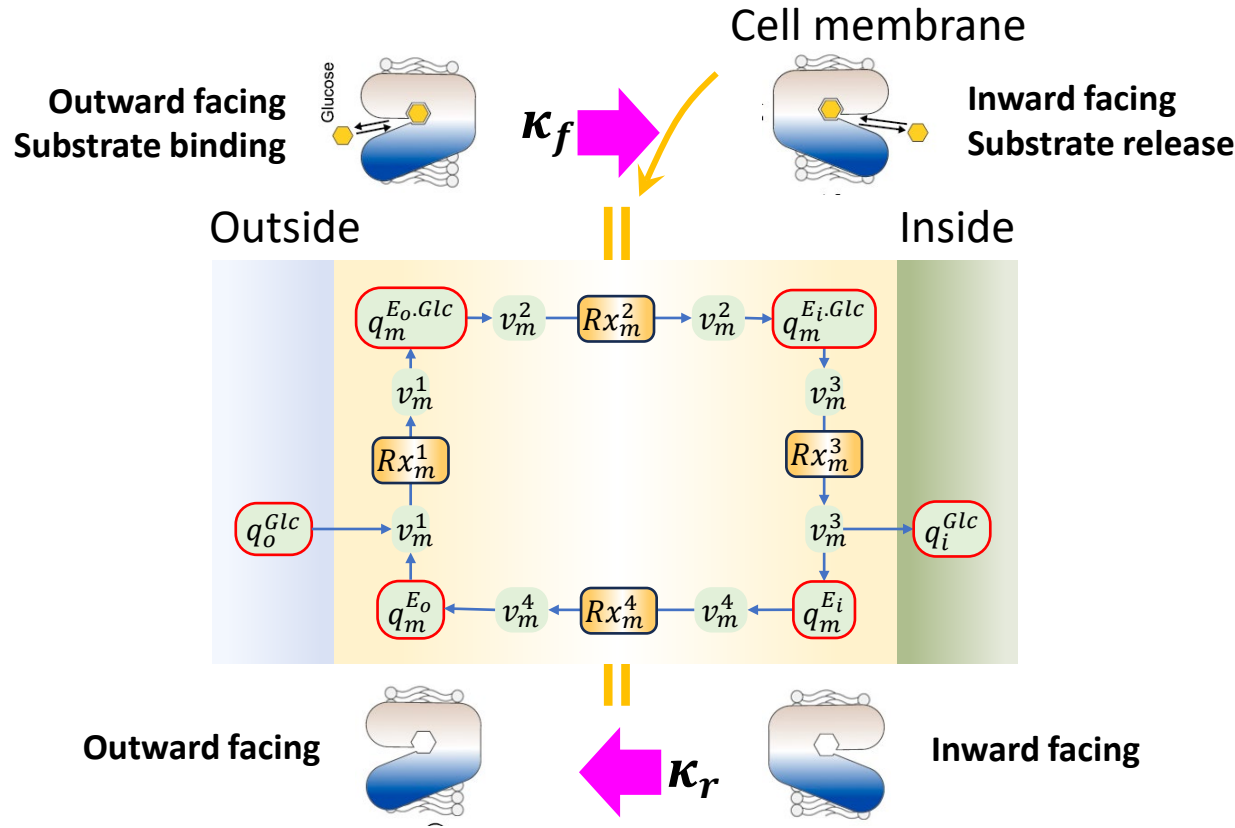
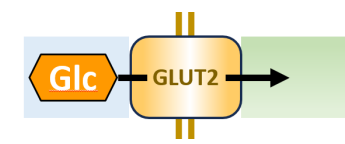
SLC5: (SGLT) Na-glucose cotransport

SLC8: (NCE) Na-Ca exchange

SLC9: (NHE) Na-H exchange



Facilitated diffusion with the glucose transporter GLUT2 (*SLC2A2*)



Mass balance (0:nodes)

$$\frac{dq_o^{Glc}}{dt} = -v_m^1$$

$$\frac{dq_i^{Glc}}{dt} = v_m^3$$

$$\frac{dq_m^{E_o}}{dt} = v_m^4 - v_m^1$$

$$\frac{dq_m^{E_i}}{dt} = v_m^3 - v_m^4$$

$$\frac{dq_m^{E_o, Glc}}{dt} = v_m^1 - v_m^2$$

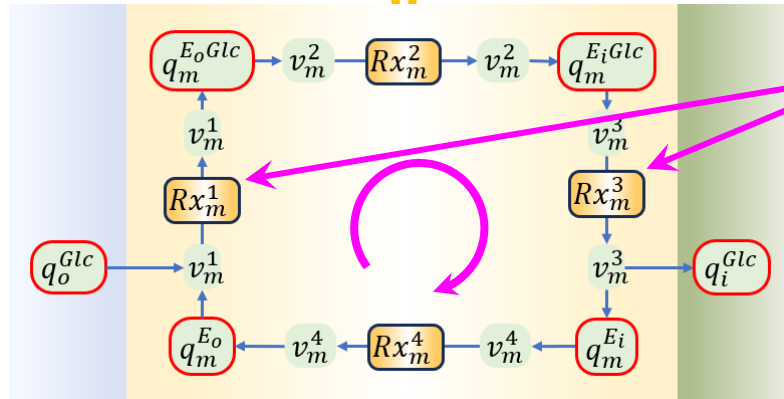
$$\frac{dq_m^{E_i, Glc}}{dt} = v_m^2 - v_m^3$$

Reaction fluxes

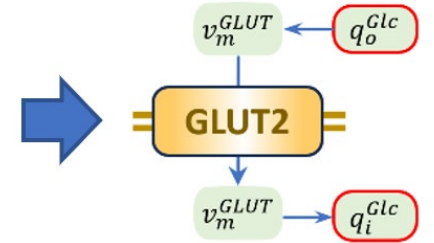
$$\begin{aligned} (Rx_m^1) \quad v_m^1 &= \kappa_1 \left(e^{u_{Rx1}^f/RT} - e^{u_{Rx1}^r/RT} \right), \text{ where } u_{Rx1}^f = u_o^{Glc} + u_m^{E_o} \text{ and } u_{Rx1}^r = u_m^{E_o, Glc}, \text{ which gives } v_m^1 = \kappa_1 \left(\bar{q}_m^{E_o} \cdot \bar{q}_o^{Glc} - \bar{q}_m^{E_o, Glc} \right) \\ (Rx_m^2) \quad v_m^2 &= \kappa_2 \left(e^{u_{Rx2}^f/RT} - e^{u_{Rx2}^r/RT} \right), \text{ where } u_{Rx2}^f = u_m^{E_o, Glc} \text{ and } u_{Rx2}^r = u_m^{E_i, Glc}, \text{ which gives } v_m^2 = \kappa_2 \left(\bar{q}_m^{E_o, Glc} - \bar{q}_m^{E_i, Glc} \right) \\ (Rx_m^3) \quad v_m^3 &= \kappa_3 \left(e^{u_{Rx3}^f/RT} - e^{u_{Rx3}^r/RT} \right), \text{ where } u_{Rx3}^f = u_m^{E_i, Glc} \text{ and } u_{Rx3}^r = u_m^{E_i} + u_i^{Glc}, \text{ which gives } v_m^3 = \kappa_3 \left(\bar{q}_m^{E_i, Glc} - \bar{q}_m^{E_i} \cdot \bar{q}_i^{Glc} \right) \\ (Rx_m^4) \quad v_m^4 &= \kappa_4 \left(e^{u_{Rx4}^f/RT} - e^{u_{Rx4}^r/RT} \right), \text{ where } u_{Rx4}^f = u_m^{E_i} \text{ and } u_{Rx4}^r = u_m^{E_o, Glc}, \text{ which gives } v_m^4 = \kappa_4 \left(\bar{q}_m^{E_i} - \bar{q}_m^{E_o} \right) \end{aligned}$$

8 parameters

Reduced models



1. Binding & unbinding is very rapid & assume constant enzyme cycling rate
2. Michaelis-Menten ($q_i^{Glc} \ll q_o^{Glc}$)



Model reduction 1: Fast binding and unbinding ($\kappa_1, \kappa_3 \rightarrow \infty$) and SS ($v_1 = v_2 = v_3 = v_4 = v$)

$$v = v_m (\bar{q}_o^{Glc} - \bar{q}_i^{Glc}) / \left(1 + \frac{\bar{q}_o^{Glc}}{k_m^1} + \frac{\bar{q}_i^{Glc}}{k_m^2} + \frac{\bar{q}_o^{Glc} \bar{q}_i^{Glc}}{k_m^3} \right); \quad 4 \text{ parameters}$$

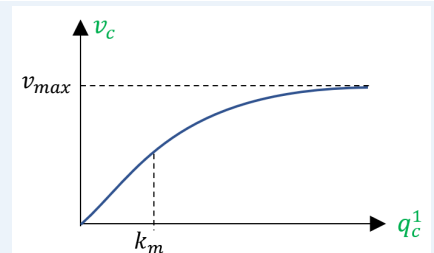
$$\text{with } k_m^1 = \frac{(K_m^{Eo})^{-1} + (K_m^{Ei})^{-1}}{(K_m^{EoGlc})^{-1} + \Lambda (K_m^{Ei})^{-1}}, \quad k_m^2 = \frac{(K_m^{Eo})^{-1} + (K_m^{Ei})^{-1}}{(K_m^{EiGlc})^{-1} + \Lambda (K_m^{Eo})^{-1}}, \quad k_m^3 = \frac{(K_m^{Eo})^{-1} + (K_m^{Ei})^{-1}}{\Lambda \left((K_m^{EoGlc})^{-1} + (K_m^{EiGlc})^{-1} \right)},$$

$$v_m = \frac{q_{tot} \kappa_2}{(K_m^{Eo})^{-1} + (K_m^{Ei})^{-1}}, \quad \Lambda = \frac{\kappa_2}{\kappa_4}$$

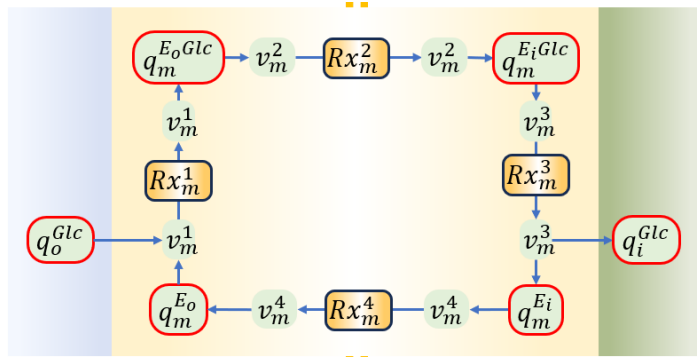
where $\bar{q}_j^i = K_j q_j^i = V_j K_j \cdot c_j^i$ (nondimensional), so k_m^1, k_m^2 & k_m^3 are also nondimensional.

Model reduction 2: Michaelis-Menten (MM) assumption ($q_i^{Glc} \ll q_o^{Glc}$)

$$v = v_{max} \frac{\bar{q}_o^{Glc}}{k_m^1 + \bar{q}_o^{Glc}} \quad 2 \text{ parameters} \quad \dots \text{but thermodynamically invalid}$$



Derivation of reduced model



Reaction fluxes

$$v_m^1 = \kappa_1 [\bar{q}_m^{E_o} \cdot \bar{q}_o^{Glc} - \bar{q}_m^{E_o.Glc}] \quad \text{As } \kappa_1 \rightarrow \infty, [\bar{q}_m^{E_o} \cdot \bar{q}_o^{Glc} - \bar{q}_m^{E_o.Glc}] \rightarrow 0 \text{ and hence } \bar{q}_m^{E_o} \cdot \bar{q}_o^{Glc} = \bar{q}_m^{E_o.Glc} \quad (1)$$

$$v_m^2 = \kappa_2 [\bar{q}_m^{E_o.Glc} - \bar{q}_m^{E_i.Glc}] \quad v_m^2 = v_m^{GLUT2} \quad \text{gives} \quad \kappa_2 [\bar{q}_m^{E_o.Glc} - \bar{q}_m^{E_i.Glc}] = v_m^{GLUT2} \quad (2)$$

$$v_m^3 = \kappa_3 [\bar{q}_m^{E_i.Glc} - \bar{q}_m^{E_i} \cdot \bar{q}_i^{Glc}] \quad \text{As } \kappa_3 \rightarrow \infty, [\bar{q}_m^{E_i.Glc} - \bar{q}_m^{E_i} \cdot \bar{q}_i^{Glc}] \rightarrow 0 \text{ and hence } \bar{q}_m^{E_i.Glc} = \bar{q}_m^{E_i} \cdot \bar{q}_i^{Glc} \quad (3)$$

$$v_m^4 = \kappa_4 [\bar{q}_m^{E_i} - \bar{q}_m^{E_o}] \quad v_m^4 = v_m^{GLUT2} \quad \text{gives} \quad \kappa_4 [\bar{q}_m^{E_i} - \bar{q}_m^{E_o}] = v_m^{GLUT2} \quad (4)$$

Mass balance:

$$\frac{d}{dt} q_m^{E_o} = v_m^4 - v_m^1 = 0 \quad \text{gives } v_m^4 = v_m^1 = v_m^{GLUT2} \quad \frac{d}{dt} q_o^{Glc} = -v_m^1 = -v_m^{GLUT2}$$

$$\frac{d}{dt} q_m^{E_o.Glc} = v_m^1 - v_m^2 = 0 \quad \text{gives } v_m^1 = v_m^2 = v_m^{GLUT2} \quad \frac{d}{dt} q_i^{Glc} = v_m^3 = v_m^{GLUT2}$$

$$\frac{d}{dt} q_m^{E_i.Glc} = v_m^2 - v_m^3 = 0 \quad \text{gives } v_m^2 = v_m^3 = v_m^{GLUT2}$$

$$\frac{d}{dt} q_m^{E_i} = v_m^3 - v_m^4 = 0 \quad \text{gives } v_m^3 = v_m^4 = v_m^{GLUT2}$$

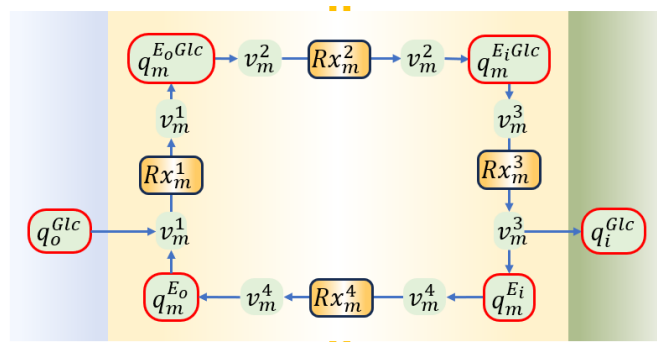
$$\text{An additional constraint is needed: } q_m^{E_o} + q_m^{E_o.Glc} + q_m^{E_i.Glc} + q_m^{E_i} = q_{tot} \quad (5)$$

Equations 1..5 are used to eliminate the enzyme states $\bar{q}_m^{E_o}$, $\bar{q}_m^{E_o.Glc}$, $\bar{q}_m^{E_i.Glc}$, $\bar{q}_m^{E_i}$ and hence obtain

$$v_m^{GLUT2} = v_{max} (\bar{q}_o^{Glc} - \bar{q}_i^{Glc}) / \left(1 + \frac{\bar{q}_o^{Glc}}{k_m^1} + \frac{\bar{q}_i^{Glc}}{k_m^2} + \frac{\bar{q}_o^{Glc} \bar{q}_i^{Glc}}{k_m^3} \right)$$

Facilitated diffusion: Comparing models

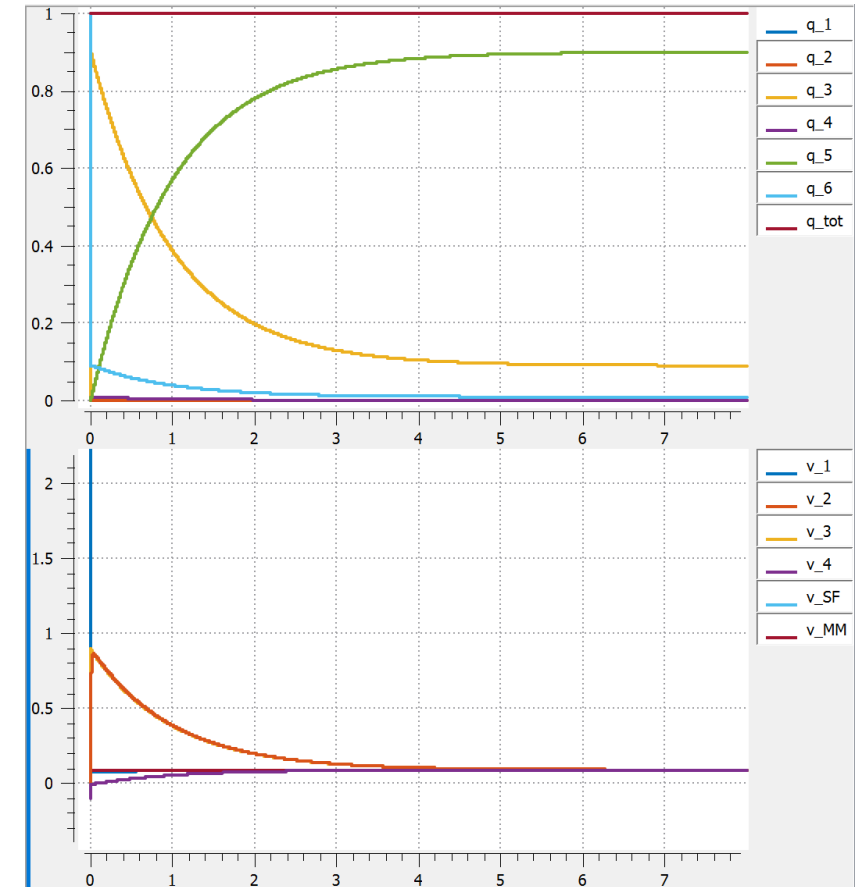
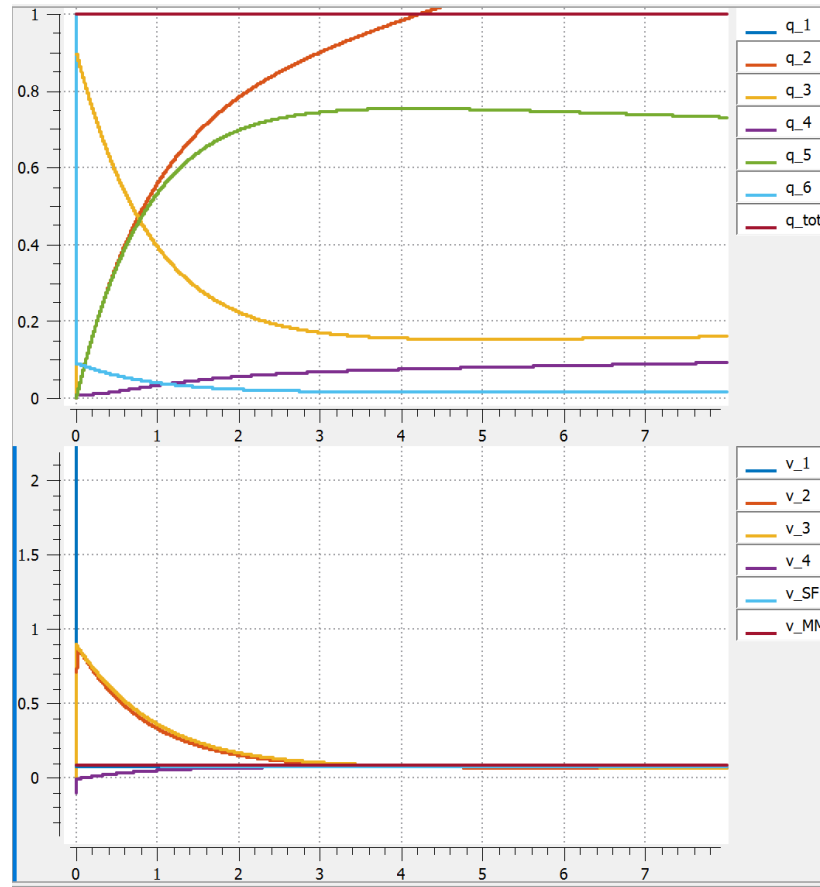
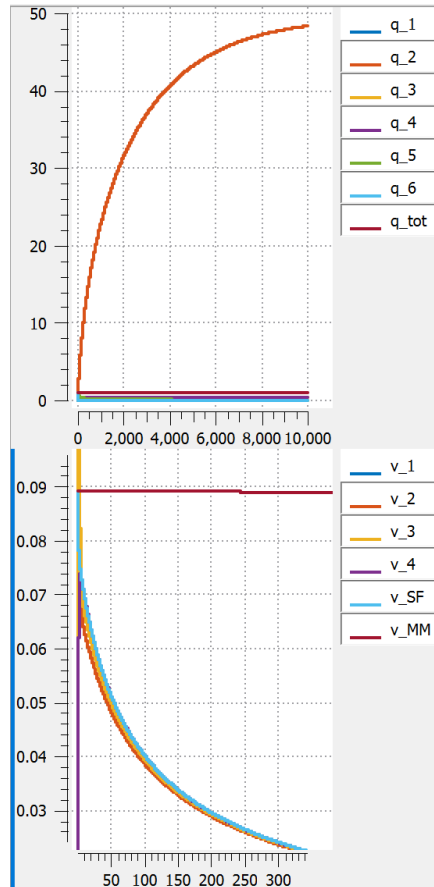
(see [model on PMR](#))



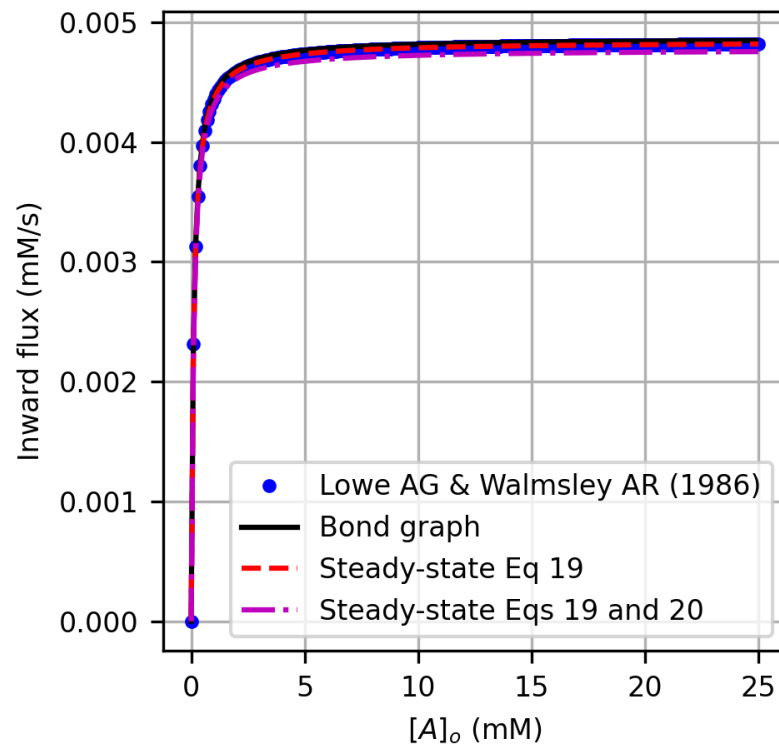
$$\begin{aligned}\kappa_1 &= 100 \\ \kappa_2 &= 1 \\ \kappa_3 &= 100 \\ \kappa_4 &= 0.1\end{aligned}$$

Initially: $q_o^{Glc} = 100$, $q_i^{Glc} = 0$, $q_{tot}^{enzyme} = 1$

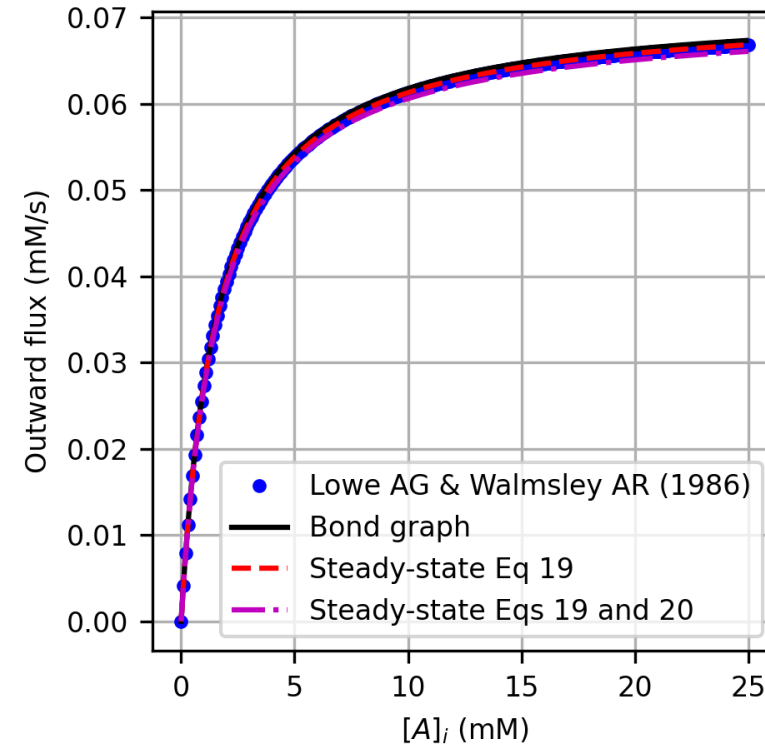
Add a sink for q_2^A ($\frac{d}{dt}q_2^A = v^A - q_2^A/100$)



GLUT2 model parameters fitted to experimental data from Lowe & Walmsley (1986)



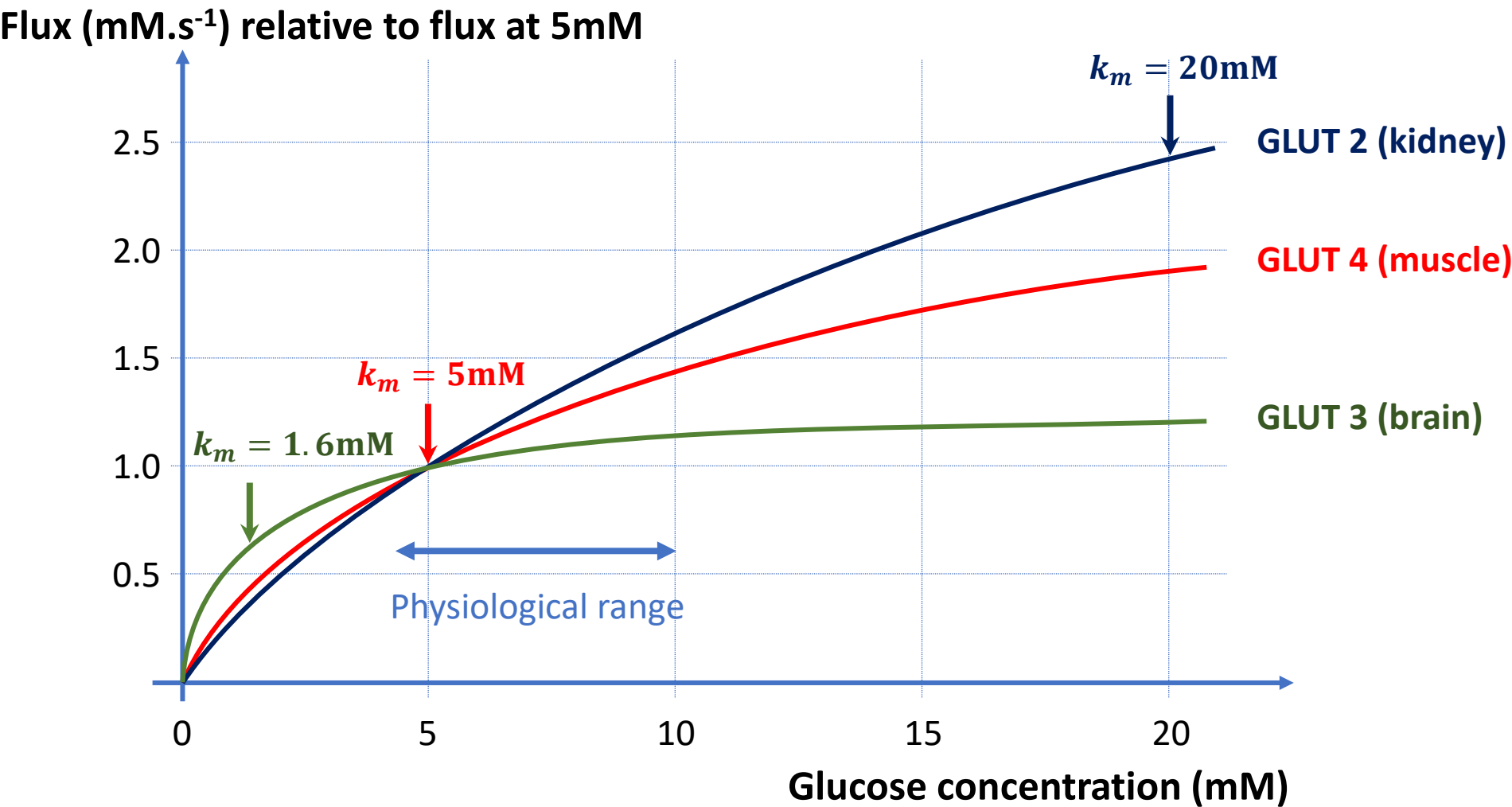
(a)



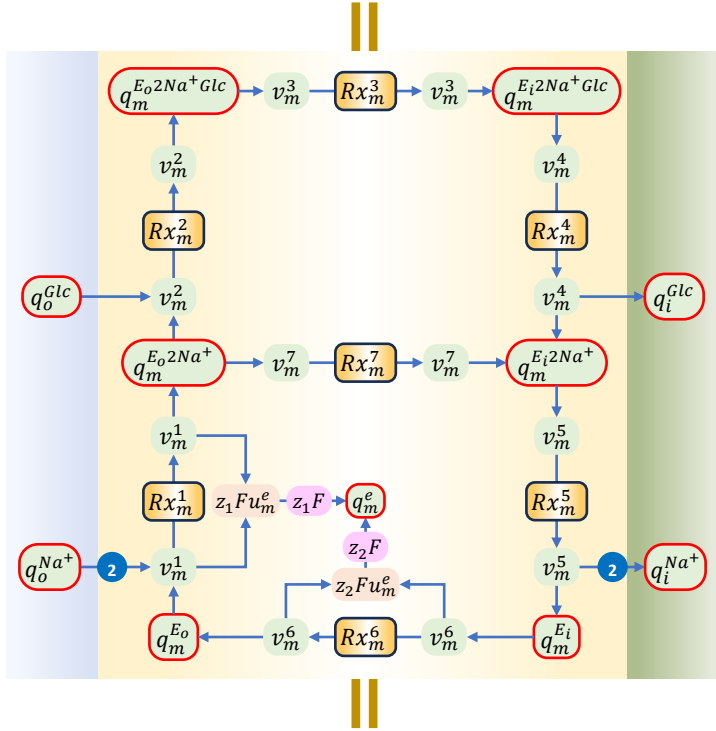
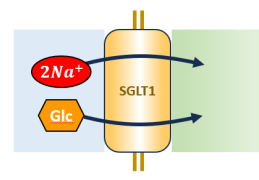
(b)

Figure (a) Inward flux as a function of $[Glc]_o$ when $[Glc]_i = 0$, and (b) outward flux as a function of $[Glc]_i$ when $[Glc]_o = 0$. Note that in order to compare with the kinetic data, the molar amount of glucose in the bond graph model was converted to glucose concentrations.

Note on tissue-specific GLUT transporters



The electrogenic, sodium-driven glucose transporter SGLT1 (*SLC5A1*)



Reaction fluxes

$$v_1 = \kappa_1 \left[(\bar{q}_o^{Na^+})^2 \cdot \bar{q}_m^{E_o} \cdot e^{\frac{-z_1 F u_m^e}{RT}} - \bar{q}_m^{E_o 2Na^+} \cdot e^{\frac{z_1 F u_m^e}{RT}} \right]$$

$$v_2 = \kappa_2 \left[\bar{q}_o^{Glc} \cdot \bar{q}_m^{E_o 2Na^+} - \bar{q}_m^{E_o 2Na^+ Glc} \right]$$

$$v_3 = \kappa_3 \left[\bar{q}_m^{E_o 2Na^+ Glc} - \bar{q}_m^{E_i 2Na^+ Glc} \right]$$

$$v_4 = \kappa_4 \left[\bar{q}_m^{E_i 2Na^+ Glc} - \bar{q}_i^{Glc} \cdot \bar{q}_m^{E_i 2Na^+} \right]$$

$$v_5 = \kappa_5 \left[\bar{q}_m^{E_i 2Na^+} - \bar{q}_i^{Na^+} \cdot \bar{q}_m^{E_i} \right]$$

$$v_6 = \kappa_6 \left[\bar{q}_m^{E_i} \cdot e^{\frac{-z_2 F u_m^e}{RT}} - \bar{q}_m^{E_o} \cdot e^{\frac{z_2 F u_m^e}{RT}} \right]$$

$$v_7 = \kappa_7 \left[\bar{q}_m^{E_o 2Na^+} - \bar{q}_m^{E_i 2Na^+} \right]$$

Mass balance (0:nodes)

$$\frac{dq_o^{Na^+}}{dt} = -2v_m^1$$

$$\frac{dq_i^{Na^+}}{dt} = 2v_m^5$$

$$\frac{dq_o^{Glc}}{dt} = -v_m^2$$

$$\frac{dq_i^{Glc}}{dt} = v_m^4$$

$$\frac{dq_m^{E_o}}{dt} = v_m^6 - v_m^1$$

$$\frac{dq_m^{E_o 2Na^+}}{dt} = v_m^1 - v_m^2 - v_m^7$$

$$\frac{dq_m^{E_o 2Na^+ Glc}}{dt} = v_m^2 - v_m^3$$

$$\frac{dq_m^{E_i 2Na^+ Glc}}{dt} = v_m^3 - v_m^4$$

$$\frac{dq_m^{E_i 2Na^+}}{dt} = v_m^4 - v_m^5 + v_m^7$$

$$\frac{dq_m^{E_i}}{dt} = v_m^5 - v_m^6$$

Energy balance (1:nodes)

$$u_{Rx1}^f = 2u_o^{Na^+} + u_m^{E_o} - z_1 F u_m^e$$

$$u_{Rx1}^r = u_m^{E_o 2Na^+} + z_1 F u_m^e$$

$$u_{Rx2}^f = u_o^{Glc} + u_m^{E_o 2Na^+}$$

$$u_{Rx2}^r = u_m^{E_o 2Na^+ Glc}$$

$$u_{Rx3}^f = u_m^{E_o 2Na^+ Glc}$$

$$u_{Rx3}^r = u_m^{E_i 2Na^+ Glc}$$

$$u_{Rx4}^f = u_m^{E_i 2Na^+ Glc}$$

$$u_{Rx4}^r = u_i^{Glc} + u_m^{E_i 2Na^+}$$

$$u_{Rx5}^f = u_m^{E_i 2Na^+}$$

$$u_{Rx5}^r = 2u_i^{Na^+} + u_m^{E_i}$$

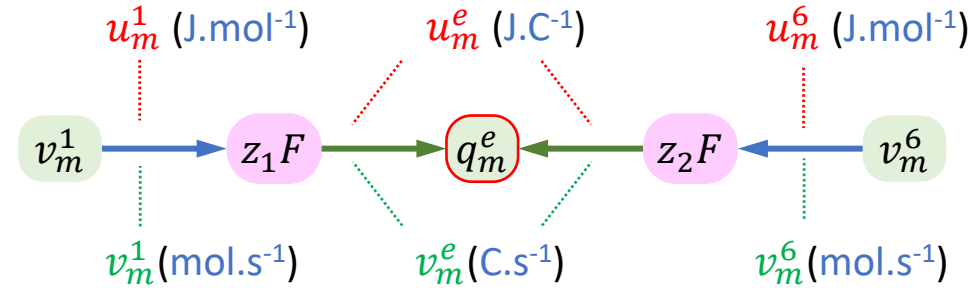
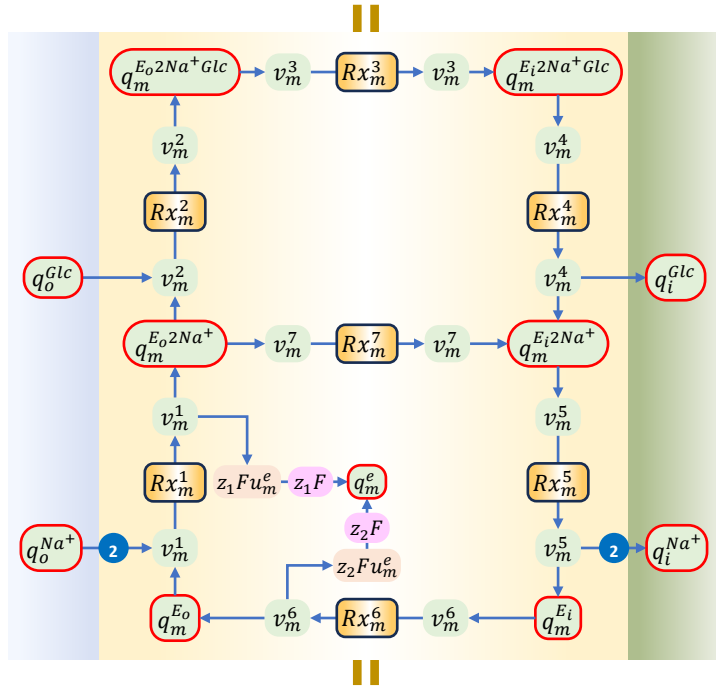
$$u_{Rx6}^f = u_m^{E_i} - z_2 F u_m^e$$

$$u_{Rx6}^r = u_m^{E_o} + z_2 F u_m^e$$

$$u_{R7}^f = u_m^{E_o 2Na^+}$$

$$u_{R7}^r = u_m^{E_i 2Na^+}$$

Modeling charge transfer in electrogenic transporters



Power: $u_m^1 \cdot v_m^1$ (J.s⁻¹) $u_m^e \cdot v_m^e$ (J.s⁻¹) $u_m^6 \cdot v_m^6$ (J.s⁻¹)

where u_m^1 (J.mol⁻¹) = $z_1 F$ (C.mol⁻¹) x u_m^e (J.C⁻¹)

and v_m^e (C.s⁻¹) = $z_1 F$ (C.mol⁻¹) x v_m^1 (mol.s⁻¹)

Note: $u_m^e = q_m^e / C$ with capacitance $C \sim 10^{-6}$ C²J⁻¹ (1μF) per m² of membrane

The sum of potentials around the loop must be zero (equivalent to adding up potential drops around a circuit).

$$\begin{aligned}
 Sum &= (u_{Rx1}^f - u_{Rx1}^r) + (u_{Rx2}^f - u_{Rx2}^r) + (u_{Rx3}^f - u_{Rx3}^r) + (u_{Rx4}^f - u_{Rx4}^r) + (u_{Rx5}^f - u_{Rx5}^r) + (u_{Rx6}^f - u_{Rx6}^r) + (u_{Rx7}^f - u_{Rx7}^r) \\
 &= (2u_o^{Na^+} + u_m^{E_o} - u_m^{E_o 2Na^+} - z_1 F u_m^e) + (u_o^{Glc} + u_m^{E_o 2Na^+} - u_m^{E_o 2Na^+ Glc}) + (u_m^{E_o 2Na^+ Glc} - u_m^{E_i 2Na^+ Glc}) + (u_m^{E_i 2Na^+ Glc} - u_i^{Glc} - u_m^{E_i 2Na^+}) \\
 &\quad + (u_m^{E_i 2Na^+} - 2u_i^{Na^+} - u_m^{E_i}) + (u_m^{E_i} - u_m^{E_o} - z_2 F u_m^e) + (u_m^{E_o 2Na^+} - u_m^{E_i 2Na^+}) \\
 &= \underbrace{2(u_o^{Na^+} - u_i^{Na^+})}_{\text{sodium transport}} + \underbrace{(u_o^{Glc} - u_i^{Glc})}_{\text{glucose transport}} + \underbrace{(u_m^{E_o 2Na^+} - u_m^{E_i 2Na^+})}_{\text{slippage}} - \underbrace{(z_1 + z_2) F u_m^e}_{\text{charge transport}}
 \end{aligned}$$

$$0 = 2 \text{ J.mol}^{-1} - 1 \text{ J.mol}^{-1} (\text{against gradient}) - (z_1 + z_2) \text{ J.mol}^{-1} \Rightarrow (z_1 + z_2) = 1 \text{ (ignoring slippage)}$$

Reduced SGLT1 model

Steady-state solution is

$$v_m^{SGLT1} = a_1 \left\{ (\bar{q}_o^{Na^+})^2 \cdot \bar{q}_o^{Glc} - (\bar{q}_i^{Na^+})^2 \cdot \bar{q}_i^{Glc} \cdot e^{2Fu_m^e/RT} \right\} / A$$

$$A = \left[e^{2z_1 Fu_m^e/RT} + (\bar{q}_o^{Na^+})^2 (a_2 + a_3 \cdot \bar{q}_o^{Glc}) \right] \left[(\bar{q}_i^{Na^+})^2 \cdot \bar{q}_i^{Glc} \cdot e^{z_2 Fu_m^e/RT} + \Lambda \right] \\ + \left[a_6 + (\bar{q}_i^{Na^+})^2 (a_5 + a_4 \cdot \bar{q}_i^{Glc}) \right] \left[(\bar{q}_o^{Na^+})^2 \cdot \bar{q}_o^{Glc} \cdot e^{z_2 Fu_m^e/RT} + \Lambda \cdot e^{2Fu_m^e/RT} \right]$$

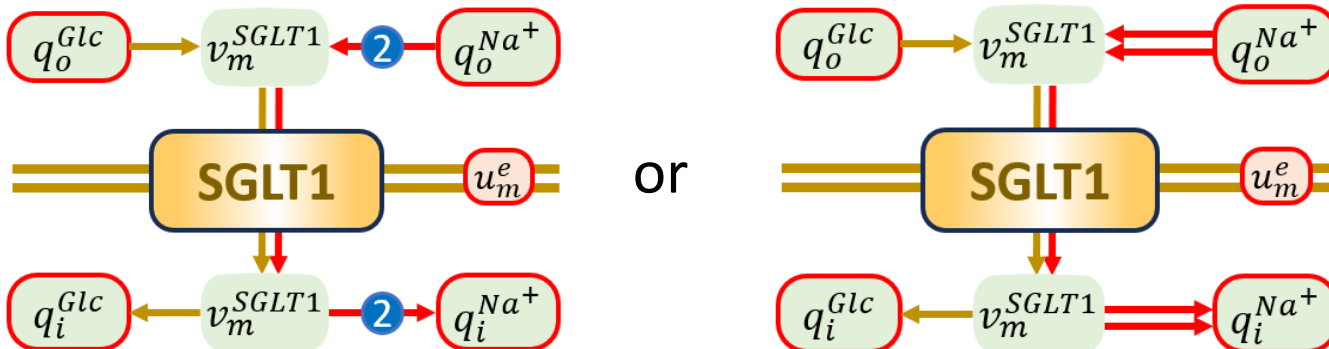
where $a_1 = q_{tot} \kappa_6 K_1$, $a_2 = \frac{K_1}{K_2}$, $a_3 = \frac{K_1}{K_3}$, $a_4 = \frac{K_1}{K_4}$, $a_5 = \frac{K_1}{K_5}$, $a_6 = \frac{K_1}{K_6}$, $\Lambda = \frac{\kappa_6}{\kappa_3}$

Note that the flux is zero when

$$(\bar{q}_o^{Na^+})^2 \cdot \bar{q}_o^{Glc} = (\bar{q}_i^{Na^+})^2 \cdot \bar{q}_i^{Glc} \cdot e^{\frac{2Fu_m^e}{RT}}$$

i.e. the equilibrium potential is

$$u_m^e = \frac{RT}{2F} \ln \left[\left(\frac{\bar{q}_o^{Na^+}}{\bar{q}_i^{Na^+}} \right)^2 \cdot \frac{\bar{q}_o^{Glc}}{\bar{q}_i^{Glc}} \right]$$



Note different colours for different chemical species.
Each arrow is carrying the same flux v_m^{SGLT1} .

SGLT1 model parameters fitted to experimental data

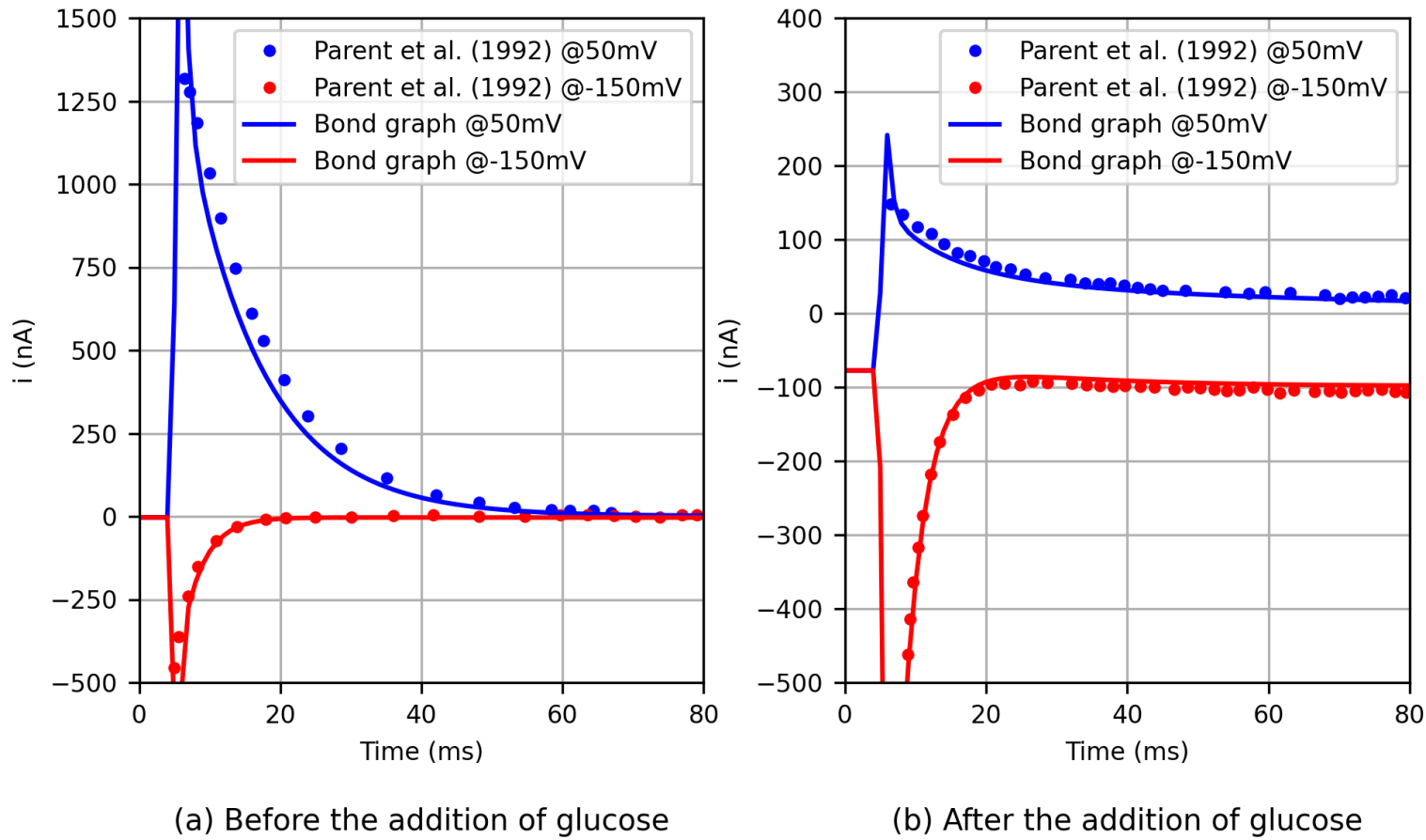


Figure 11 The time course of the carrier-mediated currents. (a) The electrical current when $[Glc]_o = 0$ mM, and (b) the current when $[Glc]_o = 1$ mM; The output of the bond graph model is the current $-I_i$, and the data of Parent et al. [19] are reproduced from Figure 10 of that paper using the digitizing software Engauge. The information needed to reproduce these results can be found in [18].

SLC transporters

In the following slides we show bond graph models for a range of SLC transporters, following the previous two examples. For each one the full kinetic model is presented, with a separate reaction for each ligand binding and unbinding event, and for the outward-facing to inward-facing transition of the protein. We also include all possible 'slippage states' where the transition may occur without completing the full cycle. Note the link to *SLC tables* in the heading for each transporter and the link to a CellML model of the transporter in the *Physiome Model Repository* (PMR).

Note that the bond graph diagrams completely define all the equations needed for the model.

For each full kinetic model we also derive a simplified model for the transporter in which the following assumptions are made:

1. The slippage reactions are ignored.
2. The states of the enzyme are assumed to be at steady-state and the enzyme therefore cycles at a constant rate (the Briggs-Haldane assumption). This is valid when the number of membrane-bound enzymes is small compared with the number of solute (ligand) molecules. The solutes can be time-varying.
3. Ligand binding and unbinding is assumed to be very rapid in comparison to the reaction rates for the membrane outward-inward-facing transitions.

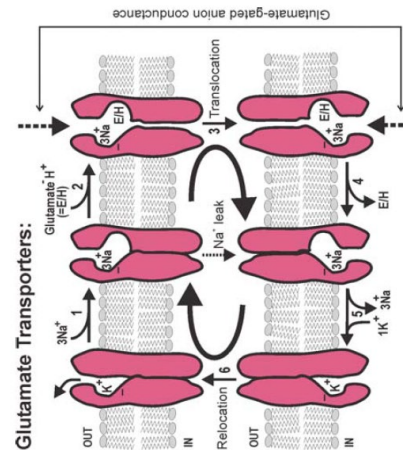
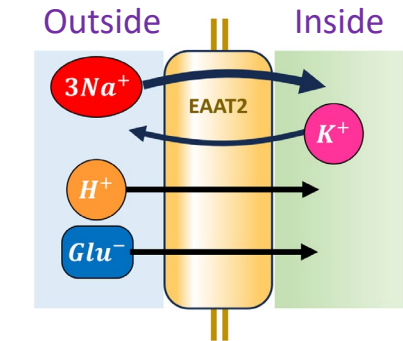
In some cases we consider different levels of complexity for the reduced models, in order to determine the simplest form that is compatible with the experimental data to an acceptable level of precision.

In each case we fit both the full kinetic model and the reduced model(s) to experimental data and compare their behaviours under various conditions.

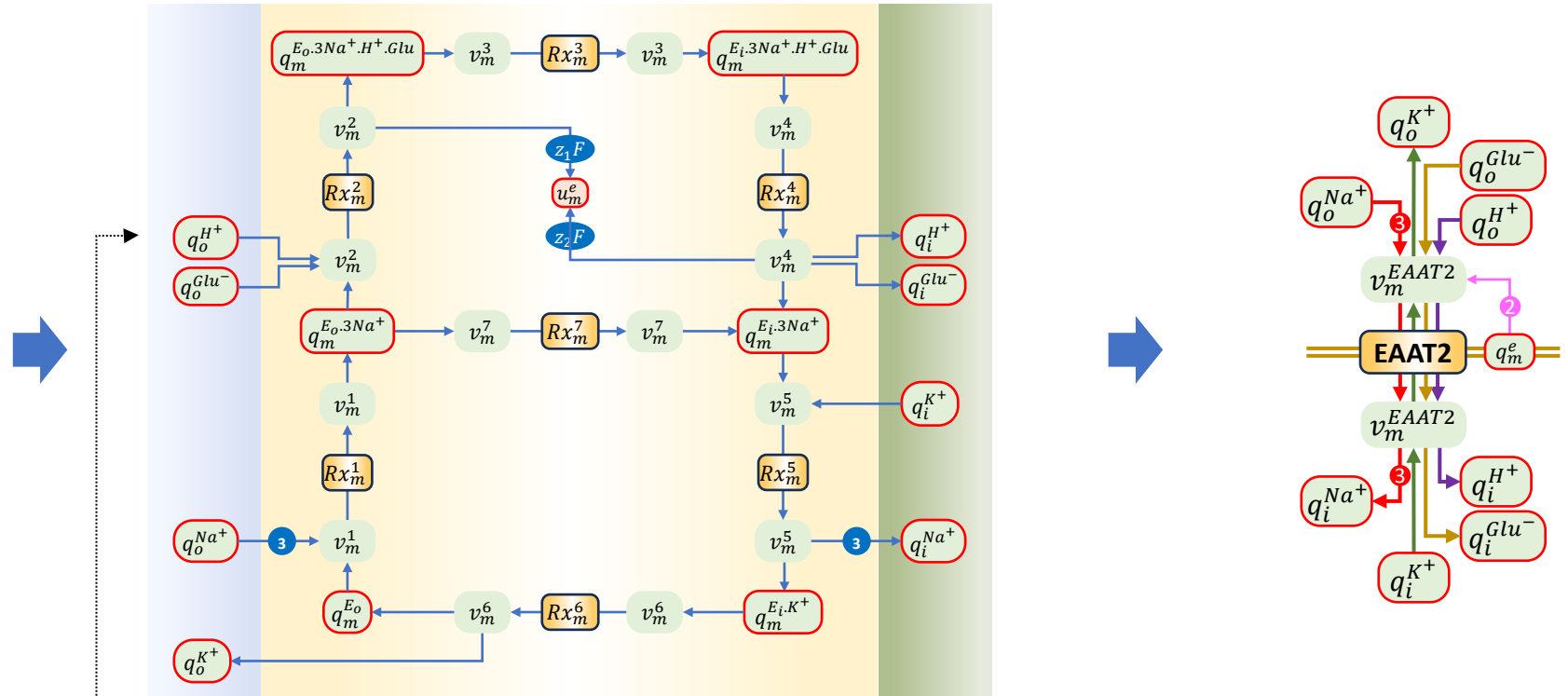
We also provide appropriate diagrams for the reduced models that can be used in the cellular models that combine multiple transporters, pumps and ion channels under dynamic changes of the solutes.

SLC1A2 (EAAT2) Glutamate transport (SLCtables)

(see [model on PMR](#))



Kanai & Hediger (2004)

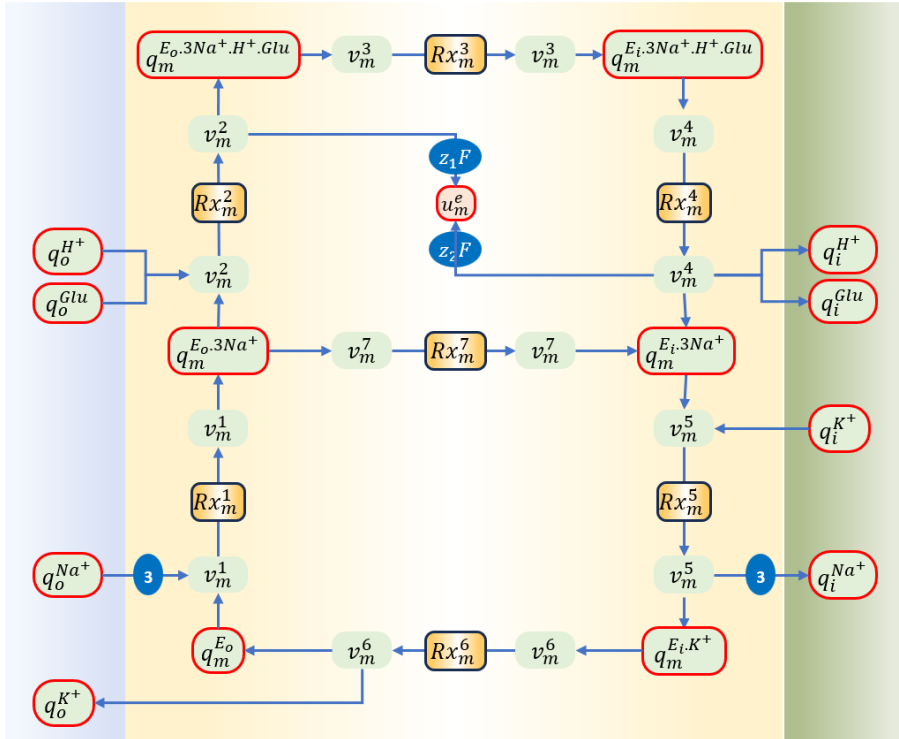


Negatively charged glutamate is often protonated and hence one binding reaction is appropriate.

Note that the order of binding and unbinding needs to be established from experimental data. Generally assume that Na^+ binds first.

Vertical orientation is used for detailed models but horizontal is useful for cell models

SLC1A2 (EAAT2) .. continued



Reaction fluxes

$$v_1 = \kappa_1 \left[\bar{q}_m^{E_o} \cdot (\bar{q}_o^{Na^+})^3 - \bar{q}_m^{E_o.3Na^+} \right]$$

$$v_2 = \kappa_2 \left[\bar{q}_m^{E_o.3Na^+} \cdot \bar{q}_o^{H^+} \cdot \bar{q}_o^{Glu} - \bar{q}_m^{E_o.3Na^+.H^+.Glu} \cdot e^{\frac{z_1 F u_m^e}{RT}} \right]$$

$$v_3 = \kappa_3 \left[\bar{q}_m^{E_o.3Na^+.H^+.Glu} - \bar{q}_m^{E_i.3Na^+.H^+.Glu} \right]$$

$$v_4 = \kappa_4 \left[\bar{q}_m^{E_i.3Na^+.H^+.Glu} - \bar{q}_m^{E_i.3Na^+} \cdot \bar{q}_i^{H^+} \cdot \bar{q}_i^{Glu} \cdot e^{\frac{z_2 F u_m^e}{RT}} \right]$$

$$v_5 = \kappa_5 \left[\bar{q}_m^{E_i.3Na^+} \cdot \bar{q}_i^{K^+} - \bar{q}_m^{E_i.K^+} \cdot (\bar{q}_i^{Na^+})^3 \right]$$

$$v_6 = \kappa_6 \left[\bar{q}_m^{E_i.K^+} - \bar{q}_m^{E_o} \cdot \bar{q}_o^{K^+} \right]$$

$$v_7 = \kappa_7 \left[\bar{q}_m^{E_o.3Na^+} - \bar{q}_m^{E_i.3Na^+} \right]$$

Mass balance:

$$\frac{d}{dt} q_m^{E_o} = v_m^6 - v_m^1$$

$$\frac{d}{dt} q_m^{E_o.3Na^+} = v_m^1 - v_m^2 - v_m^7$$

$$\frac{d}{dt} q_m^{E_o.3Na^+.H^+.Glu} = v_m^2 - v_m^3$$

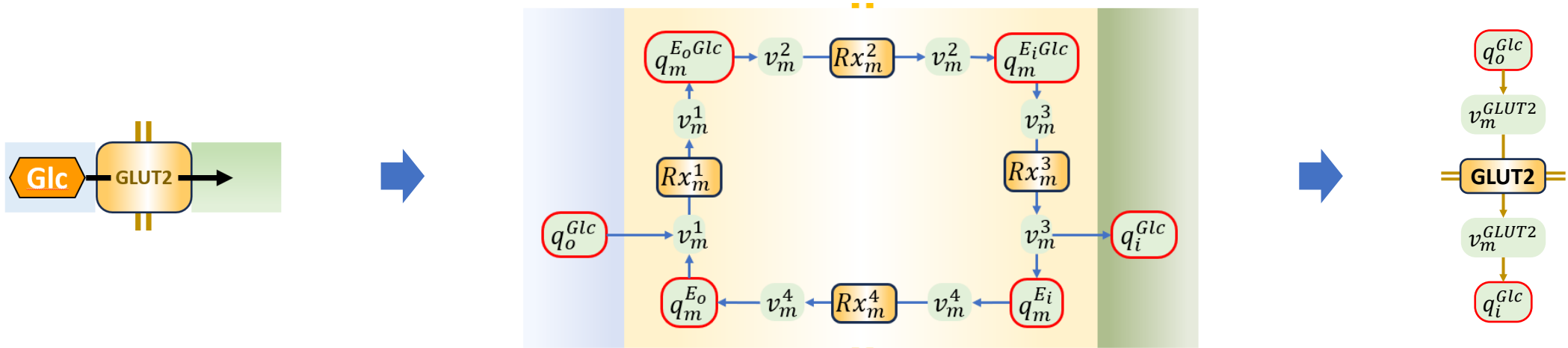
$$\frac{d}{dt} q_m^{E_i.3Na^+.H^+.Glu} = v_m^3 - v_m^4$$

$$\frac{d}{dt} q_m^{E_i.3Na^+} = v_m^4 + v_m^7 - v_m^5$$

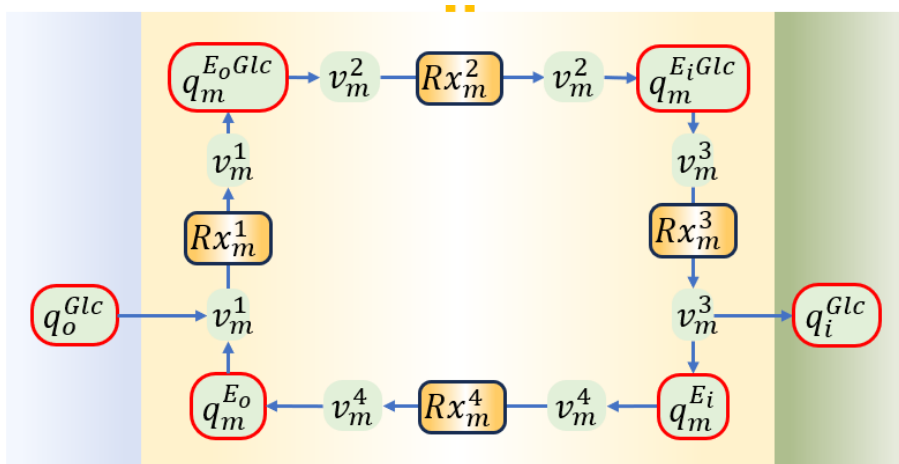
$$\frac{d}{dt} q_m^{E_i.K^+} = v_m^5 - v_m^6$$

SLC2A2 (GLUT2) Glucose transport (SLCtables)

(see [model on PMR](#))



SLC2A2 (GLUT2) .. continued



Reaction fluxes

$$v_m^1 = \kappa_1 \left[\bar{q}_m^{E_o} \cdot \bar{q}_o^{Glc} - \bar{q}_m^{E_o \cdot Glc} \right]$$

$$v_m^2 = \kappa_2 \left[\bar{q}_m^{E_o \cdot Glc} - \bar{q}_m^{E_i \cdot Glc} \right]$$

$$v_m^3 = \kappa_3 \left[\bar{q}_m^{E_i \cdot Glc} - \bar{q}_m^{E_i} \cdot \bar{q}_i^{Glc} \right]$$

$$v_m^4 = \kappa_4 \left[\bar{q}_m^{E_i} - \bar{q}_m^{E_o} \right]$$

Mass balance:

$$\frac{d}{dt} q_m^{E_o} = v_m^4 - v_m^1$$

$$\frac{d}{dt} q_m^{E_o \cdot Glc} = v_m^1 - v_m^2$$

$$\frac{d}{dt} q_m^{E_i \cdot Glc} = v_m^2 - v_m^3$$

$$\frac{d}{dt} q_m^{E_i} = v_m^3 - v_m^4$$

$$\frac{d}{dt} q_o^{Glc} = -v_m^1$$

$$\frac{d}{dt} q_i^{Glc} = v_m^3$$

SLC2A2 (GLUT2) .. Reduced model

$$v_m^{GLUT2} = v_m (\bar{q}_o^{Glc} - \bar{q}_i^{Glc}) / \left(1 + \frac{\bar{q}_o^{Glc}}{k_m^1} + \frac{\bar{q}_i^{Glc}}{k_m^2} + \frac{\bar{q}_o^{Glc} \bar{q}_i^{Glc}}{k_m^3} \right);$$

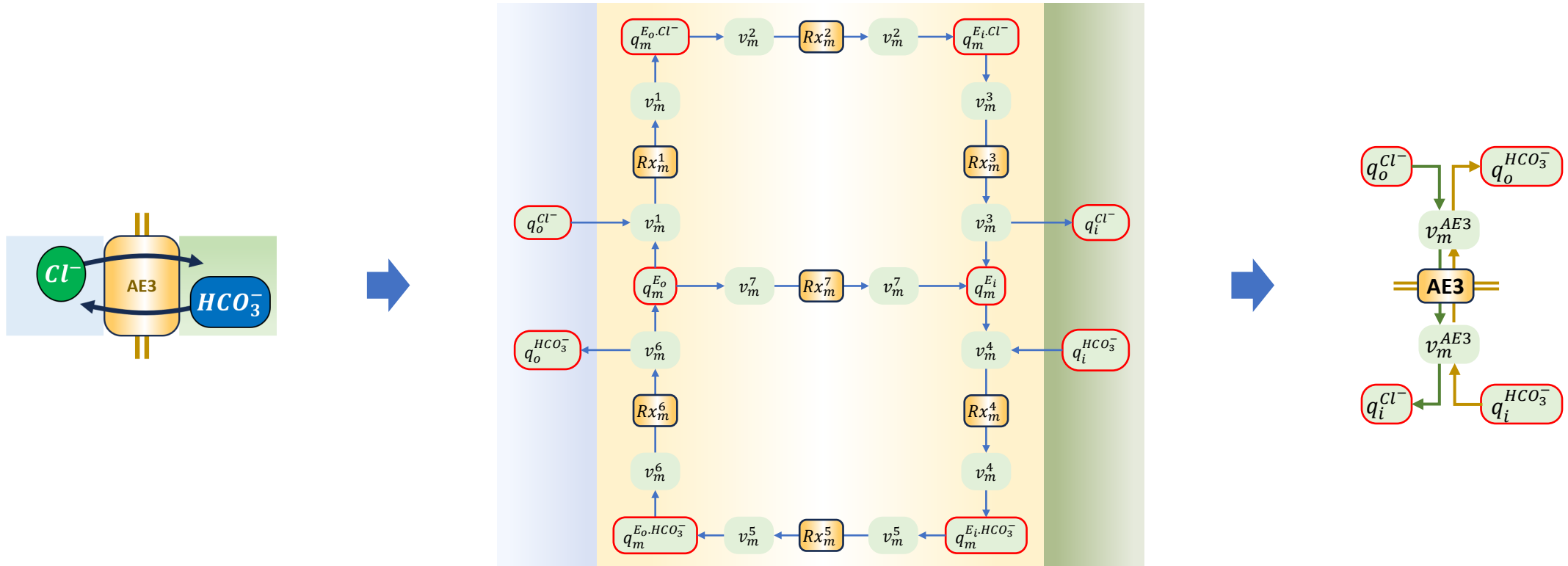
4 parameters

$$\text{with } k_m^1 = \frac{(K_m^{Eo})^{-1} + (K_m^{Ei})^{-1}}{(K_m^{EoGlc})^{-1} + \Lambda \cdot (K_m^{Ei})^{-1}}, \quad k_m^2 = \frac{(K_m^{Eo})^{-1} + (K_m^{Ei})^{-1}}{(K_m^{EiGlc})^{-1} + \Lambda \cdot (K_m^{Eo})^{-1}}, \quad k_m^3 = \frac{(K_m^{Eo})^{-1} + (K_m^{Ei})^{-1}}{\Lambda \cdot [(K_m^{EoGlc})^{-1} + (K_m^{EiGlc})^{-1}]}, \quad v_m = \frac{q_{tot} \kappa_2}{(K_m^{Eo})^{-1} + (K_m^{Ei})^{-1}}, \quad \Lambda = \frac{\kappa_2}{\kappa_4}$$

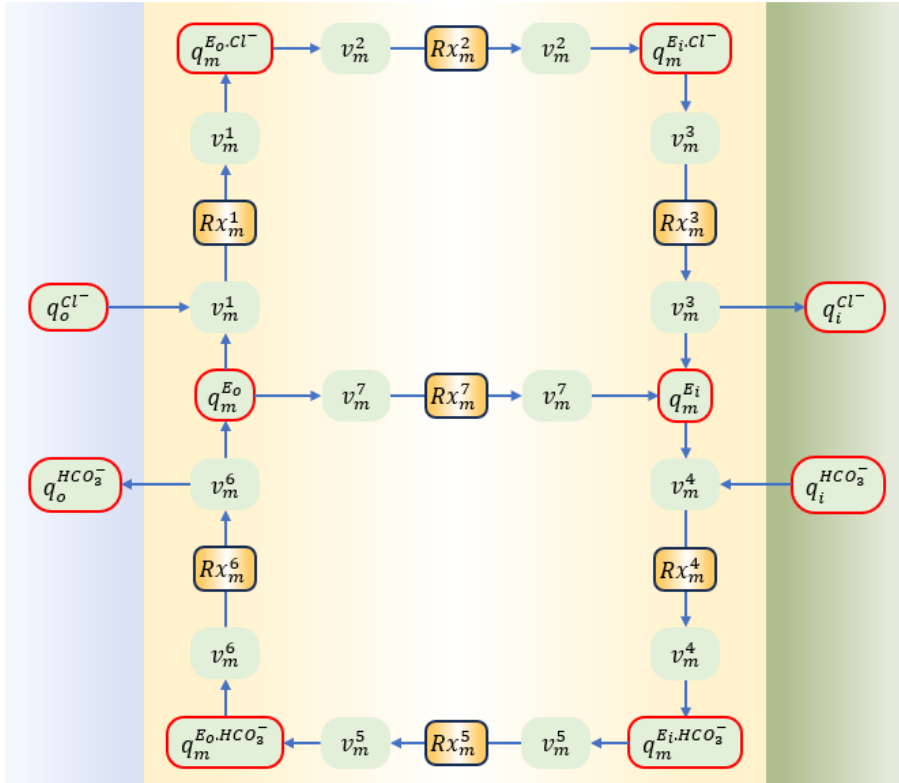
SLC4A3 (AE3) Chloride-bicarbonate exchange (SLCtables)

(see [model on PMR](#))

AE = 'anion exchanger'



SLC4A3 (AE3) .. continued



Reaction fluxes

$$v_m^1 = \kappa_1 \left[\bar{q}_m^{E_o} \cdot \bar{q}_o^{\text{Cl}^-} - \bar{q}_m^{E_o, \text{Cl}^-} \right]$$

$$v_m^2 = \kappa_2 \left[\bar{q}_m^{E_o, \text{Cl}^-} - \bar{q}_m^{E_i, \text{Cl}^-} \right]$$

$$v_m^3 = \kappa_3 \left[\bar{q}_m^{E_i, \text{Cl}^-} - \bar{q}_m^{E_i} \cdot \bar{q}_i^{\text{Cl}^-} \right]$$

$$v_m^4 = \kappa_4 \left[\bar{q}_m^{E_i} \cdot \bar{q}_o^{\text{HCO}_3^-} - \bar{q}_i^{E_i, \text{HCO}_3^-} \right]$$

$$v_m^5 = \kappa_5 \left[\bar{q}_i^{E_i, \text{HCO}_3^-} - \bar{q}_i^{E_o, \text{HCO}_3^-} \right]$$

$$v_m^6 = \kappa_6 \left[\bar{q}_i^{E_o, \text{HCO}_3^-} - \bar{q}_m^{E_o} \cdot \bar{q}_o^{\text{HCO}_3^-} \right]$$

$$v_m^7 = \kappa_7 \left[\bar{q}_m^{E_o} - \bar{q}_m^{E_i} \right]$$

Mass balance:

$$\frac{d}{dt} q_m^{E_o} = v_m^6 - v_m^1 - v_m^7$$

$$\frac{d}{dt} q_m^{E_o, \text{Cl}^-} = v_m^1 - v_m^2$$

$$\frac{d}{dt} q_m^{E_i, \text{Cl}^-} = v_m^2 - v_m^3$$

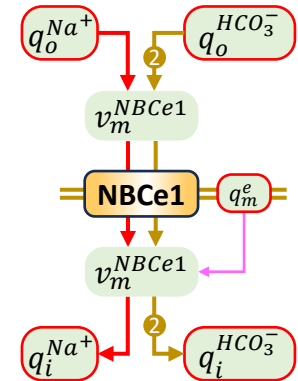
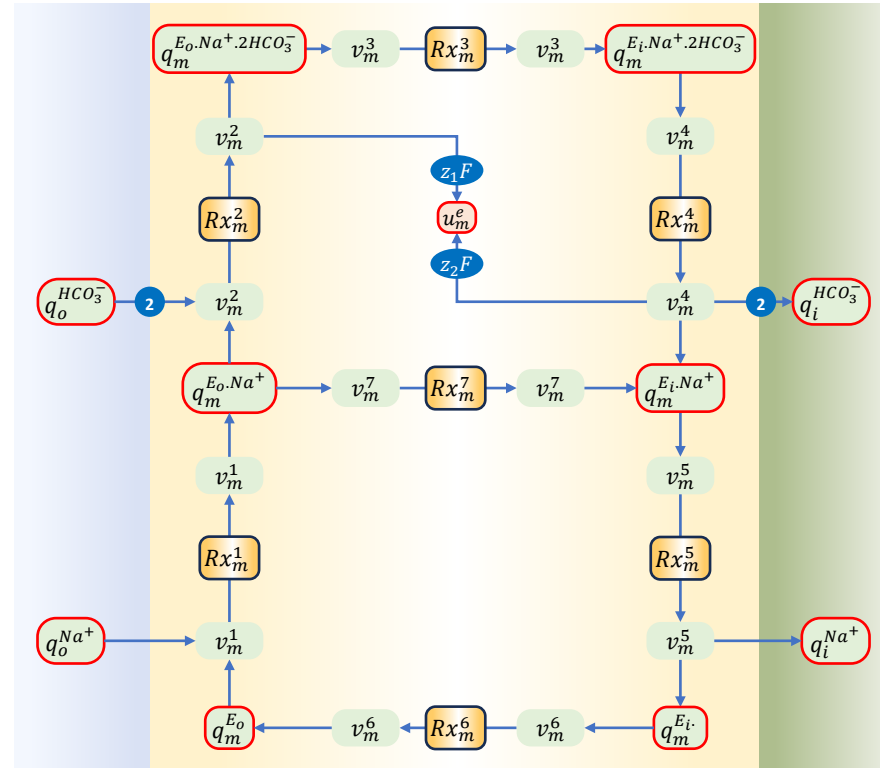
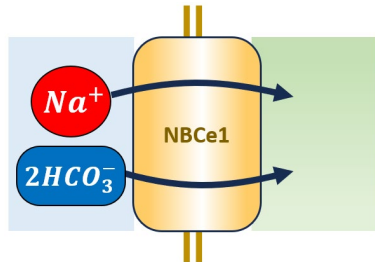
$$\frac{d}{dt} q_m^{E_i} = v_m^3 + v_m^7 - v_m^4$$

$$\frac{d}{dt} q_m^{E_i, \text{HCO}_3^-} = v_m^4 - v_m^5$$

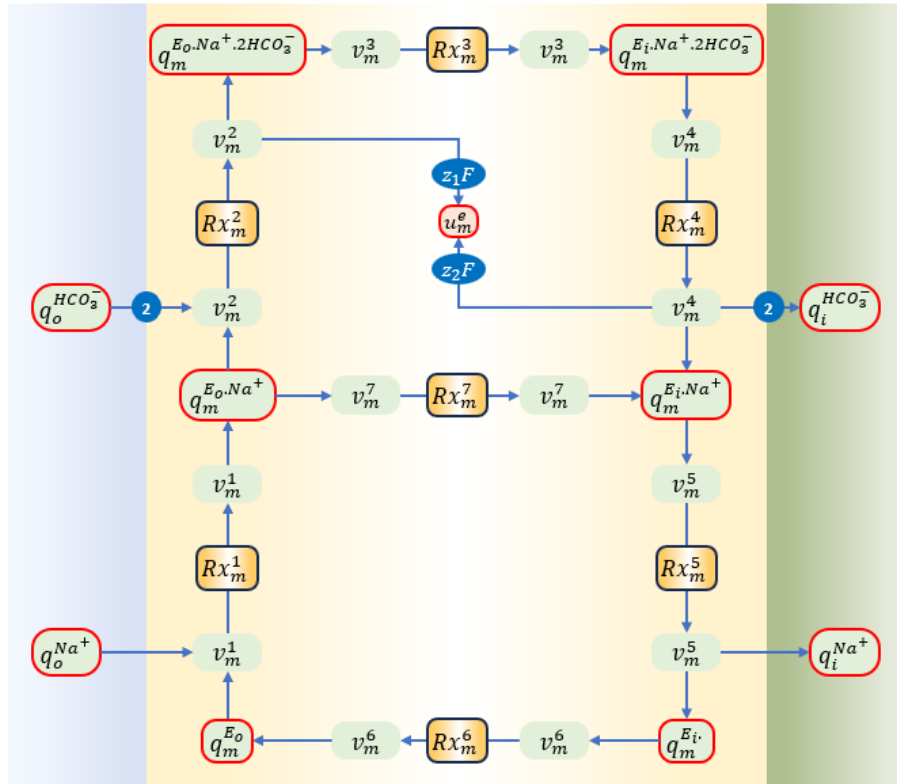
$$\frac{d}{dt} q_m^{E_o, \text{HCO}_3^-} = v_m^5 - v_m^6$$

SLC4A4 (NBCe1) Sodium-bicarbonate cotransport (SLCtables)

(see [model on PMR](#))



SLC4A4 (NBCe1) .. continued



Reaction fluxes

$$v_m^1 = \kappa_1 [\bar{q}_m^{E_o} \cdot \bar{q}_o^{Na^+} - \bar{q}_m^{E_o.Na^+}]$$

$$v_m^2 = \kappa_2 \left[\bar{q}_m^{E_o.Na^+} \cdot \left(\bar{q}_o^{HCO_3^-} \right)^2 - \bar{q}_m^{E_o.Na^+.2HCO_3^-} \cdot e^{\frac{z_1 F u_m^e}{RT}} \right]$$

$$v_m^3 = \kappa_3 \left[\bar{q}_m^{E_o.Na^+.2HCO_3^-} - \bar{q}_m^{E_i.Na^+.2HCO_3^-} \right]$$

$$v_m^4 = \kappa_4 \left[\bar{q}_m^{E_i.Na^+.2HCO_3^-} - \bar{q}_m^{E_i.Na^+} \cdot \left(\bar{q}_i^{HCO_3^-} \right)^2 \cdot e^{\frac{z_2 F u_m^e}{RT}} \right]$$

$$v_m^5 = \kappa_5 \left[\bar{q}_m^{E_i.Na^+} - \bar{q}_m^{E_i} \cdot \bar{q}_i^{Na^+} \right]$$

$$v_m^6 = \kappa_6 \left[\bar{q}_m^{E_i} - \bar{q}_m^{E_o} \right]$$

$$v_m^7 = \kappa_7 \left[\bar{q}_m^{E_o.Na^+} - \bar{q}_m^{E_i.Na^+} \right]$$

Mass balance:

$$\frac{d}{dt}q_m^{E_o} = v_m^6 - v_m^1$$

$$\frac{d}{dt}q_m^{E_o.Na^+} = v_m^1 - v_m^2 - v_m^7$$

$$\frac{d}{dt}q_m^{E_o.Na^+.2HCO_3^-} = v_m^2 - v_m^3$$

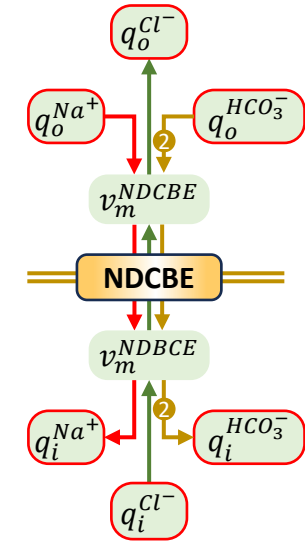
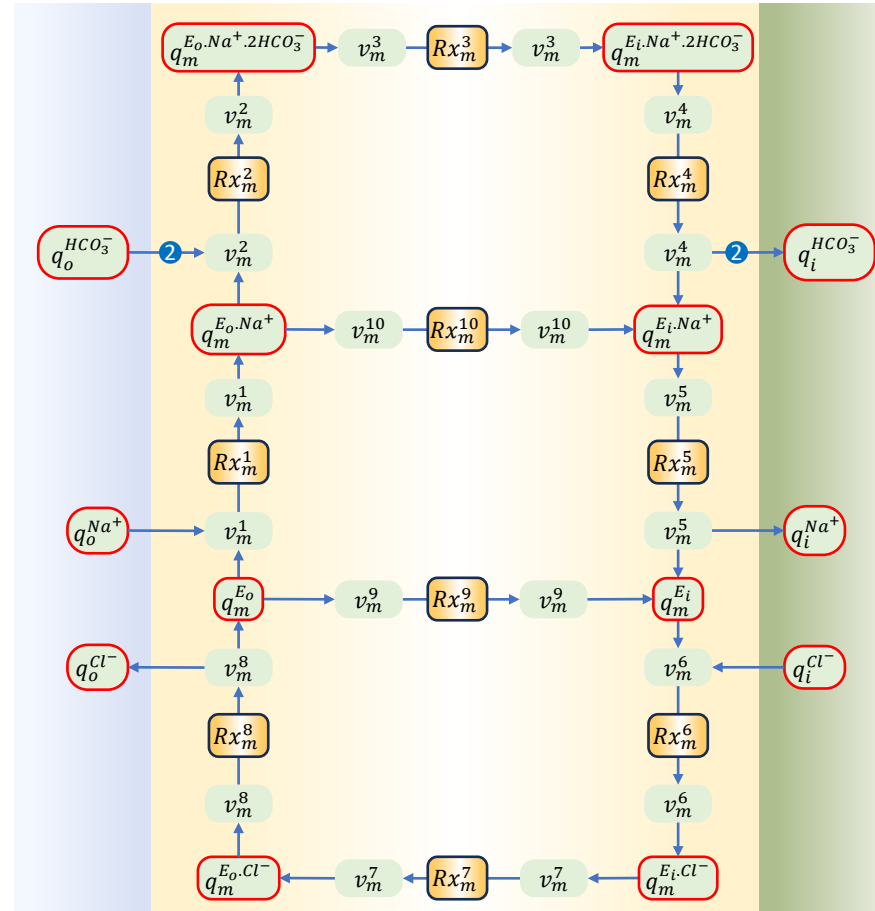
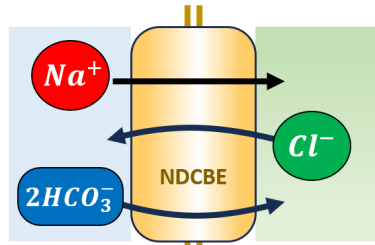
$$\frac{d}{dt}q_m^{Ei.Na^+.2HCO_3^-} = v_m^3 - v_m^4$$

$$\frac{d}{dt}q_m^{Ei.Na^+} = v_m^4 + v_m^7 - v_m^5$$

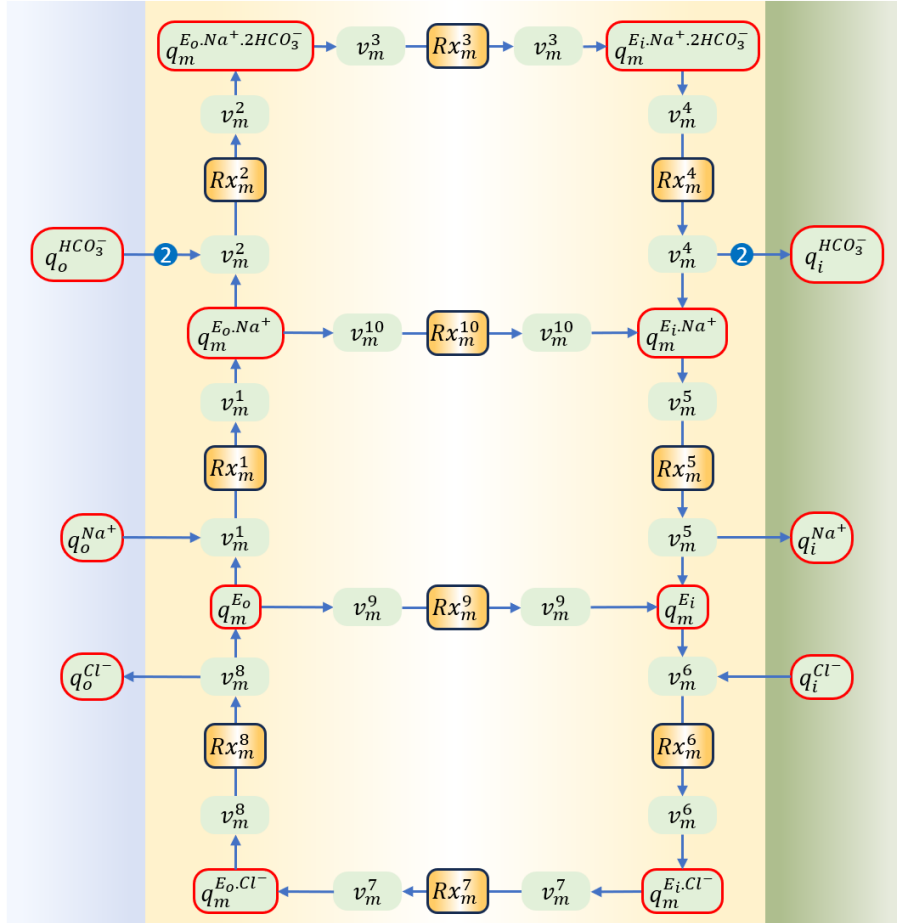
$$\frac{d}{dt}q_m^{E_i} = v_m^5 - v_m^6$$

SLC4A8 (NDCBE) Sodium-driven chloride-bicarbonate exchange ([SLCtables](#))

(see [model on PMR](#))



SLC4A8 (NDCBE) .. continued



Reaction fluxes

$$v_m^1 = \kappa_1 \left[\bar{q}_m^{E_o} \cdot \bar{q}_o^{Na^+} - \bar{q}_m^{E_o.Na^+} \right]$$

$$v_m^2 = \kappa_2 \left[\bar{q}_m^{E_o.Na^+} \cdot (q_o^{HCO_3^-})^2 - \bar{q}_m^{E_o.Na^+.2HCO_3^-} \right]$$

$$v_m^3 = \kappa_3 \left[\bar{q}_m^{E_o.Na^+.2HCO_3^-} - \bar{q}_m^{E_i.Na^+.2HCO_3^-} \right]$$

$$v_m^4 = \kappa_4 \left[\bar{q}_m^{E_i.Na^+.2HCO_3^-} - \bar{q}_m^{E_i.Na^+} \cdot (q_i^{HCO_3^-})^2 \right]$$

$$v_m^5 = \kappa_5 \left[\bar{q}_m^{E_i.Na^+} - \bar{q}_m^{E_i} \cdot \bar{q}_i^{Na^+} \right]$$

$$v_m^6 = \kappa_6 \left[\bar{q}_m^{E_i} \cdot \bar{q}_i^{Cl^-} - \bar{q}_m^{E_i.Cl^-} \right]$$

$$v_m^7 = \kappa_7 \left[\bar{q}_m^{E_i.Cl^-} - \bar{q}_m^{E_o.Cl^-} \right]$$

$$v_m^8 = \kappa_8 \left[\bar{q}_m^{E_o.Cl^-} - \bar{q}_m^{E_o} \cdot \bar{q}_o^{Cl^-} \right]$$

$$v_m^9 = \kappa_9 \left[\bar{q}_m^{E_o} - \bar{q}_m^{E_i} \right]$$

$$v_m^{10} = \kappa_{10} \left[\bar{q}_m^{E_o.Na^+} - \bar{q}_m^{E_i.Na^+} \right]$$

Mass balance:

$$\frac{d}{dt} q_m^{E_o} = v_m^8 - v_m^1 - v_m^9$$

$$\frac{d}{dt} q_m^{E_o.Na^+} = v_m^1 - v_m^2 - v_m^{10}$$

$$\frac{d}{dt} \bar{q}_m^{E_o.Na^+.2HCO_3^-} = v_m^2 - v_m^3$$

$$\frac{d}{dt} \bar{q}_m^{E_i.Na^+.2HCO_3^-} = v_m^3 - v_m^4$$

$$\frac{d}{dt} q_m^{E_i.Na^+} = v_m^4 + v_m^{10} - v_m^5$$

$$\frac{d}{dt} q_m^{E_i} = v_m^5 + v_m^9 - v_m^6$$

$$\frac{d}{dt} q_m^{E_i.Cl^-} = v_m^6 - v_m^7$$

$$\frac{d}{dt} q_m^{E_o.Cl^-} = v_m^7 - v_m^8$$

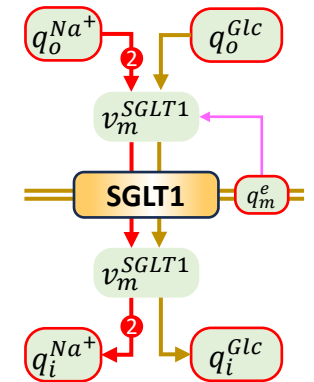
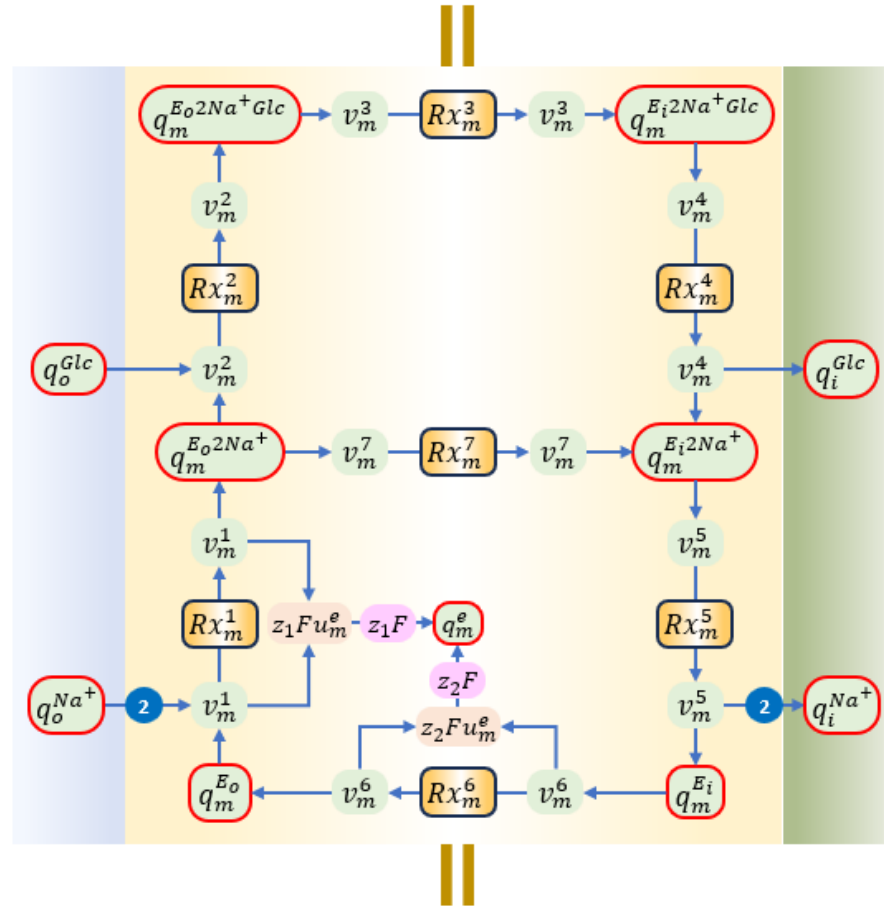
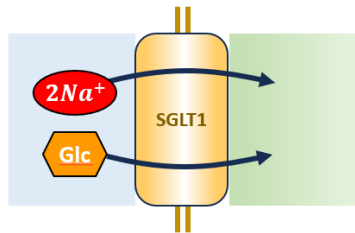
$$\frac{d}{dt} q_o^{Cl^-} = v_m^8, \quad \frac{d}{dt} q_i^{Cl^-} = -v_m^6$$

$$\frac{d}{dt} q_o^{Na^+} = -v_m^1, \quad \frac{d}{dt} q_i^{Na^+} = v_m^5$$

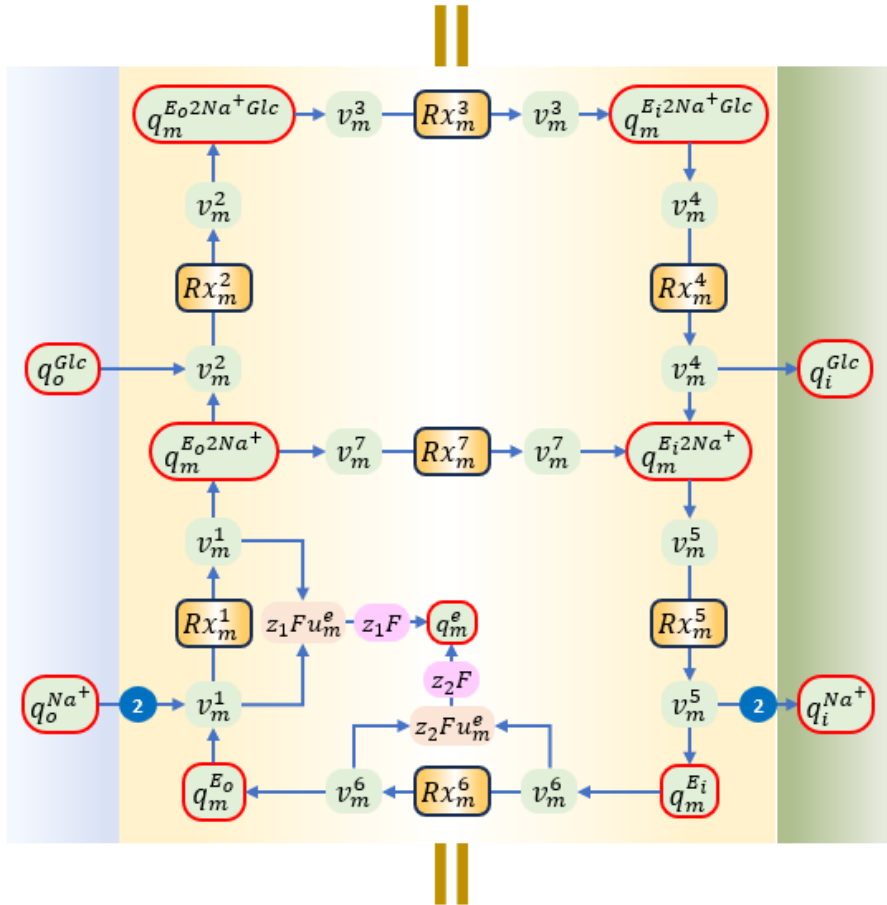
$$\frac{d}{dt} q_o^{HCO_3^-} = -2v_m^2, \quad \frac{d}{dt} q_i^{HCO_3^-} = 2v_m^4$$

SLC5A1 (SGLT1) Sodium-driven glucose transport (SLCtables)

(see [model on PMR](#))



SLC5A1 (SGLT1) .. continued



Reaction fluxes

$$v_m^1 = \kappa_1 \left[\bar{q}_m^{E_o} \cdot (\bar{q}_o^{Na^+})^2 \cdot e^{\frac{-z_1 F u_m^e}{RT}} - \bar{q}_m^{E_o.2Na^+} \cdot e^{\frac{z_1 F u_m^e}{RT}} \right]$$

$$v_m^2 = \kappa_2 \left[\bar{q}_m^{E_o.Na^+} \cdot \bar{q}_o^{Glc} - \bar{q}_m^{E_o.2Na^+.Glc} \right]$$

$$v_m^3 = \kappa_3 \left[\bar{q}_m^{E_o.2Na^+.Glc} - \bar{q}_m^{E_i.2Na^+.Glc} \right]$$

$$v_m^4 = \kappa_4 \left[\bar{q}_m^{E_i.2Na^+.Glc} - \bar{q}_m^{E_i.2Na^+} \cdot \bar{q}_i^{Glc} \right]$$

$$v_m^5 = \kappa_5 \left[\bar{q}_m^{E_i.2Na^+} - \bar{q}_m^{E_i} \cdot (\bar{q}_i^{Na^+})^2 \right]$$

$$v_m^6 = \kappa_6 \left[\bar{q}_m^{E_i} \cdot e^{\frac{-z_2 F u_m^e}{RT}} - \bar{q}_m^{E_o} \cdot e^{\frac{z_2 F u_m^e}{RT}} \right]$$

$$v_m^7 = \kappa_7 \left[\bar{q}_m^{E_o.2Na^+} - \bar{q}_m^{E_i.2Na^+} \right]$$

Mass balance:

$$\frac{d}{dt} q_m^{E_o} = v_m^6 - v_m^1$$

$$\frac{d}{dt} q_m^{E_o.Na^+} = v_m^1 - v_m^2 - v_m^7$$

$$\frac{d}{dt} q_m^{E_o.2Na^+.Glc} = v_m^2 - v_m^3$$

$$\frac{d}{dt} q_m^{E_i.2Na^+.Glc} = v_m^3 - v_m^4$$

$$\frac{d}{dt} q_m^{E_i.Na^+} = v_m^4 + v_m^7 - v_m^5$$

$$\frac{d}{dt} q_m^{E_i} = v_m^5 - v_m^6$$

$$\frac{d}{dt} q_o^{Na^+} = -2v_m^1$$

$$\frac{d}{dt} q_i^{Na^+} = 2v_m^5$$

$$\frac{d}{dt} q_o^{Glc} = -v_m^2$$

$$\frac{d}{dt} q_i^{Glc} = v_m^4$$

SLC5A1 (SGLT1) .. Reduced model

$$v_m^{SGLT1} = \frac{A_0}{A_1 A_2 + A_3 A_4} \cdot \left\{ \frac{\bar{q}_o^{Glc}}{\bar{q}_i^{Glc}} - \left(\frac{\bar{q}_i^{Na^+}}{\bar{q}_o^{Na^+}} \right)^2 \cdot e^{2Fu_m^e/RT} \right\}$$

$$A_0 = q_{tot} \kappa_3 K_1 \quad A_1 = \lambda_2 + \lambda_3 \cdot \bar{q}_o^{Glc} + \frac{e^{2z_1 Fu_m^e/RT}}{(\bar{q}_o^{Na^+})^2} \quad A_2 = \frac{1}{\bar{q}_i^{Glc}} + \lambda_1 \cdot (\bar{q}_i^{Na^+})^2 e^{z_2 Fu_m^e/RT}$$

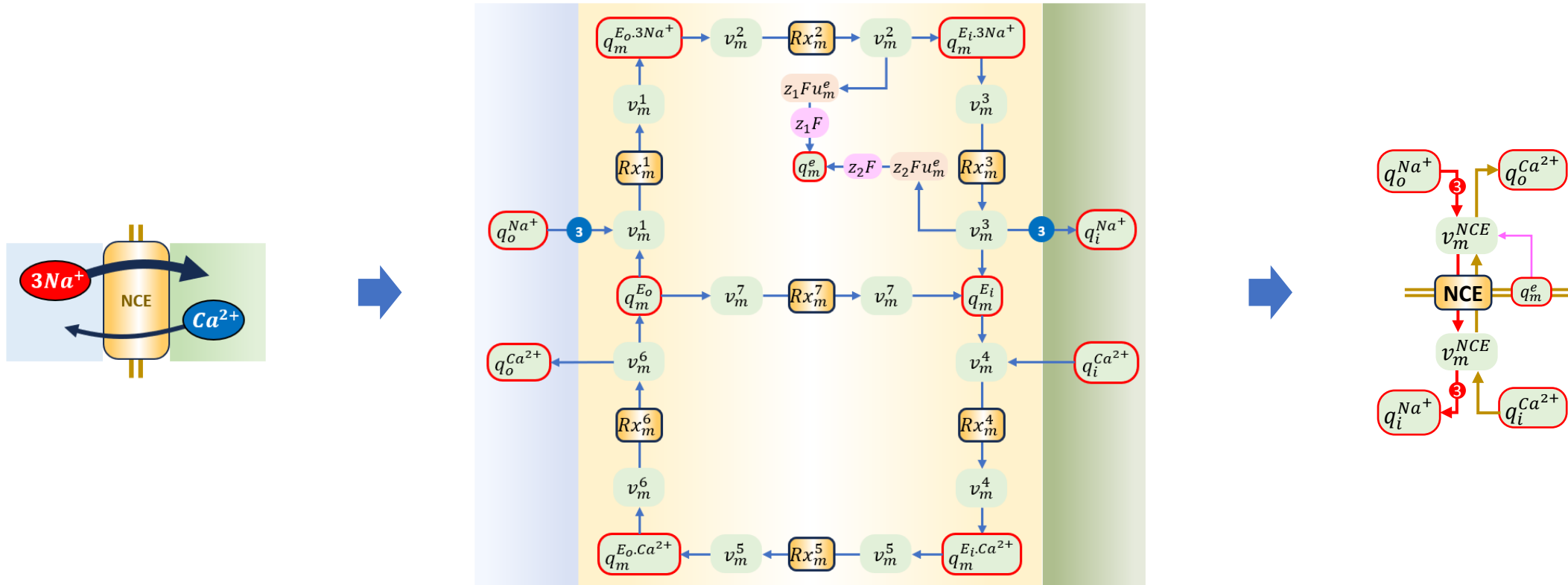
$$A_3 = \left(\lambda_4 + \frac{\lambda_5}{\bar{q}_i^{Glc}} \right) \cdot (\bar{q}_i^{Na^+})^2 + \frac{\lambda_6}{\bar{q}_i^{Glc}} \quad A_4 = \lambda_1 \bar{q}_o^{Glc} \cdot e^{z_2 Fu_m^e/RT} + \frac{e^{2Fu_m^e/RT}}{(\bar{q}_o^{Na^+})^2}$$

$$\lambda_1 = \frac{\kappa_3}{\kappa_6}, \lambda_2 = \frac{K_1}{K_2}, \lambda_3 = \frac{K_1}{K_3}, \lambda_4 = \frac{K_1}{K_4}, \lambda_5 = \frac{K_1}{K_5}, \lambda_6 = \frac{K_1}{K_6},$$

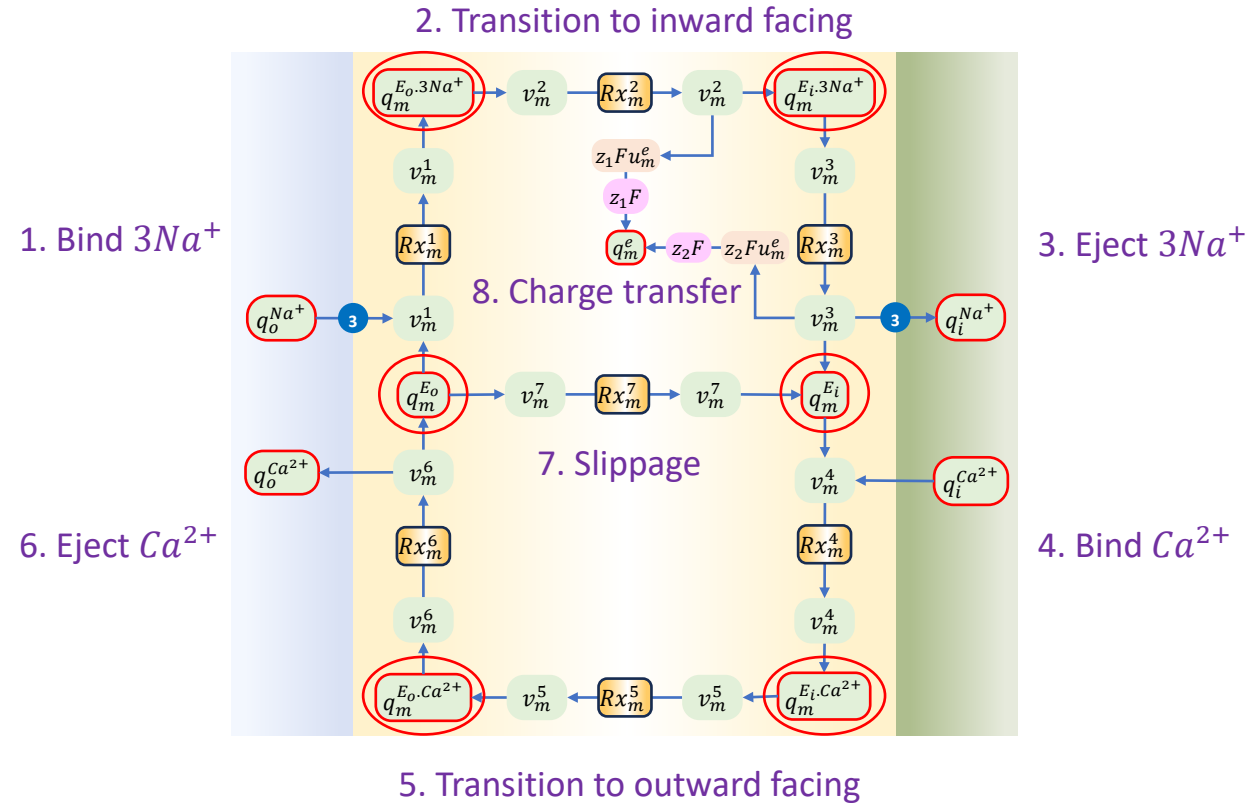
$$K_1 = K_m^{E_o}, K_2 = K_m^{E_o 2Na^+}, K_3 = K_m^{E_o 2Na^+ Glc}, K_4 = K_m^{E_i 2Na^+ Glc}, K_5 = K_m^{E_i 2Na^+}, K_6 = K_m^{E_i}$$

SLC8A1 (NCE/NCX) Sodium-calcium exchange (SLCtables)

(see [model on PMR](#))



SLC8A1 (NCE) .. continued



Reaction fluxes

$$v_m^1 = \kappa_1 \left[\bar{q}_m^{E_o} \cdot (\bar{q}_o^{Na^+})^3 - \bar{q}_m^{E_o \cdot 3Na^+} \right]$$

$$v_m^2 = \kappa_2 \left[\bar{q}_m^{E_o \cdot 3Na^+} - \bar{q}_m^{E_i \cdot 3Na^+} \cdot e^{\frac{z_1 F u_m^e}{RT}} \right]$$

$$v_m^3 = \kappa_3 \left[\bar{q}_m^{E_i \cdot 3Na^+} - \bar{q}_m^{E_i} \cdot (\bar{q}_i^{Na^+})^3 \cdot e^{\frac{z_2 F u_m^e}{RT}} \right]$$

$$v_m^4 = \kappa_4 \left[\bar{q}_m^{E_i} \cdot \bar{q}_o^{Ca^{2+}} - \bar{q}_i^{E_i \cdot Ca^{2+}} \right]$$

$$v_m^5 = \kappa_5 \left[\bar{q}_i^{E_i \cdot Ca^{2+}} - \bar{q}_i^{E_o \cdot Ca^{2+}} \right]$$

$$v_m^6 = \kappa_6 \left[\bar{q}_i^{E_o \cdot Ca^{2+}} - \bar{q}_m^{E_o} \cdot \bar{q}_o^{Ca^{2+}} \right]$$

$$v_m^7 = \kappa_7 \left[\bar{q}_m^{E_o} - \bar{q}_m^{E_i} \right]$$

Mass balance:

$$\frac{d}{dt} q_m^{E_o} = v_m^6 - v_m^1 - v_m^7$$

$$\frac{d}{dt} q_m^{E_o \cdot 3Na^+} = v_m^1 - v_m^2$$

$$\frac{d}{dt} q_m^{E_i \cdot 3Na^+} = v_m^2 - v_m^3$$

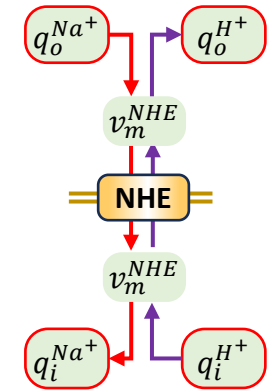
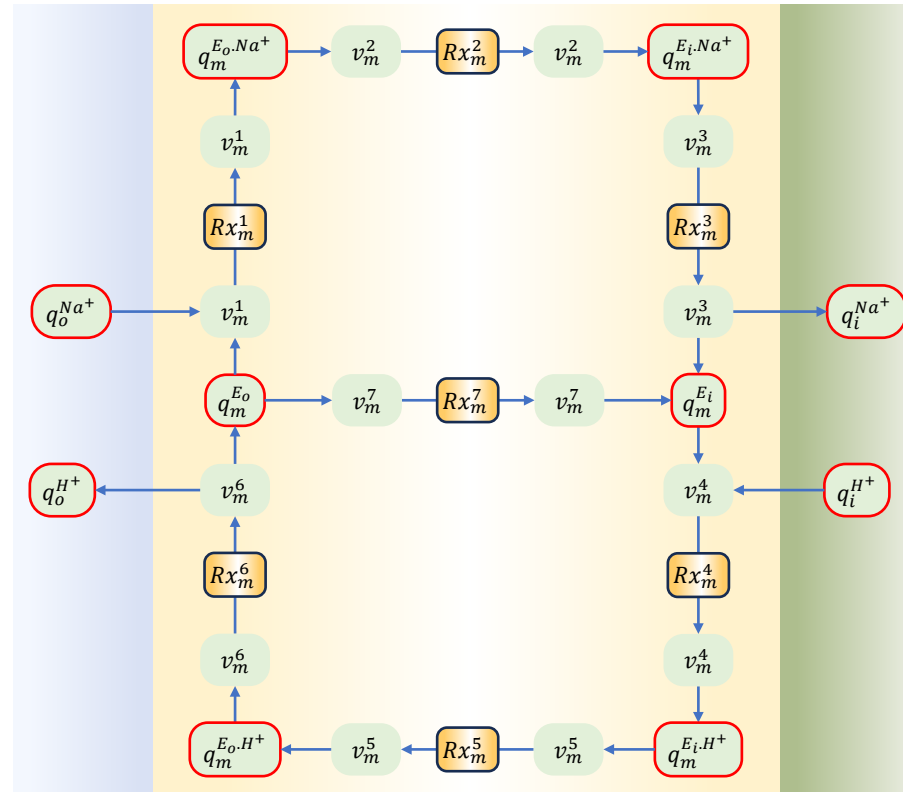
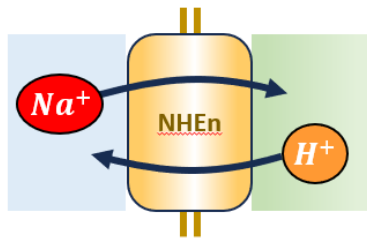
$$\frac{d}{dt} q_m^{E_i} = v_m^3 + v_m^7 - v_m^4$$

$$\frac{d}{dt} q_m^{E_i \cdot Ca^{2+}} = v_m^4 - v_m^5$$

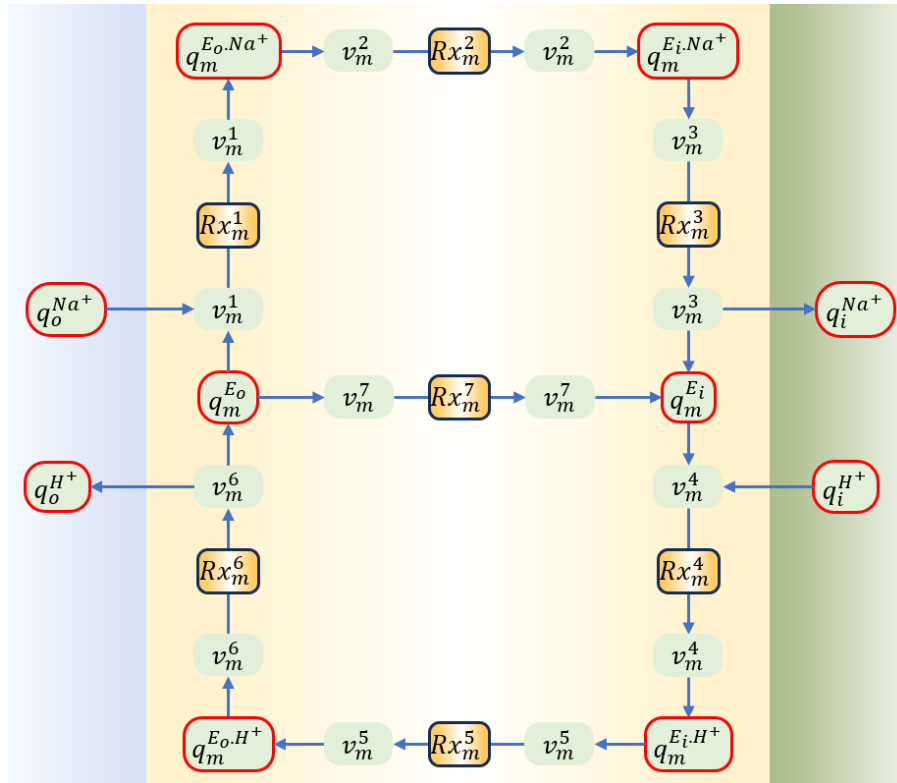
$$\frac{d}{dt} q_m^{E_o \cdot Ca^{2+}} = v_m^5 - v_m^6$$

SLC9A1 (NHE1) Sodium-hydrogen exchange (SLCtables)

(see [model on PMR](#))



SLC9A1 (NHE1) .. continued



Reaction fluxes

$$v_m^1 = \kappa_1 [\bar{q}_m^{E_o} \cdot \bar{q}_o^{Na^+} - \bar{q}_m^{E_o.Na^+}]$$

$$v_m^2 = \kappa_2 [\bar{q}_m^{E_o.Na^+} - \bar{q}_m^{E_i.Na^+}]$$

$$v_m^3 = \kappa_3 [\bar{q}_m^{E_i.Na^+} - \bar{q}_m^{E_i} \cdot \bar{q}_i^{Na^+}]$$

$$v_m^4 = \kappa_4 [\bar{q}_m^{E_i} \cdot \bar{q}_i^{H^+} - \bar{q}_i^{E_i.H^+}]$$

$$v_m^5 = \kappa_5 [\bar{q}_i^{E_i.H^+} - \bar{q}_i^{E_o.H^+}]$$

$$v_m^6 = \kappa_6 [\bar{q}_i^{E_o.H^+} - \bar{q}_m^{E_o} \cdot \bar{q}_o^{H^+}]$$

$$v_m^7 = \kappa_7 [\bar{q}_m^{E_o} - \bar{q}_m^{E_i}]$$

Mass balance:

$$\frac{d}{dt} q_m^{E_o} = v_m^6 - v_m^1 - v_m^7$$

$$\frac{d}{dt} q_m^{E_o.Na^+} = v_m^1 - v_m^2$$

$$\frac{d}{dt} q_m^{E_i.Na^+} = v_m^2 - v_m^3$$

$$\frac{d}{dt} q_m^{E_i} = v_m^3 + v_m^7 - v_m^4$$

$$\frac{d}{dt} q_m^{E_i.H^+} = v_m^4 - v_m^5$$

$$\frac{d}{dt} q_m^{E_o.H^+} = v_m^5 - v_m^6$$